MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

CONVERTER, SERIAL/PARALLEL, DIGITAL, KEY CODE EQF

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic digital serial/parallel converter module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table II.

Name	Converter, serial/parallel
Family	Digital
Key code	EQF
Size	1A
Weight	35 grams maximum
Failure rate	0.513 failures/10^6 hours maxim
Environment	Class I as specified in MIL-STD-1389
Supply voltage	5 Vdc ±10 percent
Maximum power dissipation	1,848 mW
Input voltage	
Low level	-0.5 V minimum, 0.7 V maximum
High level	2.0 V minimum, V_{CC} maximum
Input current, SDU, SDL	
Low level (V_{IL} = 0.0 V)	-0.8 mA maximum
(V_{IL} = 0.4 V)	-0.72 mA maximum
High level (V_{IH} = 2.5 V)	40 µA maximum
(V_{IH} = 5.5 V)	200 µA maximum
Input current, URC, LRC	
Low level (V_{IL} = 0.0 V)	-3.91 mA maximum
(V_{IL} = 0.4 V)	-3.52 mA maximum
High level (V_{IH} = 2.5 V)	160 µA maximum
(V_{IH} = 5.5 V)	800 µA maximum
Input current, HRM	
Low level (V_{IL} = 0.0 V)	-3.20 mA maximum
(V_{IL} = 0.4 V)	-2.88 mA maximum
High level (V_{IH} = 2.5 V)	160 µA maximum
(V_{IH} = 5.5 V)	800 µA maximum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Output voltage, SDO	
Low level (I_{OL} = 3.64 mA)	0.4 V maximum
High level (I_{OH} = -380 µA)	2.5 V minimum
Output voltage, D3, D7, D11, D15, D19, D23, D27	
Low level (I_{OL} = 3.28 mA)	0.4 V maximum
High level (I_{OH} = -360 µA)	2.5 V minimum
Output voltage, D0-D2, D4-D6, D8-D10, D12-D14, D16-D18, D20-D22, D24-D26, D28-D31	
Low level (I_{OL} = 4.0 mA)	0.4 V maximum
High level (I_{OH} = -400 µA)	2.5 V minimum
Output drive current maximum capability, SDO	
Low level (V_{OL} = 0.4 V)	3.64 mA
High level (V_{OH} = 2.5 V)	-380 µA
Output drive current maximum capability, D3, D7, D11, D15, D19, D23, D27	
Low level (V_{OL} = 0.4 V)	3.28 mA
High level (V_{OH} = 2.5 V)	-360 µA
Output drive current maximum capability, D0-D2, D4-D6, D8-D10, D12-D14, D16-D18, D20-D22, D24-D26, D28-D31	
Low level (V_{OL} = 0.4 V)	4.0 mA
High level (V_{OH} = 2.5 V)	-400 µA
Propagation delay time (t_{PLH})	
LRC to D0-D31, URC to SDO	
50 pF load	10 ns minimum, 40 ns maximum
150 pF load	10 ns minimum, 55 ns maximum
Propagation delay time (t_{PHL})	
LRC to D0-D31, URC to SDO	
50 pF load	10 ns minimum, 45 ns maximum
150 pF load	10 ns minimum, 55 ns maximum
Setup time (t_{setup}), high level and low level	
SDU, SDL	20 ns minimum
HRM	40 ns minimum
Hold time (t_{hold})	
SDU, SDL	5 ns minimum
HRM	5 ns minimum
Pulse width (t_{PW})	
URC, LRC	20 ns minimum
Maximum clock frequency (f_{max})	
URC, LRC	
50 pF load	15 MHz
Clock delay	
URC to LRC	65 ns minimum

1.3 Module function. The 32-bit serial/parallel converter module, designated as key code EQF, is a Standard Electronic Modules Program (SEMP) electronic module with 0.100 inch (2.54 mm) grid center contacts. This module contains two 32-bit ranks to perform serial to parallel conversion. The upper rank clock (URC) synchronously shifts data on serial data upper (SDU) into the upper rank 32-bit shift register. Serial output data appears at the upper rank output SDO. The lower rank clock (LRC) shifts data on the serial data lower (SDL) into the lower rank 32-bit holding register when the holding register mode (HRM) is high or loads the contents of the upper rank register when HRM is low. Serial shifting or parallel loading at the lower rank occurs after the positive transition of the URC or LRC.

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1.3.1 Input to output relationship. Each 32-bit serial/parallel converter has the input to output relationship of table I.

1.3.2 Parallel outputs. Paralleling outputs is not permissible.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on dat of invitation for bids or request for proposal, form a part of this specification t the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Snecification for

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(All military standards and military specifications shall be ordered from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

TABLE I. Input to output relationship.

Upper Rank

Input at t_n	Output at t_n+32
SDU	SDO
H L	H L

Lower Rank

Input			Upper Rank Contents	Output at t_n	Output at t_n+1
HRM	LRC	SDL	Q_0 - - Q_{31}	D_0 D_1 D_2 - - D_{31}	D_0 D_1 D_2 - - D_{31}
L	↑	X	q_0 - - q_{31}	X - - - - - - - X	q_0 - - - - - q_{31}
H	↑	L	X - - - X	d_0 - - - - d_{31}	L d_0 d_1 - d_{30}
H	↑	H	X - - - X	d_0 - - - - d_{31}	H d_0 d_1 - d_{30}

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NOTES:

1. H = High-voltage level (steady state).
2. L = Low-voltage level (steady state).
3. X = Do not care.
4. ↑ = Transition from low-to-high level.
5. q_0-q_{31} = The level of steady-state lower rank input at Q_0-Q_{31} respectively.
6. d_0-d_{31} = The level of steady-state lower rank contents.
7. t_n = Reference bit time, clock low.
8. t_n+1 = Bit time after one low-to-high clock transition.
9. t_n+32 = Bit time after 32 low-to-high clock transitions.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.513 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be EQF.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A. The cross sectional area of the entire fin shall not exceed 57 percent of the area designated in MIL-STD-1389, figure 6.

3.5.6 Depth. The module shall have no additional depth requirements.

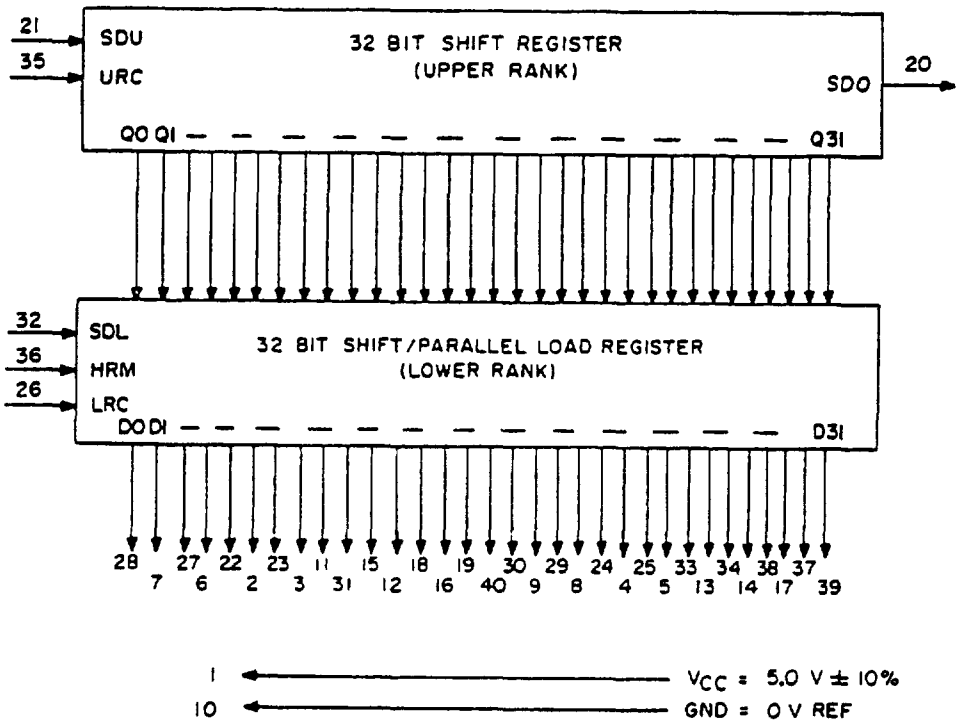
3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/214, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table II.

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NOTES
1. D0 - D31 = Parallel data outputs
2. SDU = Serial data upper input.
3. SDL = Serial data lower input
4. URC = Upper rank clock input.
5. LRC = Lower rank clock input.
6. HRM = Holding register mode input.
7. SDO = Serial data output.

FIGURE 1 EQF module function diagram, 32 bit serial/parallel converter.

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TABLE II. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
B1-B33	V_{OUT} (low-loaded)	IV	S	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	V
C1,C2	Input breakdown current, SDU, SDL	V	S	0	200	0	200	0	200	0	200	µA
C3-C5	Input breakdown current, HRM, URC, LRC	V'	S	0	800	0	800	0	800	0	800	µA
D1-D33	Output leakage current, D0-D31, SDO	VI	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
L1-L33	Output short circuit current, D0-D31, SDO	VII	X	-6	-130	-6	-130	-6	-130	-6	-130	mA
F1-F53	Functional	VIII	X	← See table VIII for limits →								
G1-G5	Input clamp diode voltage	IX	X	-	-1.5	-	-1.5	-	-1.5	-	-1.5	V
H1-H33	Static output voltage, low level	X	X	0	0.4	0	0.4	0	0.4	0	0.4	V
J1-J33	Static output voltage, high level	XI	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
K1,K2	Input current, high level, SDU, SDL	XII	X	0	40	0	40	0	40	0	40	µA
K3-K5	Input current, high level, HRM, URC, LRC	XII	X	0	160	0	160	0	160	0	160	µA
L1,L2	Input current, low level, SDU, SDL	XIII	X	-0.1	-0.8	-0.1	-0.8	-0.1	-0.8	-0.1	-0.8	mA

See notes at end of table

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TABLE II. Module electrical requirements -Continued

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° +5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
I3,I4	Input current, low level, IRC , IRC	XIII	X	-0.1	-3.91	-0.1	-3.91	-0.1	-3.91	-0.1	-3.91	mA
I5	Input current, low level, IIRM	XIII	X	-0.1	-3.20	-0.1	-3.20	-0.1	-3.20	-0.1	-3.20	mA
M	Power supply current	XIV	X	50	340	50	340	50	340	50	340	mA
N1-N33	Transition time (t_{TLH}), C_L = 50 pF	XV	S	2	25	2	25	2	25	2	25	ns
N34-N66	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	6	35	6	35	6	35	6	35	ns
P1-P33	Transition time (t_{THL}), C_L = 50 pF	XVI	S	1	10	1	10	1	10	1	10	ns
P34-P66	Transition time (t_{THL}), C_L = 150 pF	XVI	Q	2	15	2	15	2	15	2	15	ns
Q1-Q33	Propagation delay time (t_{PLH}) and SDU input setup and hold time check, C_L = 50 pf	XVII	S	10	40	10	40	10	40	10	40	ns
Q34-Q66	Propagation delay time (t_{PLH}) and SDU input setup and hold time check, C_L = 150 pf	XVII	Q	10	55	10	55	10	55	10	55	ns

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TABLE II. Module electrical requirements -Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R1-R33	Propagation delay time (t_{PHL}) and SDO input setup and hold time check, C_L = 50 pF	XVIII	S	10	45	10	45	10	45	10	45	ns
R34-R66	Propagation delay time (t_{PHL}) and SDO input setup and hold time check, C_L = 150 pF	XVIII	Q	10	55	10	55	10	55	10	55	ns
S1-S32	URC to IRC clock delay (measuring t_{PD}) check	XIX	S	450	550	450	550	450	550	450	550	ns
F1	SDO maximum operating frequency check, minimum URC pulse width check, SDO high level data setup and hold time check (measuring SDO output pulse period t_{PD})	XX	S	130	150	130	150	130	150	130	150	ns
F2	SDO maximum operating frequency check, minimum URC pulse width check, SDO high level data setup and hold time check (measuring SDO output pulse width t_{PW})	XX	S	60	80	60	80	60	80	60	80	ns
F3	SDO maximum operating frequency check, minimum URC pulse width check, SDO low level data setup and hold time check (measuring SDO output pulse period t_{PD})	XX	S	130	150	130	150	130	150	130	150	ns

See notes at end of table.

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TABLE II Module electrical requirements -Continued

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° +5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
14	SDO maximum operating frequency check, minimum ORC pulse width check, SDO low level data setup and hold time check (measuring SDO output pulse width t_{PW})	XX	S	60	80	60	80	60	80	60	80	ns
15	D31 maximum operating frequency check, minimum IRC pulse width check, SDI high level data setup and hold time check (measuring D31 output pulse period t_{PD})	XX	S	130	150	130	150	130	150	130	150	ns
16	D31 maximum operating frequency check, minimum IRC pulse width check, SDI high level data setup and hold time check (measuring D31 output pulse width t_{PW})	XX	S	60	80	60	80	60	80	60	80	ns
17	D31 maximum operating frequency check, minimum IRC pulse width check, SDI high level data setup and hold time check (measuring D31 output pulse period t_{PD})	XX	S	130	150	130	150	130	150	130	150	ns
18	D31 maximum operating frequency check, minimum LRC pulse width check, SDI low level data setup and hold time check (measuring D31 output pulse width t_{PW})	XX	S	60	80	60	80	60	80	60	80	ns

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TABLE II. Module electrical requirements -Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
T9	HRM setup and hold time check for parallel load	XX	S	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
T10	HRM setup and hold time check for serial shift	XX	S	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
U1-U66	Output crosstalk, low level, C_L = 50 pF	XXI	Q	0	800	0	800	0	800	0	800	mV
V1-V66	Output crosstalk, high level, C_L = 50 pF	XXII	Q	0	400	0	400	0	400	0	400	mV
W	Power supply transient amplitude	XXIII	Q	0	600	0	600	0	600	0	600	mV
X	Power supply filter capacitance	XXIV	S	0.14	0.26	0.14	0.26	0.14	0.26	0.14	0.26	µF
Y1,Y2	Input capacitance, SDU, SDL	XXV	Q	2	30	2	30	2	30	2	30	pF
Y3-Y5	Input capacitance, HRM, URC, IRC	XXV	Q	2	90	2	90	2	90	2	90	pF

NOTES

1 (X) parameters shall be 100 percent tested during module acceptance tests

2 (S) parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3 (Q) parameters shall be tested during qualification testing only.

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3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed two years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table II.

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table II for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent and the limits for V_{OUT} (low-loaded) (a maximum stress test) apply only for a power supply voltage of 7.0 Vdc $^{+0}_{-1}$ percent.

3.10 Functional tests. All modules shall be subjected to functional tests which conform to the requirements specified in 3.10.1 through 3.10.3.

3.10.1 Test patterns. This specification includes a set of unique input patterns (table VIII) capable of detecting all nonredundant failures in the module logic networks. These test patterns are generated by assuming "stuck at H state" and "stuck at L state" in each of the basic logic functions and applying the necessary input patterns to detect these failures. A "stuck at H state" indicates a failure resulting in a steady high-level output. A "stuck at L state" indicates a failure resulting in a steady low-level output. The unique test patterns necessary to satisfy these requirements shall be submitted with the qualification samples. If the test patterns submitted do not meet these criteria, the additional tests required shall be added to table II. Prior to any change in module design which would necessitate a different set of test patterns to detect a "stuck at L state" or "stuck at H state", the vendor shall be required to submit to the procuring activity and SEMP Quality Assurance Activity (SEMP-QAA) the proposed new set of test patterns for approval.

3.10.2 Input requirements. The following input voltage levels shall be used for test patterns

Low level - Input voltage shall be 0.7 V ±1 percent.
High level - Input voltage shall be 2.0 V ±1 percent.

3.10.3 Output requirements. The output requirements for the test patterns shall be as follows (per figure 19 output loads):

Low level - Output voltage shall be 0 V minimum and 0.3 V maximum.
High level - Output voltage shall be 2.4 V minimum and V_{CC} maximum.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below

Equipment	MIL-STD-1665 Designator	No. required (maximum per test figure
DC power supply	Item P01	4
Pulse generator	Item J01	3
Oscilloscope	Item D05	1
Oscilloscope probe	Item Y01	2
Megohm bridge	Item R06	1
DC voltmeter	Item V03	1
DC ammeter	Item X01	1
Oscilloscope	Item D03	1
Oscilliscope probe	Item Y04	1
Capacitance meter	Item C05	1
Capacitance bridge	Item C01	1

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Resistors	No. required (maximum per test figure
51.1Ω ±1%, 1/4 W	3
1.58 kΩ ±1%, 1/8 W	8
1.3 kΩ ±1%, 1/8 W	25
1.4 kΩ ±1%, 1/8 W	1
51Ω ±5%, 1/2 W	2
750Ω ±1%, 1/4 W	1
Value as required for figure 19, note 2b, 1/8 W	25
6.19 kΩ ±1%, 1/8 W	25
Value as required for figure 19, note 3b, 1/8 W	7
6.98 kΩ ±1%, 1/8 W	7
Value as required for figure 19, note 4b, 1/8 W	1
6.65 kΩ ±1%, 1/8 W	1
1 kΩ ±5%, 1/2 W	1
Values as required for figure 35, note 4, 1/8 W	3
Capacitors	
Value as required for figure 19, note 2d	33
Value as required for figure 19, note 2e	33
Diode	
1N3600, 1N4150, or equivalent	33
Switch	
SPST	1
Integrated circuits	
5476, or equivalent	1
74195, or equivalent	8
54191, or equivalent	2
54S140, or equivalent	3

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table II.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the SEMP Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows. A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Y in table II shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through X shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

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4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance wi MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table II which are marked with an (X) in the test type column, and test condition X shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the lif test, electrical tests D through X shall be performed in the order specified in 4.4 using the 0° to 60°C operating temperature end-of-life requirements of table II. After the module has been returned to 25°C, electrical tests D through Y shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table II.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table II, the module is considered to have completed its useful life.

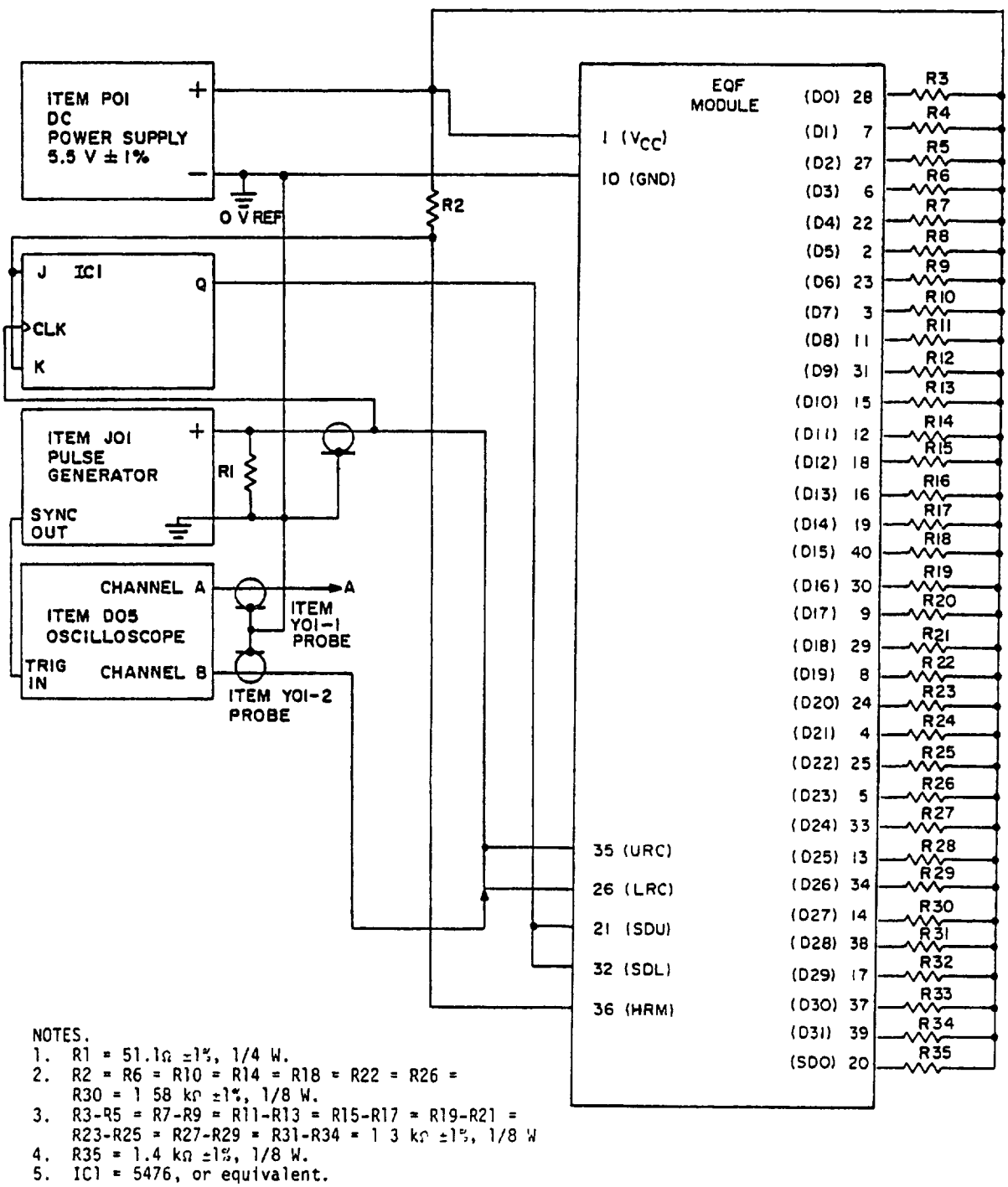
6.3 Functional tests. A set of test vectors that will check for a logic failur condition for every module input combination.

Preparing activity
Navy - EC

Agent
DLA - ES

(Project MISC-NB90-81)

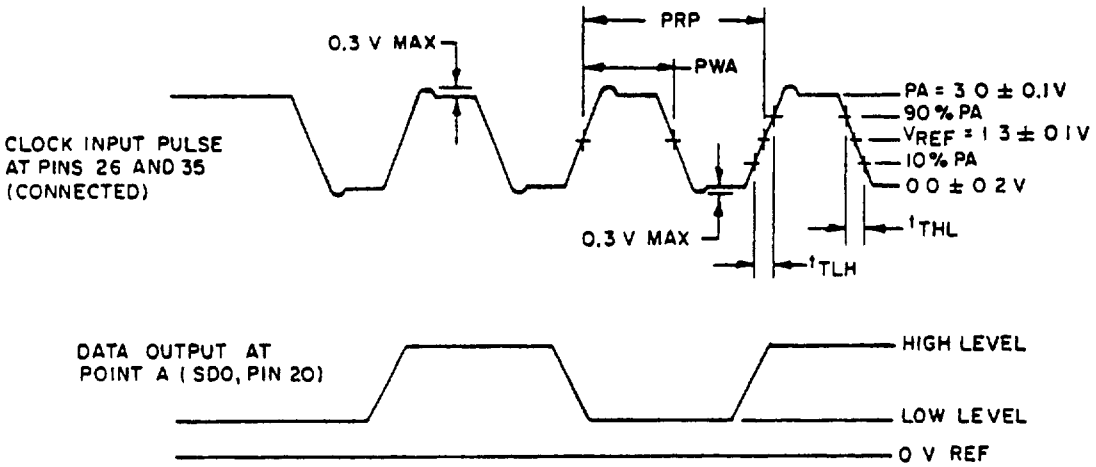
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NOTES.
1. R1 = 51.1Ω ±1%, 1/4 W.
2. R2 = R6 = R10 = R14 = R18 = R22 = R26 = R30 = 1 58 kΩ ±1%, 1/8 W.
3. R3-R5 = R7-R9 = R11-R13 = R15-R17 = R19-R21 = R23-R25 = R27-R29 = R31-R34 = 1 3 kΩ ±1%, 1/8 W
4. R35 = 1.4 kΩ ±1%, 1/8 W.
5. IC1 = 5476, or equivalent.

FIGURE 2. Operating temperature and life test circuit

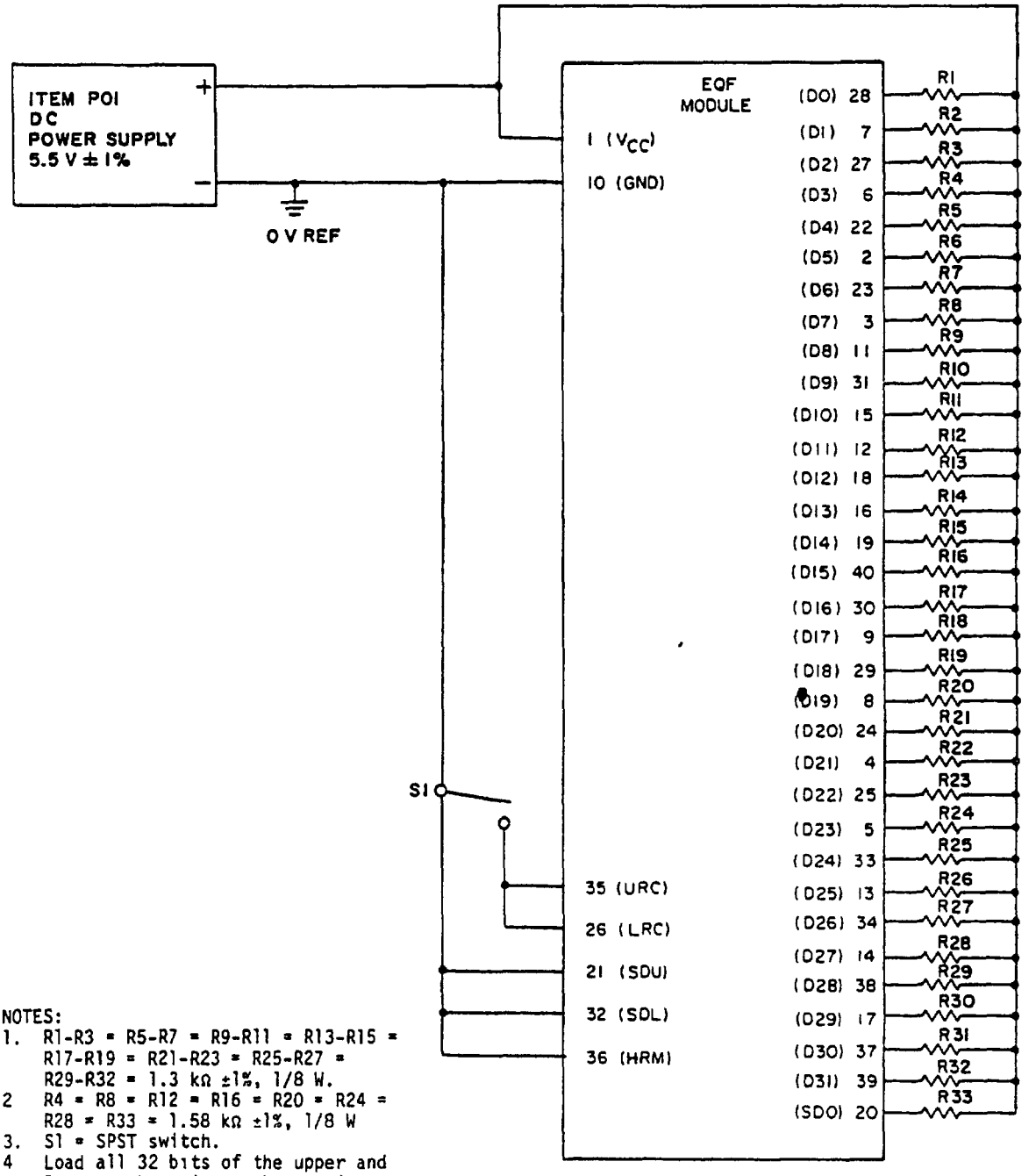
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NOTES:
1. Clock input pulse parameters:
   PRP = 500 ±50 ns.
   PWA = 200 ±20 ns.
   $t_{TLH} = t_{THL} = 15 \pm 10$ ns
2. Verify that outputs have a high level $\geq$ 2.5 V and a low level $\leq$ 0.4 V.
3. All voltages are referenced to 0 V reference unless otherwise specified.

FIGURE 3. Operating temperature and life test circuit waveforms.

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NOTES:
1. R1-R3 = R5-R7 = R9-R11 = R13-R15 = R17-R19 = R21-R23 = R25-R27 = R29-R32 = 1.3 kΩ ±1%, 1/8 W.
2 R4 = R8 = R12 = R16 = R20 = R24 = R28 = R33 = 1.58 kΩ ±1%, 1/8 W
3. S1 = SPST switch.
4 Load all 32 bits of the upper and lower rank registers by opening and closing switch S1 for 33 cycles. Open S1 for a duration of 25 ns minimum.

FIGURE 4. Component temperature test circuit.

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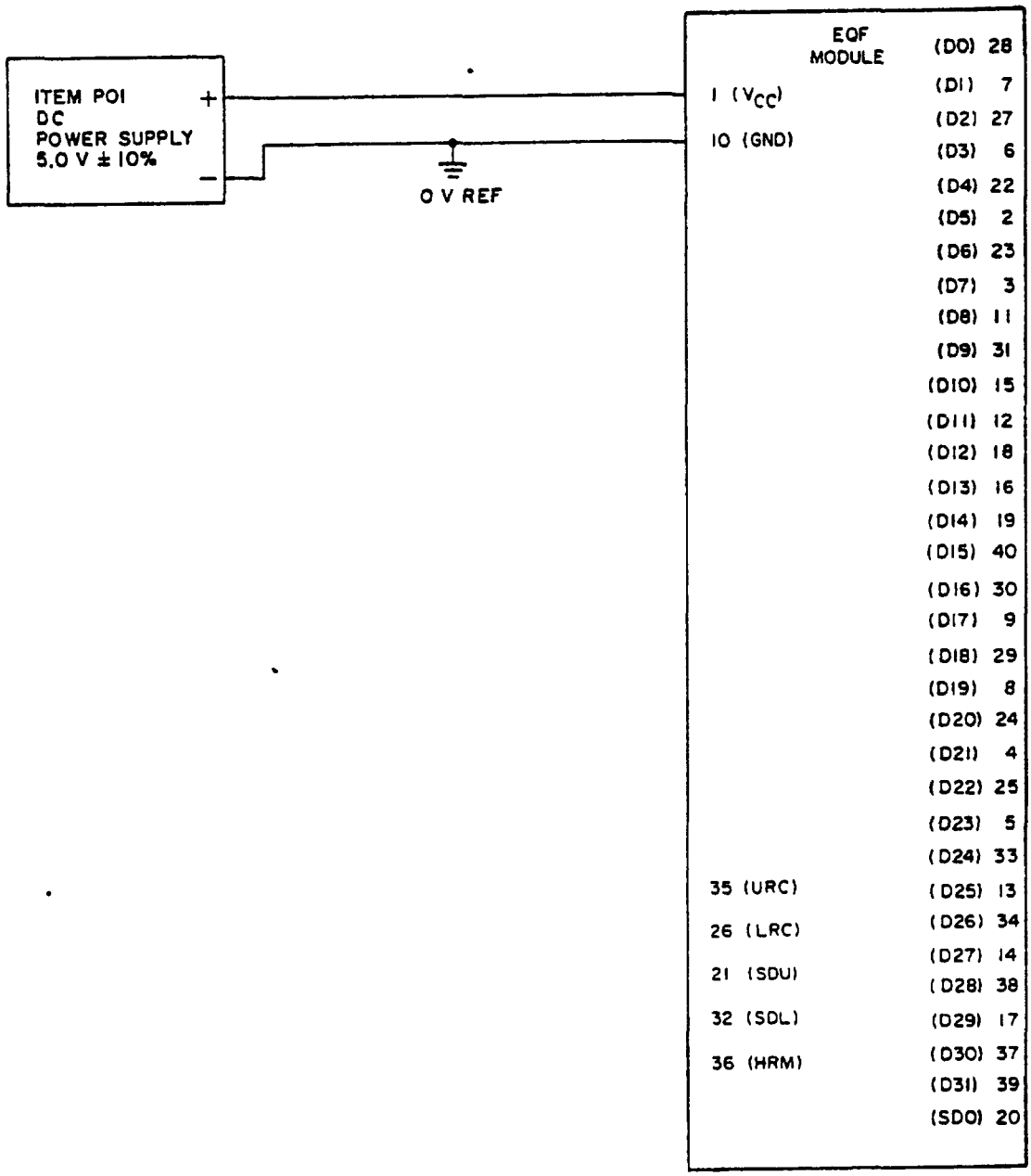


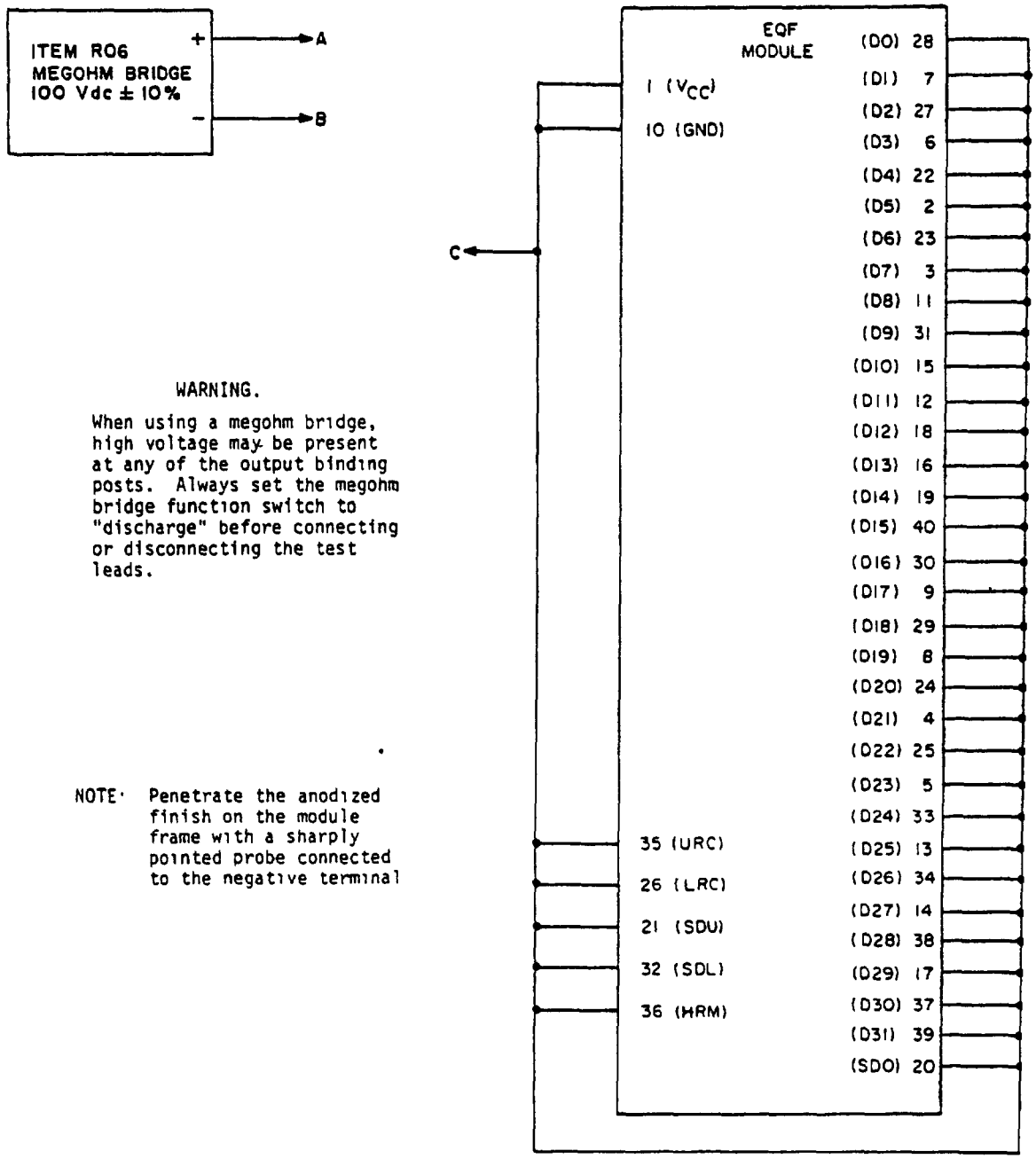
FIGURE 5 Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect megohm bridge item R06		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to point	B to	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	6

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WARNING.

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal

FIGURE 6 Isolation test circuit for table III.

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TABLE IV. V_{OUT} (low-loaded) test.

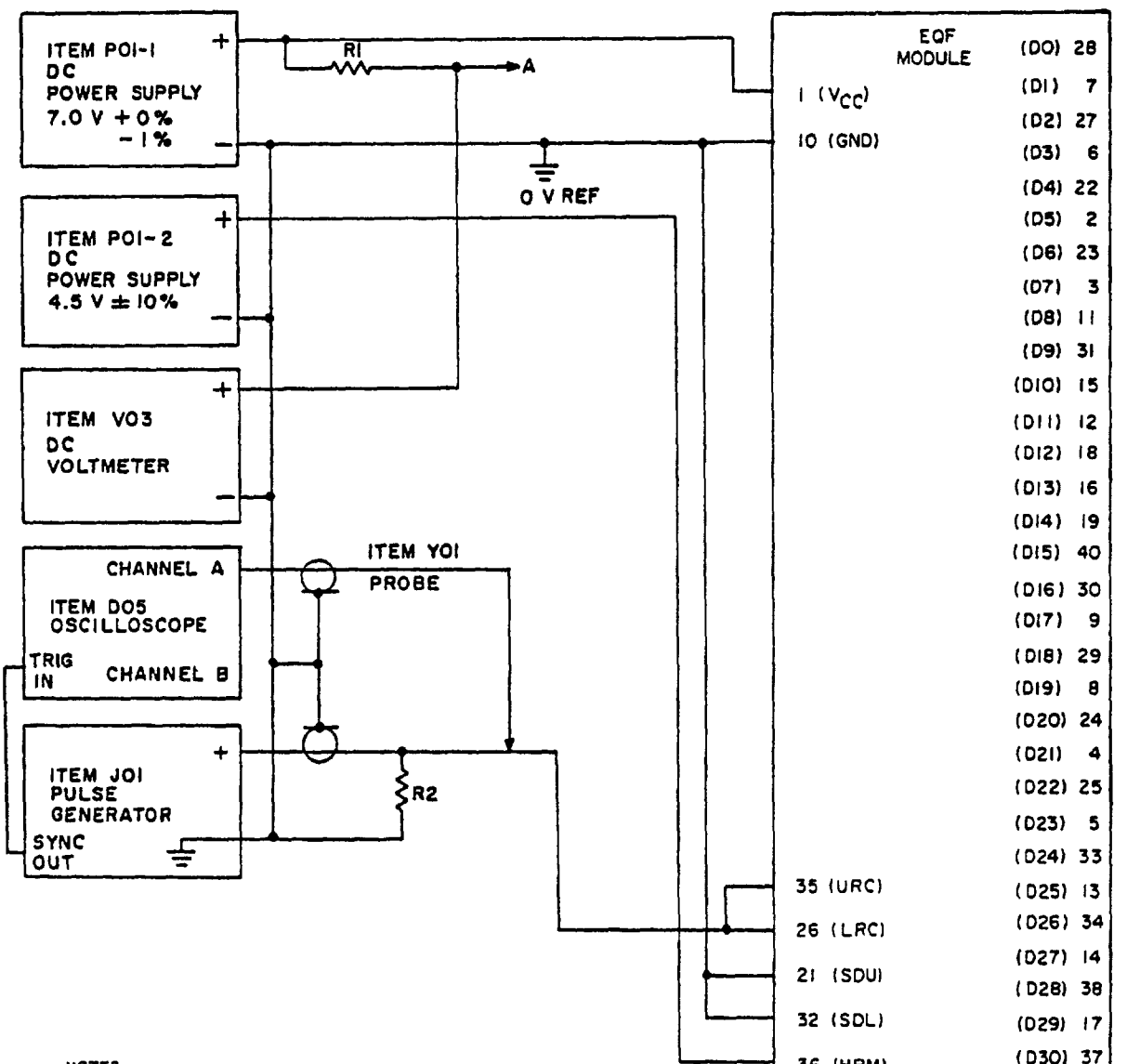
Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
B1	D0 V_{OUT} (low loaded)	28	0.01	0.8	V	7,8
B2	D1	7				
B3	D2	27				
B4	D3	6				
B5	D4	22				
B6	D5	2				
B7	D6	23				
B8	D7	3				
B9	D8	11				
B10	D9	31				
B11	D10	15				
B12	D11	12				
B13	D12	18				
B14	D13	16				
B15	D14	19				
B16	D15	40				
B17	D16	30				
B18	D17 V_{OUT} (low loaded)	9	0.01	0.8	V	7,8

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TABLE IV. V_{OUT} (low-loaded) test -Continued.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
B19	D18 V_{OUT} (low-loaded)	29	0.01	0.8	V	7,8
B20	D19	8				
B21	D20	24				
B22	D21	4				
B23	D22	25				
B24	D23	5				
B25	D24	33				
B26	D25	13				
B27	D26	34				
B28	D27	14				
B29	D28	38				
B30	D29	17				
B31	D30	37				
B32	D31	39				
B33	SDO V_{OUT} (low-loaded)	20	0.01	0.8	V	7,8

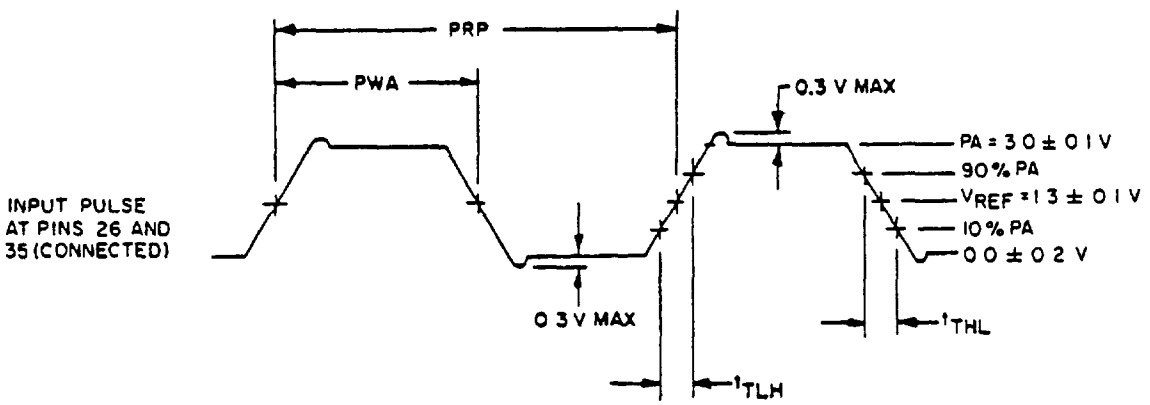
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NOTES:
1. R1 = 1.58 kΩ ±1%, 1/8 W.
2. R2 = 51Ω ±5%, 1/2 W.
3. A minimum of 33 clock pulses are required before point A is connected.
4. After point A has been connected to the appropriate module pin, the output voltage shall be monitored for a minimum of 5 seconds before the measurement is recorded.
5. If tests are successively repeated, a time interval of 10 seconds minimum between tests must be allowed to preclude detrimental effects due to module heating.

FIGURE 7. V_{out} (low-loaded) test circuit for table IV.

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NOTE. Input pulse parameters:
PRP = 1,000 ±100 ns.
PWA = 500 ±50 ns.
$t_{TLH} = t_{THL}$ = 50 ±25 ns.

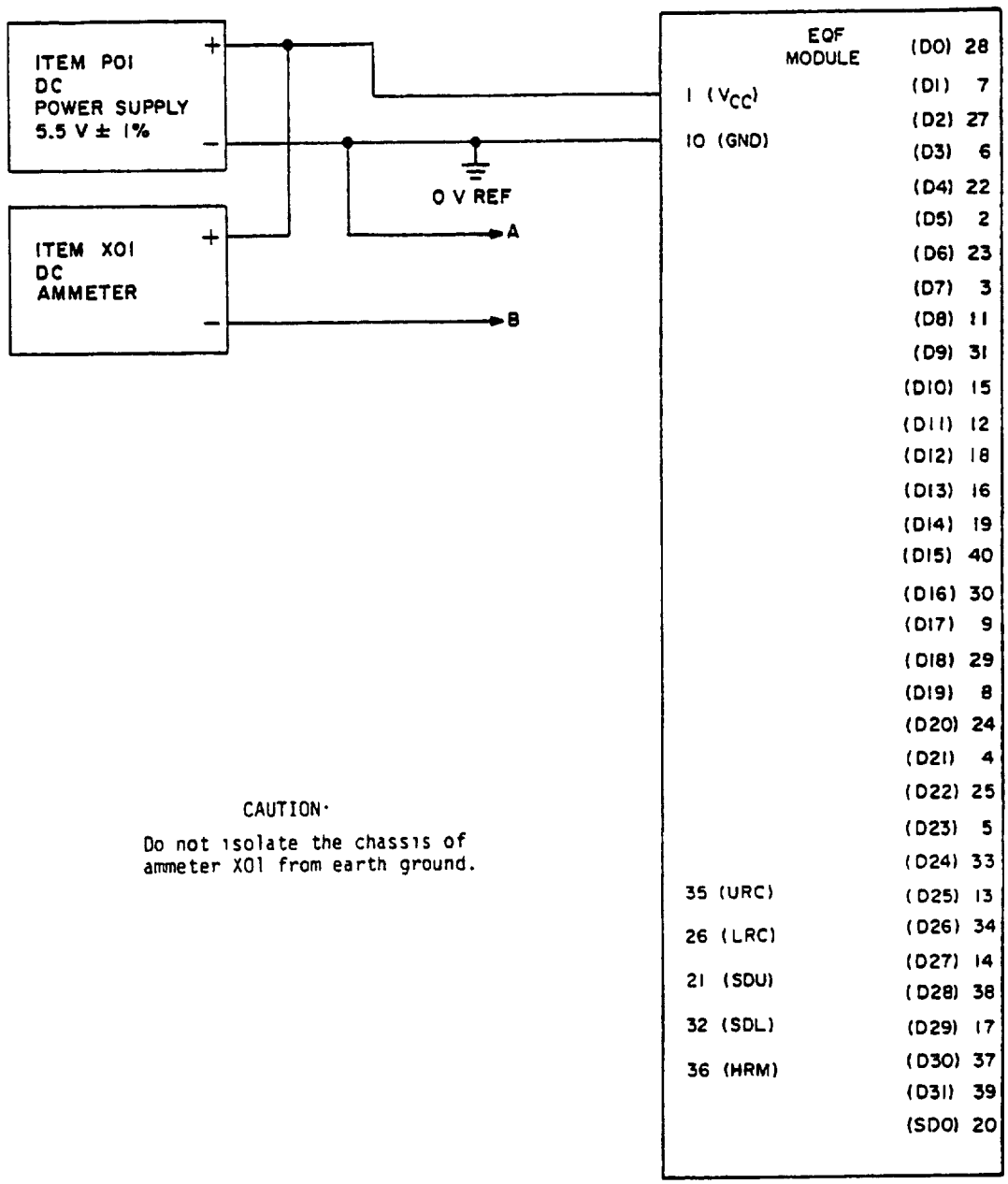
FIGURE 8. V_{out} (low-loaded) test circuit waveform for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	SDU Input breakdown current	35,26, 32,36	21	0	200	µA	9
C2	SDL	35,26, 21,36	32		200		
C3	HRM	35,26, 21,32	36		800		
C4	URC	26,21, 32,36	35		800		
C5	LRC Input breakdown current	35,21, 32,36	26	0	800	µA	9

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 9 Input breakdown current test circuit for table V.

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TABLE VI. Output leakage current test.

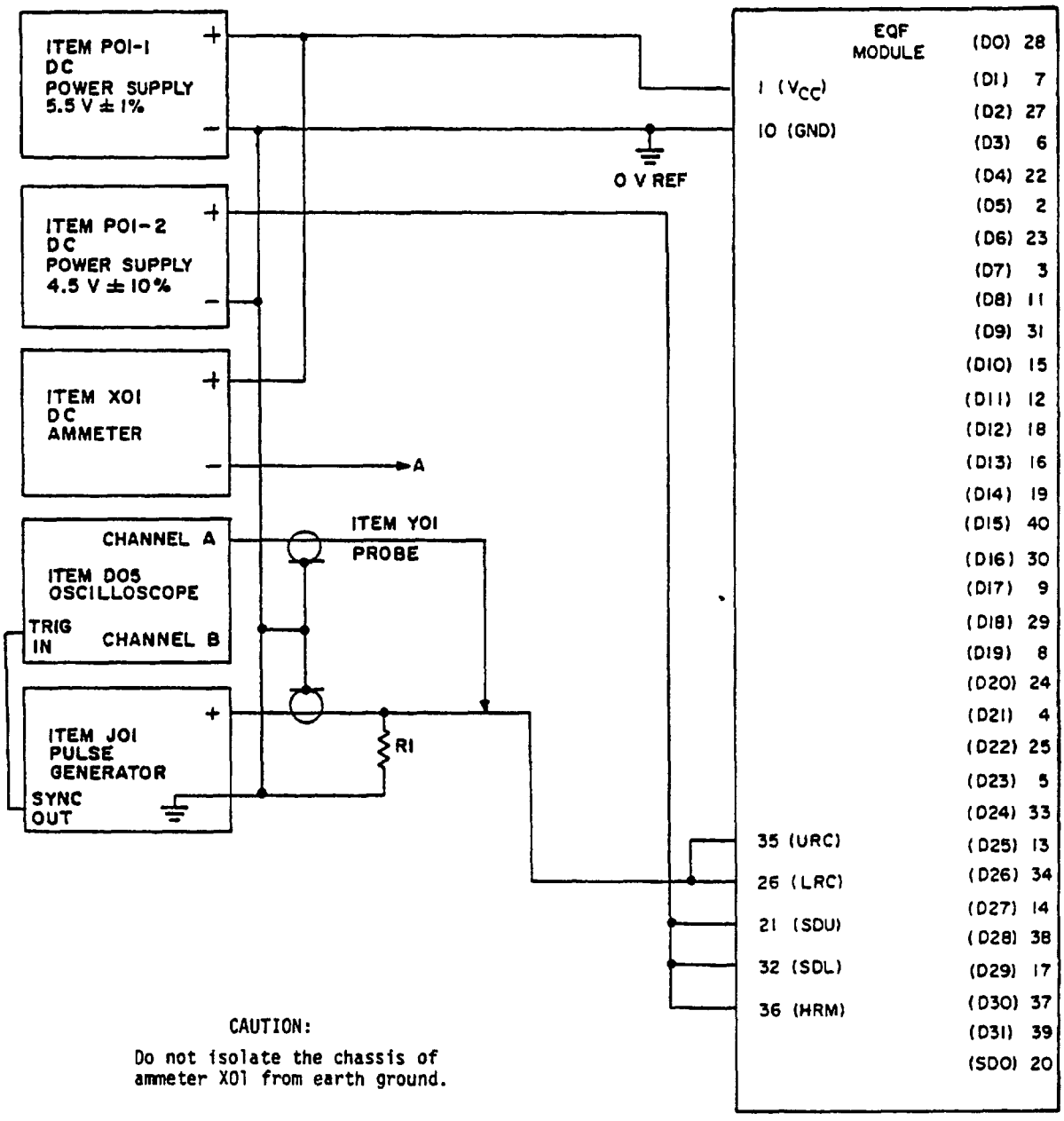
Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
D1	D0	Output leakage current	28	0.0	1.0	mA	10,11
D2	D1		7				
D3	D2		27				
D4	D3		6				
D5	D4		22				
D6	D5		2				
D7	D6		23				
D8	D7		3				
D9	D8		11				
D10	D9		31				
D11	D10		15				
D12	D11		12				
D13	D12		18				
D14	D13		16				
D15	D14		19				
D16	D15		40				
D17	D16		30				
D18	D17	Output leakage current	9	0.0	1.0	mA	10,11

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:43Z
Check the source to verify that this is the current version before use.

TABLE VI. Output leakage current test -Continued.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
D19	D18 Output leakage current	29	0.0	1.0	mA	10,11
D20	D19	8				
D21	D20	24				
D22	D21	4				
D23	D22	25				
D24	D23	5				
D25	D24	33				
D26	D25	13				
D27	D26	34				
D28	D27	14				
D29	D28	38				
D30	D29	17				
D31	D30	37				
D32	D31	39				
D33	SDO Output leakage current	20	0.0	1.0	mA	10,11

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:43Z
Check the source to verify that this is the current version before use.



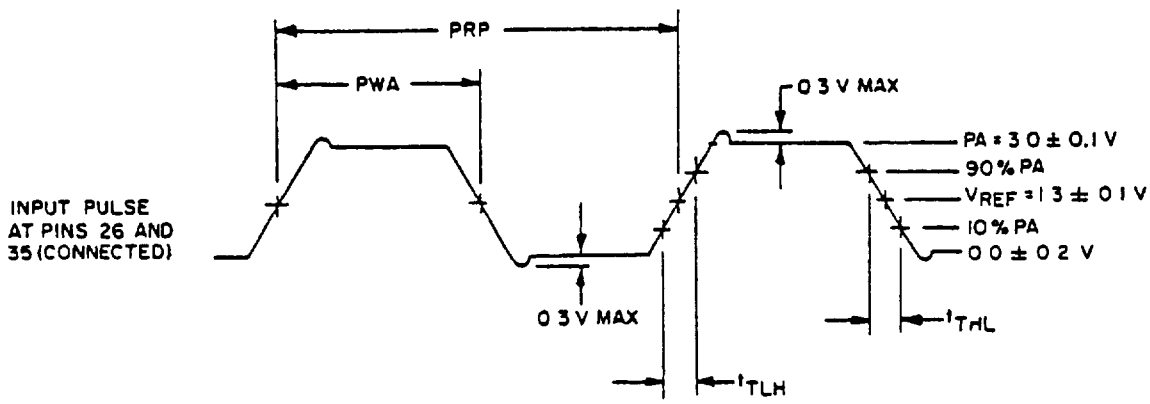
CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Before making point A connection, allow a minimum of 33 clock pulses.

FIGURE 10. Output leakage current test circuit for table VI.

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NOTE Input pulse parameters:
PRP = 1,000 ±100 ns.
PWA = 500 ±50 ns.
$t_{TLH} = t_{THL} = 50 \pm 25$ ns.

FIGURE 11 Output leakage current test circuit waveform for table VI.

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TABLE VII. Output short circuit current test.

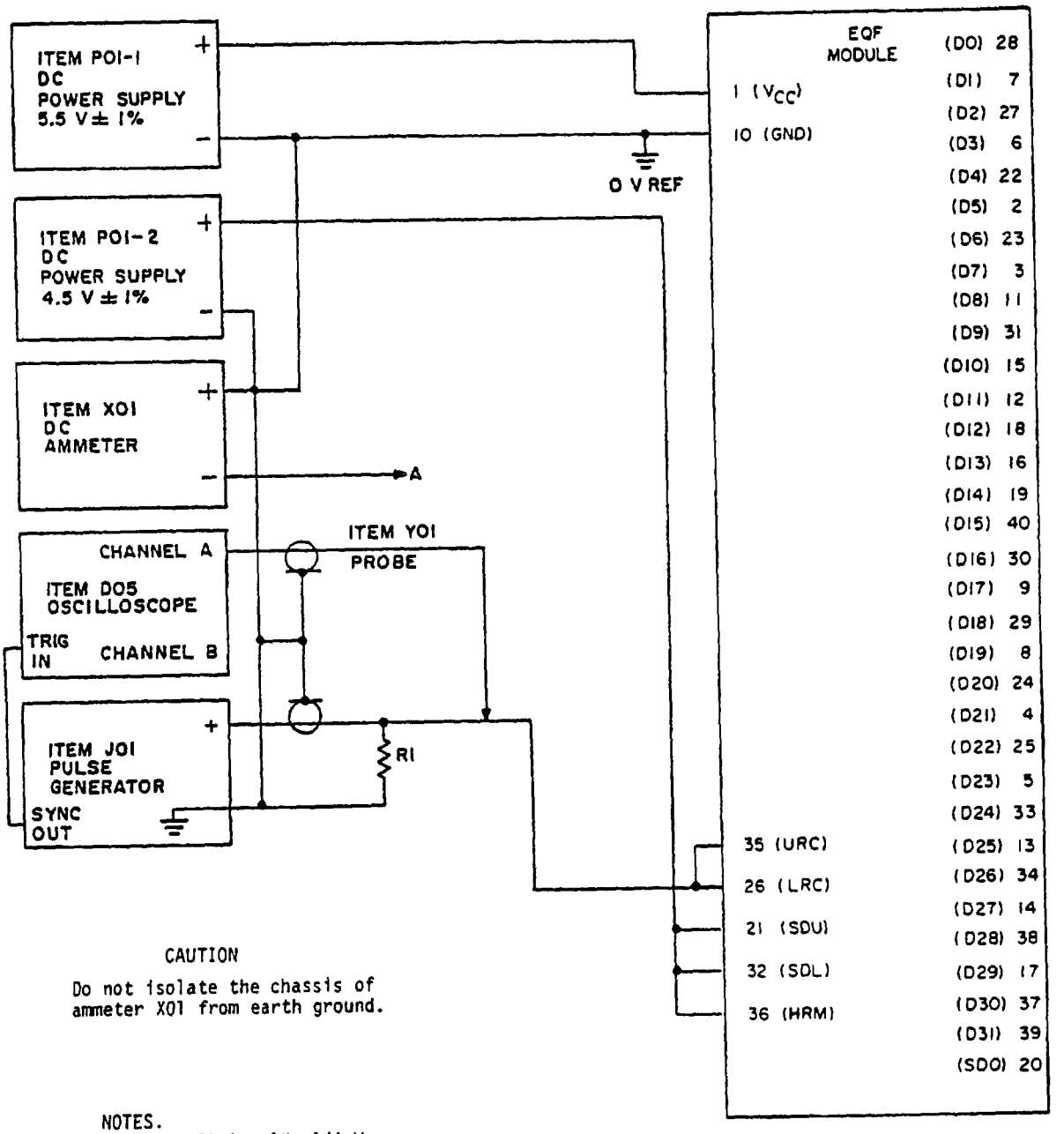
Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
E1	D0	Output short circuit current	28	-6	-130	mA	12,13
E2	D1		7				
E3	D2		27				
E4	D3		6				
E5	D4		22				
E6	D5		2				
E7	D6		23				
E8	D7		3				
E9	D8		11				
F10	D9		31				
E11	D10		15				
E12	D11		12				
E13	D12		18				
E14	D13		16				
E15	D14		19				
E16	D15		40				
E17	D16		30				
E18	D17	Output short circuit current	9	-6	-130	mA	12,13

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TABLE VII. Output short circuit current test -Continued.

Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
I 19	D18	Output short circuit current	29	-6	-130	mA	12,13
I 20	D19		8				
I 21	D20		24				
I 22	D21		4				
I 23	D22		25				
I 24	D23		5				
I 25	D24		33				
I 26	D25		13				
I 27	D26		34				
I 28	D27		14				
I 29	D28		38				
I 30	D29		17				
I 31	D30		37				
I 32	D31		39				
I 33	S00	Output short circuit current	20	-6	-130	mA	12,13

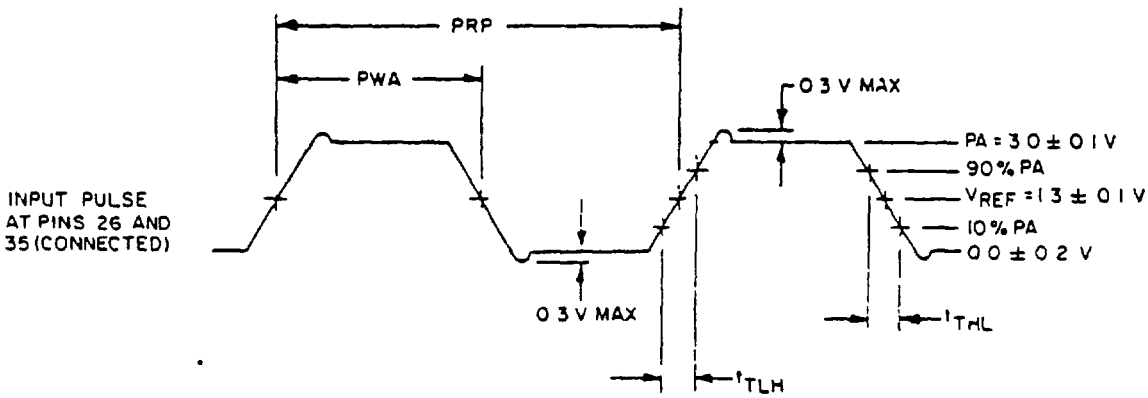
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NOTES.
1. RI = 51.1Ω ±1%, 1/4 W.
2. Before connecting point A allow a minimum of 33 clock pulses. Before making measurement, disable clock (to high level).
3. When performing output short circuit current test, only one output shall be shorted at a time, and the duration of the test shall not exceed 1 second

FIGURE 12. Output short circuit current test circuit for table VII.

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NOTE Input pulse parameters
PRP = 1,000 ±100 ns
PWA = 500 ±50 ns
$t_{TLH} = t_{THL}$ = 50 ±25 ns

FIGURE 13 Output short circuit current test circuit waveform for table VII

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TABLE VIII. Functional test.

Test condition	Parameter	Input				Output																												Test figure
		Pin numbers 21 to 32	Pin numbers 36 to 35	Pin numbers 26 to	Pin numbers 28	Pin numbers 7	Pin numbers 27	Pin numbers 6	Pin numbers 22	Pin numbers 2	Pin numbers 23	Pin numbers 3	Pin numbers 11	Pin numbers 31	Pin numbers 15	Pin numbers 12	Pin numbers 18	Pin numbers 16	Pin numbers 19	Pin numbers 40	Pin numbers 30	Pin numbers 9	Pin numbers 29	Pin numbers 8	Pin numbers 24	Pin numbers 4	Pin numbers 25	Pin numbers 5	Pin numbers 33	Pin numbers 13	Pin numbers 34	Pin numbers 14	Pin numbers 38	Pin numbers 17	Pin numbers 37	Pin numbers 39	Pin numbers 20	
	Functional	B	B	A	X	X	X	X	X	A	A	X	X	X	X	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	14, 15
		B	B	A	X	X	X	X	X	A	A	X	X	X	X	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

TABLE VIII. Functional test - Continued.

Test condition	Parameter	Input: P1 to P21	Input: P22 to P32	Input: P33 to P35	Input: P36 to P38	Output: P39	Output: P37	Output: P17	Output: P38	Output: P14	Output: P34	Output: P13	Output: P33	Output: P5	Output: P25	Output: P4	Output: P21	Output: P8	Output: P29	Output: P9	Output: P30	Output: P10	Output: P19	Output: P16	Output: P18	Output: P12	Output: P15	Output: P31	Output: P11	Output: P23	Output: P22	Output: P6	Output: P27	Output: P7	Output: P26	Output: P20	Test figure
	Functional	B	B	A	1	1	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	14, 15
		B	B	B	A	1	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	B	1	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	1	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	1	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	1	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	1	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	1	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	1	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	1	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	1	A	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	1	A	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	1	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	1	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	1	B	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	1	B	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	1	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		1	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		1	A	A	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		1	A	A	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		1	A	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		1	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		1	A	B	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		1	A	B	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		1	A	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		1	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		1	B	A	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		1	B	A	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		1	B	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		1	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		1	B	B	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		1	B	B	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		1	B	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
	Functional	B	B	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	14, 15
I1 I1 I1							L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	
I2							X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	I2

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Input pins 21 to 32	Input pins 36 to 35	Input pins 26 to	Output pins 28	7	27	6	22	2	23	3	11	31	15	12	18	16	19	40	30	9	29	8	24	4	25	5	33	13	34	14	38	17	37	39	20	Test figure
F2	Functional	B	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	14, 15
F2	Functional	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	14, 15
F2	Functional	B	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	14, 15
F2	Functional	B	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	14, 15
F2	Functional	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	14, 15
F2	Functional	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	14, 15
F2	Functional	A	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	14, 15
F2	Functional	A	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	14, 15
F2	Functional	A	A

TABLE VIII. Functional test - Continued.

Test condition	Parameter	Input: Pin 21 to 32	Input: Pin 36 to 35	Input: Pin 26 to	Output: Pin 28	Output: Pin 7	Output: Pin 27	Output: Pin 6	Output: Pin 22	Output: Pin 2	Output: Pin 23	Output: Pin 3	Output: Pin 11	Output: Pin 31	Output: Pin 15	Output: Pin 12	Output: Pin 18	Output: Pin 16	Output: Pin 19	Output: Pin 40	Output: Pin 30	Output: Pin 9	Output: Pin 29	Output: Pin 8	Output: Pin 24	Output: Pin 4	Output: Pin 25	Output: Pin 5	Output: Pin 33	Output: Pin 13	Output: Pin 34	Output: Pin 14	Output: Pin 38	Output: Pin 17	Output: Pin 37	Output: Pin 39	Output: Pin 20	Test figure
12	Functional	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	14, 15
		A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		A	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		A	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
		A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
		A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
		A	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
		B	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
		B	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
12	Functional	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	14, 15

TABLE VIII Functional test -Continued

Test condition	Parameter	Input					Output																														Test figure
		Pin 21 to	Pin 32 to	Pin 36 to	Pin 35 to	Pin 26 to	Pin 28	Pin 7	Pin 27	Pin 6	Pin 22	Pin 2	Pin 23	Pin 5	Pin 11	Pin 31	Pin 15	Pin 12	Pin 18	Pin 16	Pin 19	Pin 40	Pin 30	Pin 9	Pin 29	Pin 8	Pin 24	Pin 4	Pin 25	Pin 5	Pin 33	Pin 13	Pin 34	Pin 14	Pin 38	Pin 17	Pin 37	Pin 39	Pin 20	
12	Functional	A	A	B	A	A	X	\	Y	X	Y	X	X	\	X	X	X	X	X	X	Y	X	X	X	\	X	X	X	X	X	X	X	X	X	X	X	X	X	14, 15
		B	B	B	A	A	X	X	\	Y	Y	X	X	X	X	X	\	X	Y	X	Y	X	X	X	Y	\	X	X	X	X	X	X	X	X	X	X	\	X	
		B	B	B	B	A	X	X	\	X	X	\	X	X	X	X	X	X	Y	X	Y	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	A	A	X	\	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	Y	X	X	X	X	X	X	X	X	X	X	X	X	X	
		A	A	B	A	A	X	X	Y	X	Y	X	X	X	\	X	X	X	X	X	X	X	X	X	X	X	X	X	\	X	X	X	X	X	X	X	X	X	
		A	A	B	B	A	X	X	Y	X	Y	X	X	X	X	X	\	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		A	A	B	A	A	Y	X	\	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	A	A	X	\	X	X	\	Y	X	\	X	X	X	X	X	X	Y	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	B	A	X	X	\	X	X	X	\	X	X	X	X	X	X	X	X	X	X	X	\	X	X	X	X	X	X	X	X	X	X	X	X	X	
		A	A	B	A	A	X	X	X	\	Y	X	Y	Y	X	X	X	X	Y	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		A	A	B	B	A	X	\	\	X	X	\	X	X	X	X	X	X	X	\	X	\	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		A	A	B	A	A	X	\	X	X	X	X	X	X	X	X	X	X	X	Y	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	A	A	X	X	\	X	X	Y	X	X	X	X	X	Y	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
12		B	B	B	B	A	X	X	\	\	Y	X	X	X	Y	X	X	X	Y	X	Y	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
13		B	B	B	\	A	I	I	I	I	I	L	I	L	L	L	I	L	L	L	L	L	I	I	I	L	L	L	L	L	L	L	L	L	L	L	H	
13		\	A	B	A	A	X	\	\	X	X	X	X	X	X	X	X	X	X	Y	\	X	X	X	\	X	Y	X	X	X	X	X	X	X	X	X	X	
13		A	A	B	A	B	X	X	\	\	X	\	X	Y	X	X	X	X	Y	\	Y	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
14		A	A	B	A	A	I	H	I	H	L	H	I	H	L	H	I	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	H	
11		A	A	A	A	\	Y	X	\	\	Y	\	\	\	X	X	X	X	X	Y	X	X	\	\	\	X	Y	X	X	Y	X	X	X	X	X	X	X	
11		A	\	A	A	B	Y	Y	\	\	Y	\	X	Y	X	X	X	X	X	Y	X	X	\	\	\	X	Y	X	X	Y	X	X	X	X	X	X	X	
15		A	A	\	A	A	H	I	H	I	H	I	H	I	H	I	H	I	H	I	H	I	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	
15		B	B	A	\	A	Y	\	\	\	Y	\	\	Y	\	Y	Y	\	X	\	X	X	X	X	Y	X	X	X	X	X	X	X	X	X	X	X	X	
15		B	B	\	A	B	X	\	\	\	Y	\	\	Y	\	\	X	Y	X	\	X	Y	X	X	X	X	\	X	X	X	X	X	X	X	X	X	X	
16		B	B	A	A	A	I	H	I	H	I	H	I	H	I	H	I	H	I	H	I	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	H	
		B	A	\	A	\	X	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	X	X	X	X	X	X	X	X	X	X	X	X	
		B	\	\	B	A	X	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	X	X	X	X	X	X	X	X	X	X	X	X	
		B	A	A	\	A	X	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	X	X	\	X	X	X	X	X	X	X	X	X	
		B	\	\	B	A	X	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	X	X	X	X	X	X	X	X	X	X	X	X	
		B	\	\	\	B	X	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	X	X	X	X	X	X	X	X	X	X	X	X	
		B	\	\	\	A	X	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	X	X	X	X	\	X	X	X	X	X	X	X	
16	Functional	B	A	\	A	\	X	X	\	X	\	\	X	\	\	\	\	\	X	\	Y	Y	Y	X	X	X	X	X	X	X	X	X	X	X	X	X	X	14, 15

See notes at end of table

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TABLE VIII. Functional test - Continued.

| Test condition | Parameter | Input | | |

TABLE VIII Functional test -Continued.

Test condition	Parameter	Input					Output																																		Test figure
		Pin 21 to	Pin 32 to	Pin 36 to	Pin 35 to	Pin 26 to	Pin 28	Pin 7	Pin 27	Pin 6	Pin 22	Pin 2	Pin 23	Pin 3	Pin 11	Pin 31	Pin 15	Pin 12	Pin 18	Pin 16	Pin 19	Pin 10	Pin 30	Pin 9	Pin 29	Pin 8	Pin 24	Pin 4	Pin 25	Pin 5	Pin 33	Pin 13	Pin 34	Pin 14	Pin 38	Pin 17	Pin 37	Pin 39	Pin 20	
F6	Functional	B	A	A	B	A	X	λ	X	X	X	λ	λ	X	λ	λ	X	X	X	Y	\	X	X	λ	Y	λ	X	X	X	X	X	X	X	X	X	X	X	X	X	X	14, 15
		B	A	A	A	A	λ	X	\	λ	X	λ	λ	X	X	X	Y	X	X	Y	X	X	X	X	Y	λ	λ	X	X	X	X	X	X	X	X	X	X	X	X		
		B	A	A	B	A	λ	X	X	λ	X	X	X	X	\	λ	\	X	X	Y	X	X	X	X	λ	X	X	X	X	X	X	X	X	X	X	X	X	X			
		B	A	A	A	A	\	X	\	λ	λ	X	X	X	X	X	λ	X	λ	X	λ	X	X	X	Y	X	X	X	X	X	X	X	X	X	X	X	X	X			
		B	A	A	B	A	X	Y	X	\	X	λ	λ	X	X	X	X	X	X	X	\	Y	X	X	\	X	X	\	X	X	Y	X	X	X	X	X	X			
		B	A	A	A	B	λ	λ	\	\	X	λ	Y	X	X	X	λ	X	X	X	Y	X	Y	X	λ	X	Y	X	X	X	X	X	X	X	X	X	X	X			
		B	A	A	A	A	λ	X	\	\	X	X	\	λ	X	λ	Y	X	\	X	Y	X	Y	X	Y	X	X	X	X	X	X	X	X	X	λ	X	X	X			
		B	A	A	B	A	X	\	\	\	\	λ	Y	\	Y	Y	λ	X	X	X	Y	λ	Y	\	\	X	X	X	X	X	X	X	X	Y	X	X	X	X			
		B	A	A	A	A	λ	X	\	\	X	X	\	\	X	X	Y	X	\	X	Y	X	Y	X	\	X	Y	X	X	X	X	X	X	X	X	X	X	X			
		B	A	A	B	A	\	X	\	Y	X	\	X	\	Y	X	X	\	\	X	X	X	X	X	\	X	Y	\	X	X	X	X	X	X	X	X	X	X			
		B	A	A	\	A	Y	Y	\	X	\	λ	X	\	Y	Y	X	Y	\	\	X	X	\	X	\	X	Y	\	\	X	X	X	X	X	X	X	X	X			
		B	A	A	B	A	X	\	\	\	\	Y	\	\	Y	\	X	\	X	\	X	X	X	X	\	X	Y	X	Y	X	X	Y	Y	X	X	X	X	X			
		B	A	A	A	A	Y	\	\	\	\	Y	\	\	Y	X	X	Y	X	\	\	Y	\	X	X	\	X	X	Y	X	X	X	X	X	X	X	X	X			
		B	A	A	B	A	Y	λ	\	λ	\	Y	λ	\	Y	X	X	Y	X	\	\	Y	X	X	\	X	X	Y	X	X	X	X	X	X	X	X	X	X			
		B	A	A	A	A	\	\	\	\	Y	Y	\	\	Y	X	X	Y	X	\	\	Y	X	\	X	\	X	X	Y	X	X	X	X	X	X	X	X	X			
		B	A	A	B	A	Y	X	\	\	\	\	Y	\	X	\	\	\	\	\	\	X	X	λ	λ	Y	X	X	Y	X	\	X	X	X	X	X	X	X			
P6		B	A	A	A	B	Y	X	\	\	\	\	Y	\	X	\	\	\	\	\	\	X	X	λ	λ	Y	X	X	Y	X	\	X	X	X	X	X	X	X			
F7		B	A	A	A	A	H	I	H	I	H	I	H	I	H	L	H	I	H	I	H	I	H	L	H	L	H	L	H	L	H	L	H	L	H	L	L			
F7		B	B	A	A	A	λ	Y	\	\	\	λ	\	λ	\	Y	λ	Y	Y	\	λ	X	λ	Y	λ	Y	Y	X	X	λ	X	Y	X	X	X	X	X			
F7		B	B	\	A	B	Y	\	\	Y	\	Y	Y	\	λ	Y	Y	Y	\	\	\	λ	Y	λ	\	λ	Y	X	Y	X	X	X	X	Y	X	X	X			
F8		B	B	A	A	\	I	H	I	H	I	H	L	H	I	H	I	H	L	H	L	H	L	H	I	H	I	H	I	H	I	H	I	H	L	H	L			
F8		B	B	B	A	\	Y	\	Y	Y	\	λ	\	λ	Y	Y	λ	Y	X	X	λ	λ	X	λ	λ	X	X	\	\	X	Y	λ	X	Y	X	X	X			
F8		B	B	B	A	B	λ	X	\	\	λ	\	\	λ	X	Y	λ	X	X	\	λ	λ	X	λ	λ	X	X	\	\	X	Y	λ	X	Y	X	X	X			
F9		B	B	B	A	A	I	I	I	I	I	I	L	L	I	L	I	I	L	L	I	L	L	L	I	L	L	I	L	I	L	L	L	L	L	L	L			
F9		\	A	B	A	A	Y	\	\	\	\	\	Y	\	Y	\	\	\	Y	\	\	X	Y	X	X	Y	\	X	X	X	X	X	Y	X	X	X	X			
F9		\	A	B	B	A	Y	\	\	\	\	λ	\	\	\	λ	\	\	\	X	Y	Y	X	λ	X	Y	\	Y	X	X	λ	Y	X	X	X	X	X			
F10		\	A	B	A	A	I	I	I	I	L	I	I	I	L	I	I	I	I	I	L	L	I	L	L	L	I	L	L	L	L	L	L	L	L	L	L			
F10		B	B	B	A	A	\	\	\	\	\	λ	Y	\	λ	X	λ	\	Y	Y	Y	Y	Y	X	X	\	\	Y	Y	X	X	X	X	X	X	\	X	X			
F10		B	B	B	B	\	λ	\	\	\	Y	\	λ	\	Y	Y	\	Y	λ	Y	\	\	X	Y	\	X	λ	\	X	X	X	X	X	X	X	X	X	X			
F11		B	b	B	A	A	I	I	I	I	I	L	L	I	L	L	I	I	L	L	I	I	L	L	L	I	L	L	L	L	L	L	L	L	L	L	L			
F11	Functional	A	A	B	A	A	\	\	\	Y	\	\	λ	λ	Y	\	X	\	\	Y	\	\	X	\	X	\	λ	X	Y	Y	X	X	X	X	X	X	X			14, 15

See notes at end of table

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Input pins 21 to 28	Input pins 32 to 36	Input pins 35 to 26	Output pins 27	Output pins 6	Output pins 22	Output pins 2	Output pins 23	Output pins 3	Output pins 11	Output pins 31	Output pins 15	Output pins 12	Output pins 18	Output pins 16	Output pins 19	Output pins 10	Output pins 30	Output pins 9	Output pins 29	Output pins 8	Output pins 24	Output pins 4	Output pins 25	Output pins 5	Output pins 33	Output pins 13	Output pins 34	Output pins 14	Output pins 38	Output pins 17	Output pins 37	Output pins 39	Output pins 20	Test figure
111	Functional	A	B	A	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	14, 15
112		A	A	A	1	1	1	1	1	1	1	1	1	1	1	1	1	L	1	1	1	1	1	1	L	L	1	1	1	L	L	L	L	L	X
113		B	B	B	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	L	L	1	1	1	L	L	L	L	L	X
114		B	A	A	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	L	L	1	1	1	L	L	L	L	L	X
115		B	B	A	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	L	L	1	1	1	L	L	L	L	L	X
116		A	A	A	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	L	L	1	1	1	L	L	L	L	L	X
117		A	B	A	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	L	L	1	1	1	L	L	L	L	L	X
118		A	B	B	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	L	L	1	1	1	L	L	L	L	L	X
119		B	B	A	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	L	L	1	1	1	L	L	L	L	L	X
120		1	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	L	1	1	1	L	L	L	L	L	X
120		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	L	1	1	1	L	L	L	L	L	X
120	Functional	B	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	L	1	1	1	L	L	L	L	L	X

See notes at end of table.

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TABLE VIII Functional test -Continued

Test condition	Parameter	Input					Output																																Test figure
		Pin 21 to	Pin 32 to	Pin 36 to	Pin 35 to	Pin 26 to	Pin 28	Pin 7	Pin 27	Pin 6	Pin 22	Pin 2	Pin 23	Pin 3	Pin 11	Pin 31	Pin 15	Pin 12	Pin 18	Pin 16	Pin 19	Pin 40	Pin 30	Pin 9	Pin 29	Pin 8	Pin 24	Pin 4	Pin 25	Pin 5	Pin 35	Pin 13	Pin 34	Pin 14	Pin 38	Pin 17	Pin 37	Pin 39	Pin 20	
F21	Functional	B	B	B	A	A	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	14, 15
F21		A	A	B	A	A	X	Y	λ	X	X	X	X	Y	X	X	X	X	X	X	Y	X	X	X	X	X	λ	X	X	X	X	X	X	X	X	X	X	X	
F21		A	A	B	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	λ	X	X	X	X	X	X	X	X	X	X	X	
F22		A	A	B	A	A	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	
F22		B	B	B	A	A	X	X	X	X	X	λ	Y	X	Y	X	X	\	X	X	λ	X	X	X	λ	X	X	X	X	X	X	X	X	X	X	X	X	X	
F22		B	B	B	B	A	X	X	X	X	X	λ	X	λ	X	X	X	λ	λ	X	X	X	λ	λ	X	λ	X	X	Y	X	X	X	X	X	X	X	X	X	
F23		B	B	B	A	A	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	
F23		A	A	B	A	A	X	X	Y	λ	λ	\	Y	X	X	X	X	X	X	X	X	X	X	Y	Y	X	X	X	X	X	X	X	X	X	X	X	X	X	
F23		A	A	B	B	A	X	Y	λ	\	\	X	λ	X	X	Y	λ	λ	Y	X	Y	λ	λ	Y	\	λ	X	X	X	λ	X	X	X	X	X	X	X	X	
F24		A	A	B	A	A	I	I	I	L	L	L	I	L	I	L	L	L	L	L	I	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	
F24		B	B	B	A	A	λ	\	\	\	Y	\	λ	Y	λ	Y	λ	\	X	X	λ	Y	Y	Y	λ	Y	\	λ	\	X	Y	X	X	X	X	X	X	X	
F24		B	B	B	B	A	X	λ	λ	\	Y	λ	\	\	X	λ	\	X	λ	λ	Y	X	Y	\	\	Y	λ	X	X	Y	Y	Y	X	X	X	X	X	X	
F25		B	B	B	\	\	L	I	I	I	L	L	L	I	I	I	L	L	L	L	I	L	I	I	I	I	L	L	I	L	L	L	L	L	L	L	L	L	
F25		A	A	B	\	A	\	\	\	\	\	λ	\	\	\	\	\	X	\	\	\	X	λ	\	\	\	Y	\	\	\	λ	X	X	X	X	X	X	X	
F25		A	A	B	B	\	λ	Y	\	\	Y	\	\	Y	\	λ	λ	\	\	\	λ	X	\	\	\	\	Y	λ	\	λ	X	X	X	Y	Y	Y	X	X	λ	
F26		\	A	B	A	A	I	I	I	I	L	I	I	I	I	I	L	I	L	I	I	L	I	L	I	L	L	I	L	I	L	I	L	L	L	L	L	L	
F26		B	B	B	A	A	\	\	\	\	\	\	\	\	\	λ	\	X	Y	\	\	Y	\	\	\	\	Y	\	\	\	Y	Y	λ	X	X	X	\	X	
F26		B	B	B	B	A	\	\	\	\	\	\	\	\	\	Y	\	\	\	\	\	\	\	\	λ	Y	\	\	\	Y	Y	X	X	\	λ	X	Y	X	λ	
F27		B	B	B	\	A	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	L	I	I	I	L	\	Y	Y	L	Y	L	I	
F27		\	\	B	\	\	\	\	\	\	\	\	\	\	\	λ	\	\	\	\	\	\	\	\	\	\	\	\	\	\	Y	Y	\	Y	X	Y	X	Y	X	
F27		\	A	B	B	A	\	\	\	\	\	\	\	\	\	\	λ	\	\	\	\	\	Y	\	\	\	\	λ	Y	\	λ	Y	\	Y	\	Y	X	Y	λ	
F28		\	\	B	\	A	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	L	I	I	I	I	L	I	I	I	I	L	I	I	L	I	I	
F28		B	B	B	\	\	\	\	\	\	\	\	Y	\	\	\	\	\	\	\	\	\	\	\	\	\	\	Y	\	\	\	\	\	Y	Y	X	X	Y	
F28		B	B	B	B	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	X	X	\	\	
F29		B	B	B	\	\	I	I	I	I	I	I	I	L	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	L	L	L	L	I	L	L	L	I	
F29		\	\	B	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	\	Y	X	\	\	\	X	\	
F29		\	\	B	B	\	\	\	\	\	\	\	\	\	\	\	\	Y	\	\	\	\	\	\	\	\	\	\	\	\	\	Y	X	\	\	X	\	X	λ	
F30		\	\	B	\	\	I	I	I	I	I	I	I	I	I	I	L	I	I	L	I	I	I	I	I	I	L	L	I	L	L	L	L	L	L	L	L	L	
F30	Functional	B	B	B	A	A	\	\	\	\	\	Y	\	X	\	\	\	Y	λ	X	\	\	X	\	X	\	λ	\	Y	X	λ	λ	Y	X	X	X	X	X	14, 15

See notes at end of table

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TABLE VIII. Functional test -Continued

Test condition	Parameter	Input					Output																																	Test figure
		Pin 21 to	Pin 32 to	Pin 36 to	Pin 35 to	Pin 26 to	Pin 28	Pin 7	Pin 27	Pin 6	Pin 22	Pin 2	Pin 23	Pin 3	Pin 11	Pin 31	Pin 15	Pin 12	Pin 18	Pin 16	Pin 19	Pin 10	Pin 30	Pin 9	Pin 29	Pin 8	Pin 24	Pin 4	Pin 25	Pin 5	Pin 33	Pin 13	Pin 34	Pin 14	Pin 38	Pin 17	Pin 37	Pin 39	Pin 20	
I 30	Functional	B	B	B	B	A	X	A	A	A	A	X	A	X	X	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	14, 15
I 31		B	B	B	A	A	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	L	I	L	L	L	L	L	L	L	L	L	L	L	L	
I 31		A	A	B	A	A	A	A	A	A	A	Y	A	A	A	A	A	A	Y	X	X	X	Y	A	A	A	Y	X	X	A	X	X	X	X	X	X	X	A	
I 31		A	A	B	B	A	A	A	A	A	A	A	Y	A	A	A	Y	A	A	A	Y	A	A	A	A	X	X	X	A	X	X	X	X	X	X	X	X	A	
I 32		A	A	B	A	A	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	L	I	L	I	I	L	I	L	I	L	L	L	I	
I 32		B	B	B	A	A	A	A	A	A	A	A	A	A	Y	A	A	A	Y	A	A	Y	A	A	A	X	A	X	X	A	A	X	A	X	X	X	X	A	
I 32		B	B	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	Y	A	A	A	X	Y	Y	X	A	A	X	A	X	X	X	X	A	
I 33		B	B	B	A	A	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	L	I	I	I	I	L	I	I	I	L	I	I	L	I	L	I	I	
I 33		A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	X	A	A	A	A	X	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	A	
I 33		A	A	B	B	A	A	A	A	A	A	X	A	A	A	A	X	A	A	A	A	Y	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	A	
I 34		A	A	B	A	A	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	L	I	I	I	I	L	L	I	L	L	L	L	L	L	L	L	L	
I 34		B	B	B	A	A	X	A	A	Y	A	X	A	A	X	X	A	Y	X	X	X	X	X	Y	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
I 34		B	B	B	B	A	A	X	X	X	Y	X	X	Y	A	X	Y	X	Y	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
I 35		B	B	B	A	A	I	I	I	I	I	I	I	L	L	I	I	I	L	L	I	L	L	I	L	L	L	L	L	L	L	L	L	L	L	L	I	I	
I 35		A	A	B	A	A	Y	X	X	Y	X	A	X	X	X	X	X	X	X	X	X	X	X	X	Y	X	X	X	X	X	X	X	X	X	X	X	X	X	
I 35		A	A	B	B	A	X	X	X	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
I 36		A	A	B	A	A	L	I	I	I	I	I	L	I	I	L	I	I	I	I	L	I	L	L	I	L	L	I	L	L	L	L	L	L	L	L	L	L	
I 36		B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
I 36		B	B	B	B	A	X	X	X	X	X	X	X	X	A	X	X	X	X	X	X	X	X	X	A	X	X	X	X	X	X	X	X	X	X	X	X	X	
I 37		B	B	B	A	A	I	I	I	L	I	L	L	I	I	I	L	I	I	I	L	L	L	I	I	L	L	L	L	L	L	L	L	L	L	L	L	L	
I 37		A	A	B	A	A	X	X	X	X	X	X	X	X	X	X	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
I 37		A	A	B	B	A	X	X	X	X	X	A	X	X	X	X	X	X	X	X	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
I 38		A	A	B	A	A	I	I	I	I	I	L	L	I	I	I	I	L	L	L	I	L	L	I	L	I	L	L	L	L	L	L	L	L	I	L	L	L	
I 38		B	B	B	A	A	X	X	X	X	X	A	X	X	X	X	X	A	X	X	X	A	X	X	X	X	A	X	X	X	X	X	X	X	X	X	X	X	
I 38		B	B	B	B	A	X	X	X	X	X	X	A	X	X	X	X	X	X	X	X	X	X	X	X	X	A	X	X	X	X	X	X	X	X	X	X	X	
I 39		B	B	B	A	A	L	L	L	L	L	L	L	L	L	L	I	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	
I 39		A	A	B	A	A	X	A	X	X	X	X	X	X	X	X	A	X	X	X	X	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
I 39	Functional	A	A	B	B	A	X	X	X	A	A	A	X	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	14, 15

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TABLE VIII. Functional test -Continued.

Test condition	Parameter	Input					Output																													Test figure
		Pin 21 to	Pin 32 to	Pin 36 to	Pin 35 to	Pin 26 to	Pin 28	Pin 7	Pin 27	Pin 6	Pin 22	Pin 2	Pin 23	Pin 3	Pin 11	Pin 31	Pin 15	Pin 12	Pin 18	Pin 16	Pin 19	Pin 40	Pin 30	Pin 9	Pin 29	Pin 8	Pin 24	Pin 4	Pin 25	Pin 5	Pin 33	Pin 13	Pin 34	Pin 14	Pin 38	Pin 17	Pin 37	Pin 39	Pin 20	
F40	Functional	A	A	B	A	A	L	L	L	I	L	I	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	14, 15
F40		B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	λ	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
F40		B	B	B	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
F41		B	B	B	A	A	L	L	L	L	L	I	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H	
F41		A	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
F41		A	A	B	B	A	X	X	X	Y	Y	X	X	λ	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
F42		A	A	B	A	A	L	L	I	L	L	L	I	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	
F42		B	B	B	A	A	X	X	X	X	X	λ	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
F42		B	B	B	A	B	X	X	X	X	λ	λ	Y	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
F43		B	B	B	A	A	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	
F43		B	B	A	A	A	X	X	λ	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	λ	X	X	λ	X	X	X	X	X	X	X	X	X	X	
F43		B	B	A	A	B	X	X	X	X	Y	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
F44		B	B	A	A	A	L	H	L	H	I	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	L	
F44		A	A	A	A	A	X	X	X	X	X	λ	λ	X	\	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	λ	X	X	X	X	X	
F44		A	A	A	A	B	X	Y	X	X	X	λ	X	X	X	X	X	X	X	X	X	X	X	X	X	X	Y	X	X	X	X	X	X	X	X	X	X	X	
F45		A	A	A	A	A	H	L	H	I	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	
		B	B	B	A	A	X	X	X	λ	Y	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	A	A	X	X	Y	λ	X	λ	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	B	A	λ	X	X	X	X	λ	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	A	A	X	X	Y	X	X	λ	X	X	X	X	X	X	X	X	X	X	X	λ	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	B	A	X	X	X	X	X	λ	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	A	A	X	Y	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	B	A	X	X	X	X	X	λ	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	A	A	X	X	X	λ	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	B	A	λ	λ	\	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	A	A	X	X	\	X	X	λ	X	X	λ	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	A	A	X	X	X	X	X	λ	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	B	A	X	X	X	X	X	X	Y	X	Y	X	X	X	X	X	X	Y	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
		B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
F45	Functional	B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	14, 15

See notes at end of table

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Input pins 21 to 28	Input pins 32 to 36	Input pins 35 to 26	Output pins 7	27	6	22	2	23	3	11	31	15	12	18	16	19	40	30	9	29	8	24	4	25	5	33	13	34	14	38	17	37	39	20	Test figure
F45	Functional	B	B	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	14, 15
		B	B	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

TABLE VIII. Functional test - (Continued

Test condition	Parameter	Input: Pin 21 to 32	Input: Pin 36 to 35	Input: Pin 26 to	Output: Pin 28	Output: Pin 7	Output: Pin 27	Output: Pin 6	Output: Pin 22	Output: Pin 2	Output: Pin 23	Output: Pin 3	Output: Pin 11	Output: Pin 31	Output: Pin 15	Output: Pin 12	Output: Pin 18	Output: Pin 16	Output: Pin 19	Output: Pin 40	Output: Pin 30	Output: Pin 9	Output: Pin 29	Output: Pin 8	Output: Pin 24	Output: Pin 4	Output: Pin 25	Output: Pin 5	Output: Pin 33	Output: Pin 13	Output: Pin 34	Output: Pin 14	Output: Pin 38	Output: Pin 17	Output: Pin 37	Output: Pin 39	Output: Pin 20	Test figure
F45	Functional	B	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	14, 15
		B	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

TABLE VIII. Functional test - Continued

| Test condition | Parameter | Input | | |

TABLE VIII Functional test -Continued.

Test condition	Parameter	Input pins 21 to 32	Input pins 36 to 35	Input pins 26 to	Output pins 28	7	27	11	31	15	12	18	16	19	40	30	9	29	8	24	4	25	5	33	13	34	14	38	17	37	39	20	Test figure
I-46	Functional	A	B	A	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	Y	L	L	L	L	L	L	L	L	L	L	H	14, 15
I-47		A	A	B	Y	X	X	A	Y	X	X	X	X	X	X	X	X	X	X	X	X	A	X	X	X	X	X	X	X	X	X	X	
I47		A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F-48		A	B	B	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	
F-48		A	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
I-48		A	B	A	X	A	A	X	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
I-49		A	A	A	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
		B	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
		B	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
		B	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
		B	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
		B	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
		B	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
I-49	Functional	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	14, 15

See notes at end of table

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TABLE VIII. Functional test - Continued

| Test condition | Parameter | Input | | | | Output | | | |

TABLE VIII Functional test -Continued.

Test condition	Parameter	Input					Output																																Test figure
		Pin 21 to	Pin 32 to	Pin 36 to	Pin 35 to	Pin 26 to	Pin 28	Pin 7	Pin 27	Pin 6	Pin 22	Pin 2	Pin 23	Pin 3	Pin 11	Pin 31	Pin 15	Pin 12	Pin 18	Pin 16	Pin 19	Pin 40	Pin 30	Pin 9	Pin 29	Pin 8	Pin 24	Pin 4	Pin 25	Pin 5	Pin 33	Pin 13	Pin 34	Pin 14	Pin 38	Pin 17	Pin 37	Pin 39	Pin 20	
FS0	Functional	A	A	B	A	A	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	14, 15
FS0		A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
FS0		A	A	A	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
FS1		A	A	A	A	A	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	
FS1		A	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
FS1		A	A	B	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
FS2		A	A	B	A	A	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	
FS2		B	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
FS2		B	B	A	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
FS3	Functional	B	B	A	A	A	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	14, 15

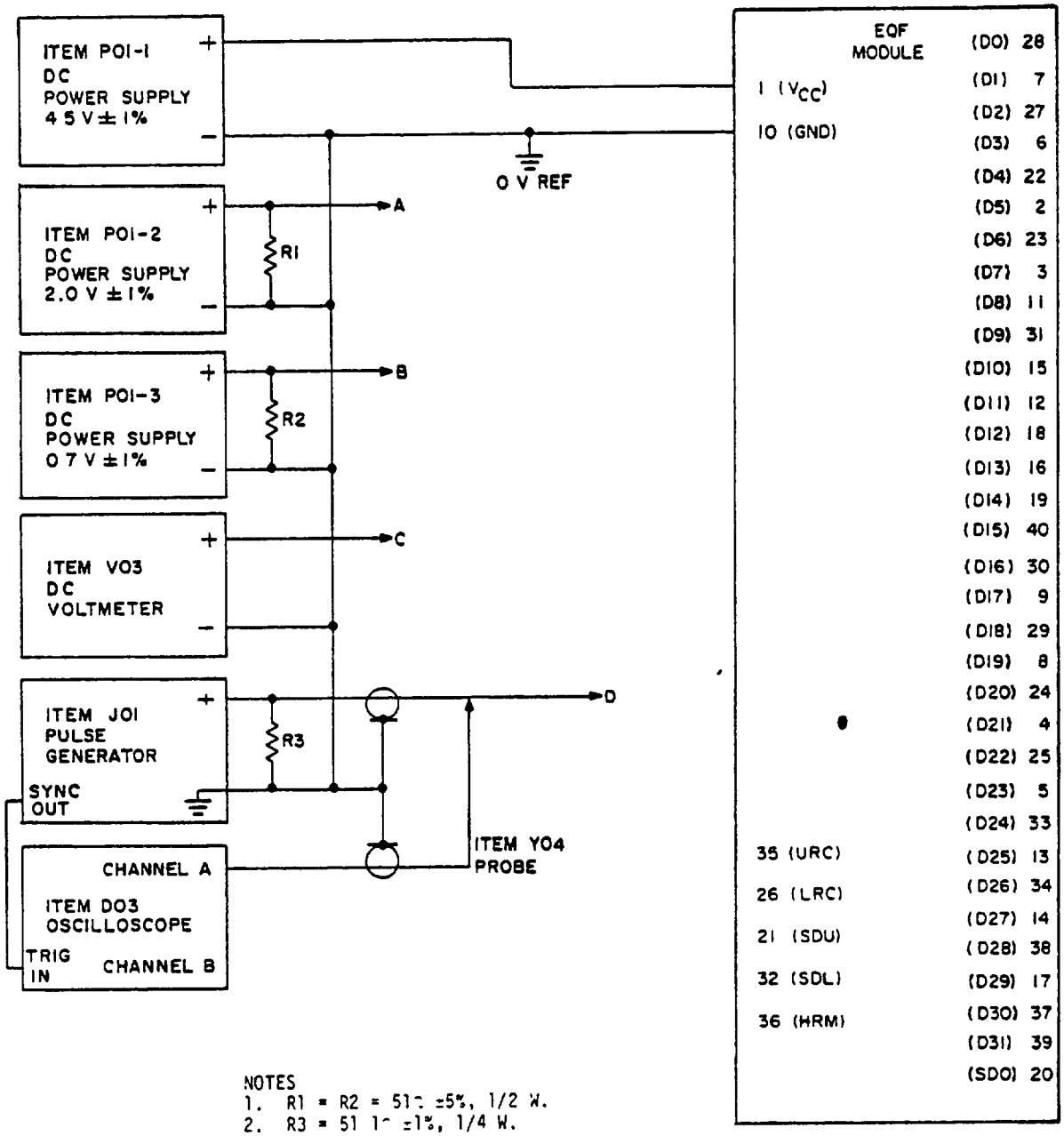
NOTES

1. Initial and end-of-life limits at 25° ±5°C ambient and 0° to 60°C operating temperature:

   L = 0.0 V minimum, 0.3 V maximum.
   H = 2 4 V minimum, 4.5 V maximum.
   X = Don't care

2. Verification of output logic levels L and H may be performed using a logic state indicator in place of item V03, figure 14.

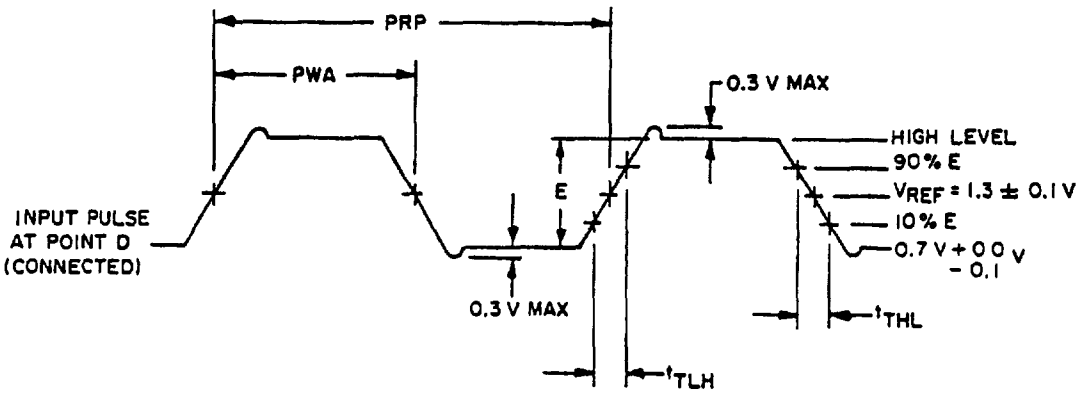
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NOTES
1. R1 = R2 = 51 Ω ±5%, 1/2 W.
2. R3 = 51 Ω ±1%, 1/4 W.

FIGURE 14 Functional test circuit for table VIII.

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NOTES:
1. Input pulse parameters:
   PRP = 1,000 ±200 ns.
   PWA = 500 ±100 ns.
   $t_{TLH} = t_{THL} = 50 \pm 25$ ns.
2. Clock threshold level setup: Make circuit connections per table VIII. Place item J01 in single pulse mode. Use item V03 and adjust dc offset of item J01 for a low level of 0.7 +0 V, -0.1 V (clock low-level threshold). Adjust item J01 to obtain the required PWA and PRP. Use item D03 and adjust item J01 amplitude for a high level of 2.0 +0.1 V, -0 V (clock high-level threshold). The high level amplitude can be adjusted to the required voltage by combining the dc offset as measured with item V03 with the pulse amplitude (PA) as measured with item D03.

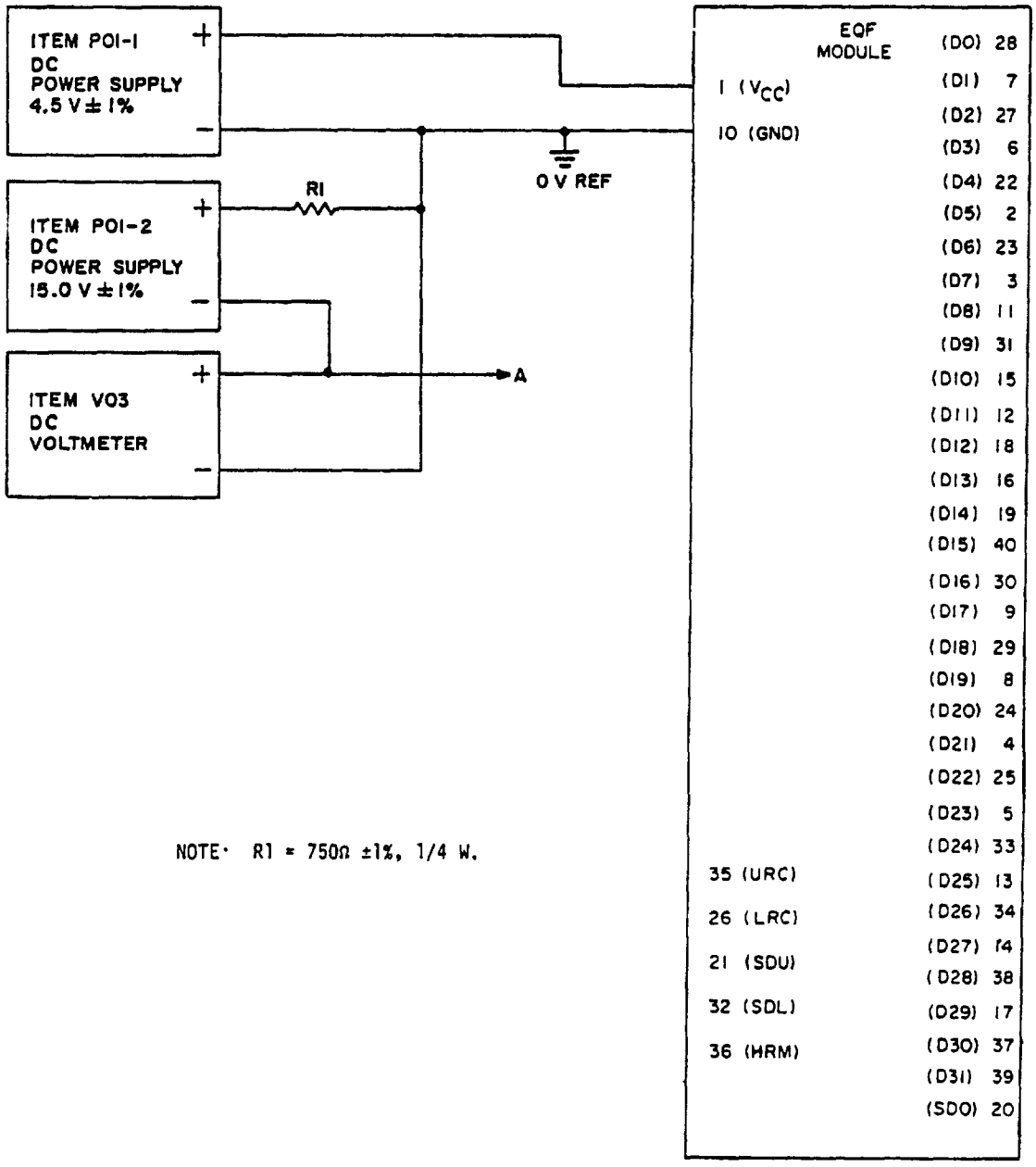
FIGURE 15. Functional test circuit waveform for table VIII.

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TABLE IX. Input clamp diode voltage test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
G1	URC Input clamp diode voltage	35	-	-1.5	V	16
G2	LRC	26				
G3	SDU	21				
G4	SDL	32				
G5	HRM Input clamp diode voltage	36		-1.5	V	16

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NOTE: R1 = 750Ω ±1%, 1/4 W.

FIGURE 16. Input clamp diode voltage test circuit for table IX

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TABLE X. Static output voltage (low-level) test.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H1	D0	Static output voltage, low level	21,32	36	28	0	0.4	V	17,18
H2	D1				7				
H3	D2				27				
H4	D3				6				
H5	D4				22				
H6	D5				2				
H7	D6				23				
H8	D7				3				
H9	D8				11				
H10	D9				31				
H11	D10				15				
H12	D11				12				
H13	D12				18				
H14	D13				16				
H15	D14				19				
H16	D15				40				
H17	D16				30				
H18	D17	Static output voltage,	21,32	36	9	0	0.4	V	17,18

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TABLE X. Static output voltage (low-level) test -Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H19	D18 Static output voltage, low level	21,32	36	29	0	0.4	V	17,18
H20	D19			8				
H21	D20			24				
H22	D21			4				
H23	D22			25				
H24	D23			5				
H25	D24			33				
H26	D25			13				
H27	D26			34				
H28	D27			14				
H29	D28			38				
H30	D29			17				
H31	D30			37				
H32	D31			39				
H33	SDO Static output voltage, low level	21,32	36	20	0	0.4	V	17,18

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TABLE XI. Static output voltage (high-level) test.

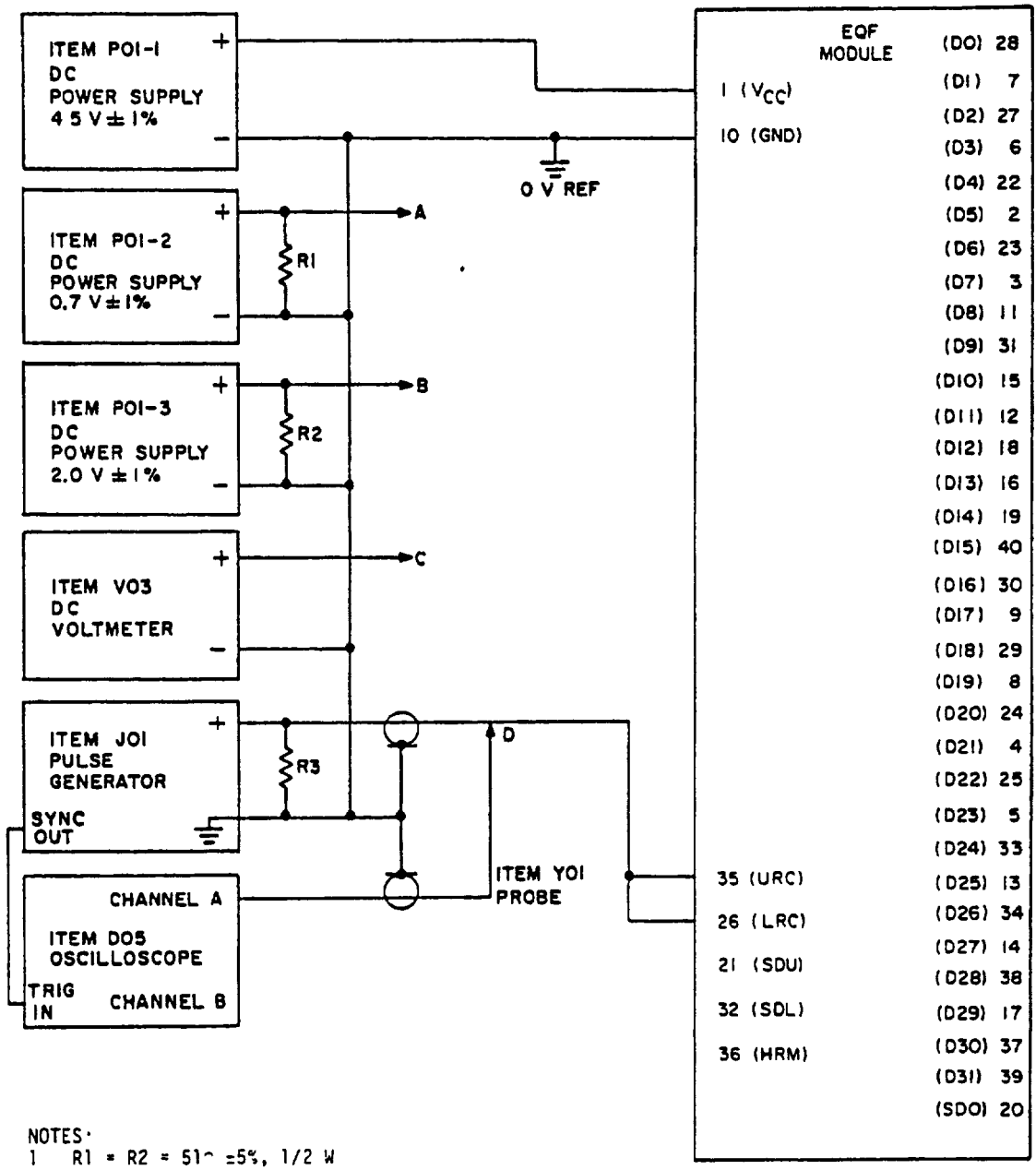
Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	D0 Static output voltage, high level	Open	21,32,36	28	2.5	4.5	V	17,18
J2	D1			7				
J3	D2			27				
J4	D3			6				
J5	D4			22				
J6	D5			2				
J7	D6			23				
J8	D7			3				
J9	D8			11				
J10	D9			31				
J11	D10			15				
J12	D11			12				
J13	D12			18				
J14	D13			16				
J15	D14			19				
J16	D15			40				
J17	D16			30				
J18	D17 Static output voltage, high level	Open	21,32,36	9	2.5	4.5	V	17,18

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TABLE XI. Static output voltage (high-level) test -Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
J19	D18	Static output voltage, high level	Open	21,32,36	29	2.5	4.5	V	17,18
J20	D19				8				
J21	D20				24				
J22	D21				4				
J23	D22				25				
J24	D23				5				
J25	D24				33				
J26	D25				13				
J27	D26				34				
J28	D27				14				
J29	D28				38				
J30	D29				17				
J31	D30				37				
J32	D31				39				
J33	SD0	Static output voltage, high level	Open	21,32,36	20	2.5	4.5	V	17,18

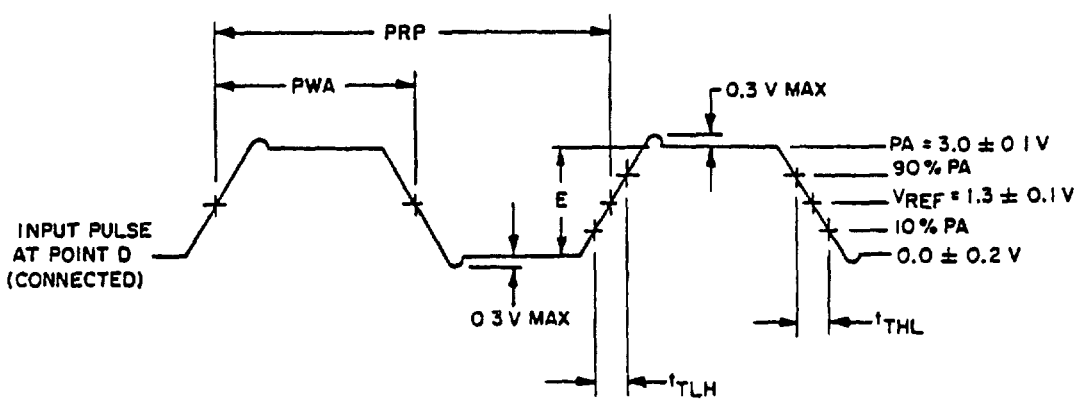
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NOTES:
1. R1 = R2 = 51^ ±5%, 1/2 W
2. R3 = 51.1^ ±1%, 1/4 W.
3. Load each output with a load as specified on figure 19
4. Allow a minimum of 33 clock pulses before making measurements.

FIGURE 17 Static output voltage (low-level and high-level) test circuit for tables X and XI

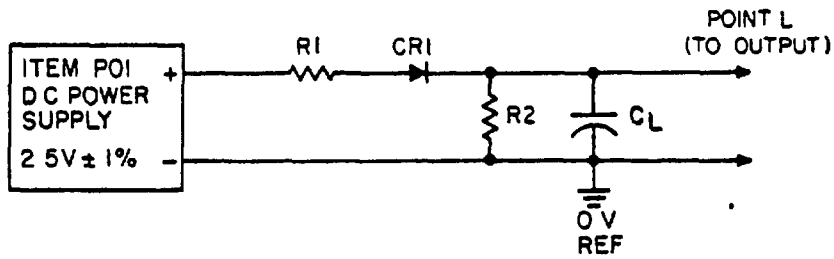
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:43Z
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NOTE: Input pulse parameters:
PRP = 1,000 ±100 ns.
PWA = 500 ±50 ns.
$t_{TLH} = t_{THL}$ = 50 ±25 ns.

FIGURE 18. Static output voltage (low-level and high-level) test circuit waveform for tables X and XI.

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NOTES:
1. The above circuit configuration shall be used for all load circuits.
2. Load circuit 1 for outputs D0-D2, D4-D6, D8-D10, D12-D14, D16-D18, D20-D22, D24-D26 and D28-D31:

   a. CR1 = 1N3600, 1N4150, or equivalent.
   b. R1 = Select value for a current flow of 4.0 mA ±5% into point L when point L is held at 0.4 V.
   c. R2 = 6.19 kΩ ±1%, 1/8 W.
   d. For test conditions N1-33, P1-33, Q1-33, R1-33, S1-32, T1-10, U1-66 and V1-66: C_L = 50 ±5 pF (including probe and parasitic capacitance).
   e. For test conditions N34-66, P34-66, Q34-66 and R34-66: C_L = 150 ±5 pF (including probe and parasitic capacitance).
   f. Capacitor C_L is not required for dc electrical test conditions F, H, and J.
   g. Lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 mm).

3. Load circuit 2 for outputs D3, D7, D11, D15, D19, D23, and D27.

   a. Same as 2a.
   b. R1 = Select value for a current flow of 3.28 mA ±5% into point L when point L is held at 0.4 V.
   c. R2 = 6.98 kΩ ±1%, 1/8 W.
   d. Same as 2d.
   e. Same as 2e.
   f. Same as 2f.
   g. Same as 2g.

4. Load circuit 3 for output SDO:

   a. Same as 2a.
   b. R1 = Select value for a current flow of 3.64 mA ±5% into point L when point L is held at 0.4 V.
   c. R2 = 6.65 kΩ ±1%, 1/8 W.
   d. Same as 2d.
   e. Same as 2e
   f Same as 2f.
   g Same as 2g.

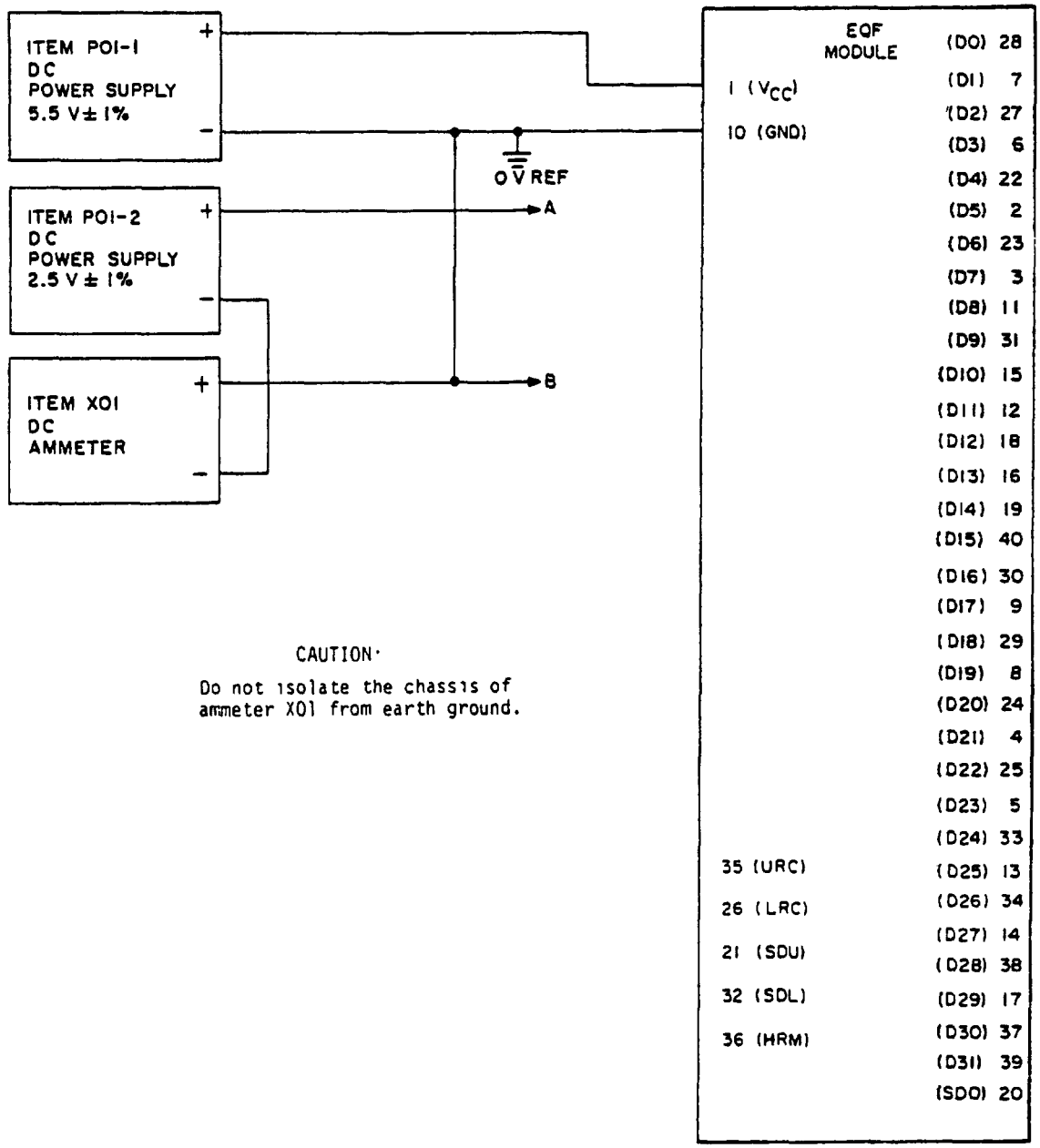
FIGURE 19 EQF logic load circuit

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TABLE XII. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	SDU Input current, high level	21	35,26,32,36	0.0	40	µA	20
K2	SDL	32	35,26,21,36		40		
K3	HRM	36	35,26,21,32		160		
K4	URC	35	26,21,32,36		160		
K5	LRC Input current, high level	26	35,21,32,36	0.0	160	µA	20

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

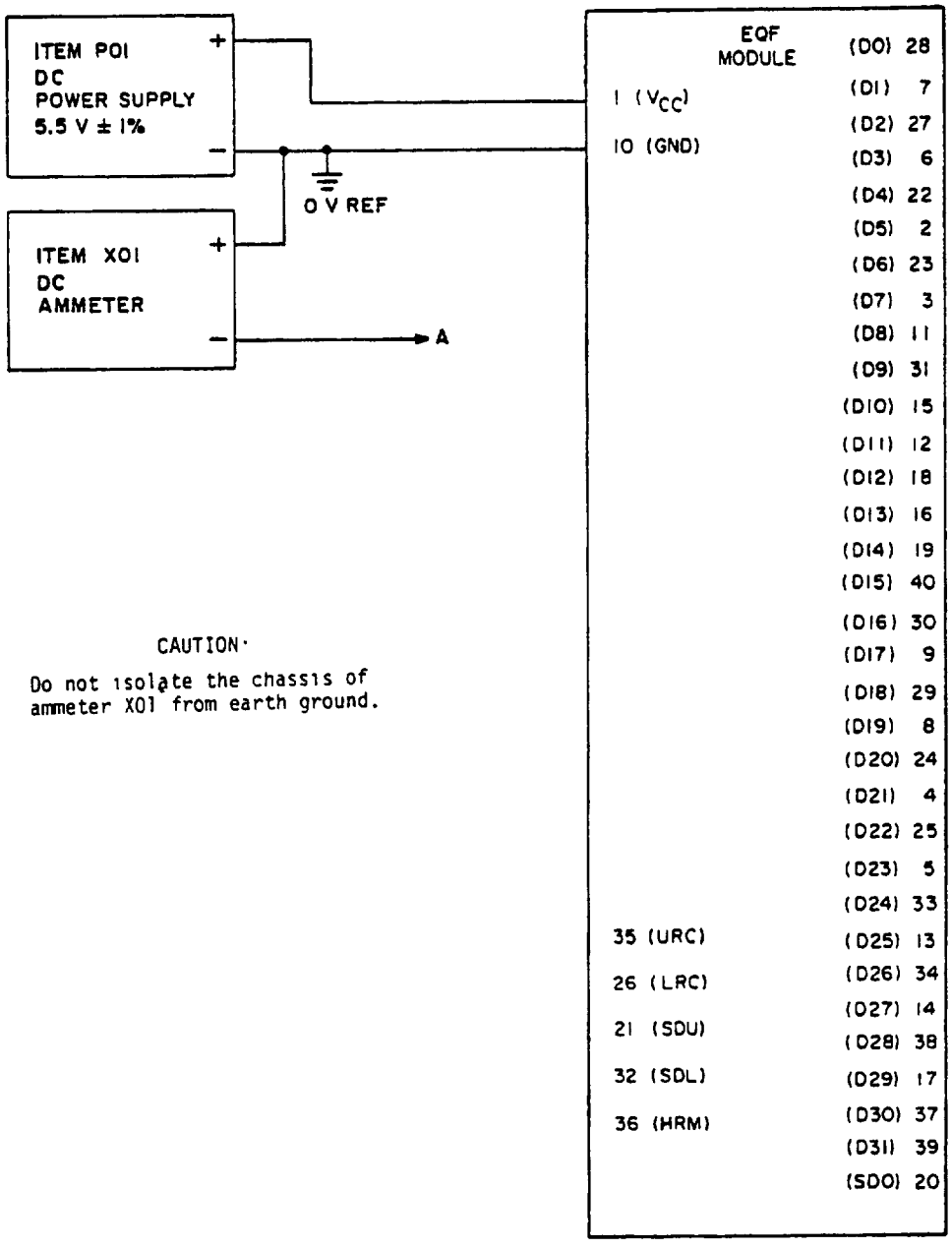
FIGURE 20 Input current (high-level) test circuit for table XII.

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TABLE XIII. Input current (low-level) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
L1	SDU Input current, low level	21	-0.1	-0.8	mA	21
L2	SDL	32		-0.8		
L3	URC	35		-3.91		
L4	LRC	26		-3.91		
L5	HRM Input current, low level	36	-0.1	-3.20	mA	21

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

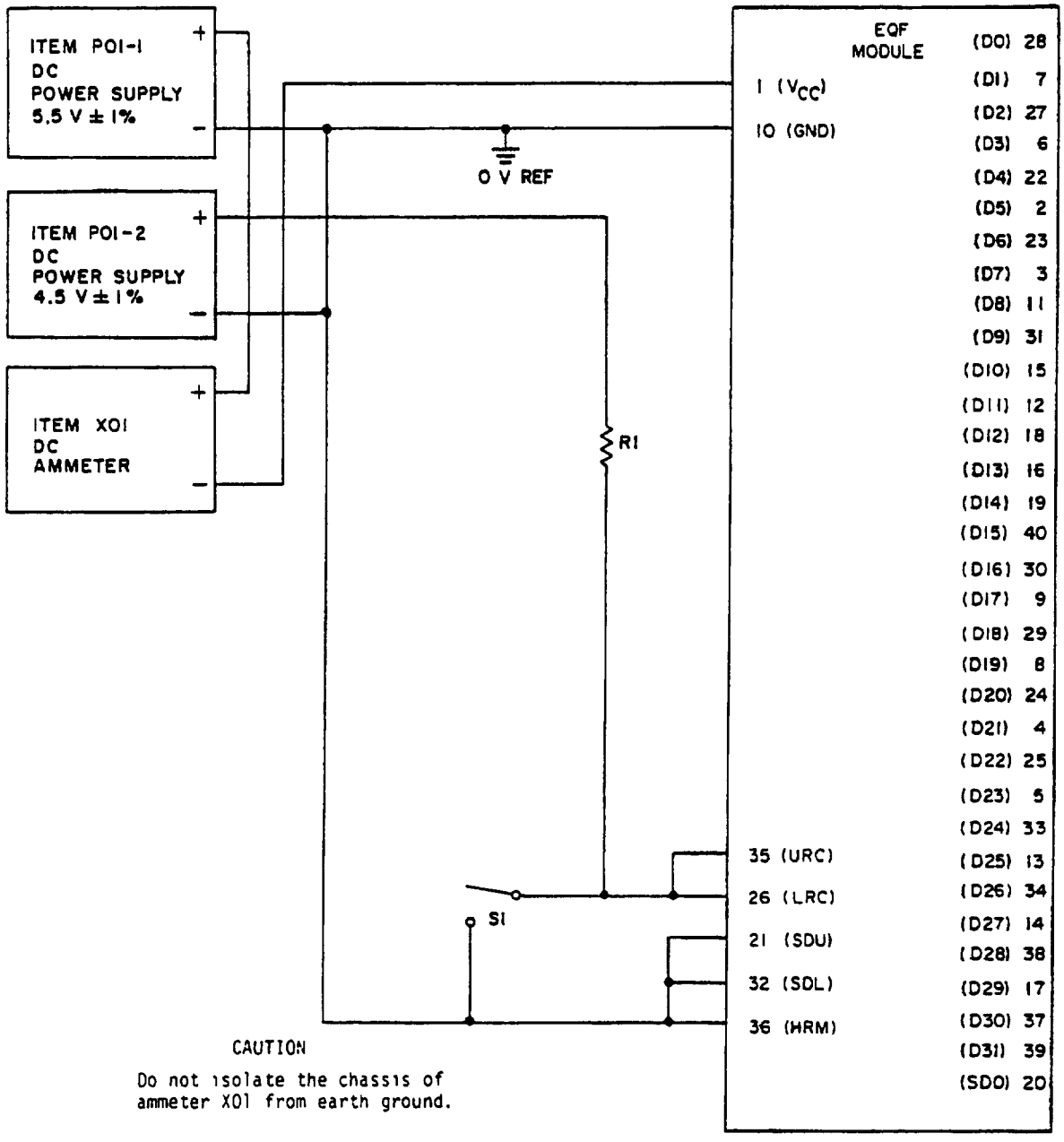
FIGURE 21 Input current (low-level) test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
M	Power supply current	50	340	mA	22

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CAUTION

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES
1 R1 = 1 58 k^ ±1^, 1/8 W
2 Open switch S1 for a minimum of 100 ns then close S1 and repeat the procedure of switching S1 32 times and then make measurement.

FIGURE 22 Power supply current test circuit for table XIV.

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TABLE XV. Transition time (t_{TLH}) test.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
N1	D0	Transition time (t_{TLH}), C_L = 50 pF	26	26	28	2	25	ns	23,24
N2	D1				7				
N3	D2				27				
N4	D3				6				
N5	D4				22				
N6	D5				2				
N7	D6				23				
N8	D7				3				
N9	D8				11				
N10	D9				31				
N11	D10				15				
N12	D11				12				
N13	D12				18				
N14	D13				16				
N15	D14				19				
N16	D15	Transition time (t_{TLH}), C_L = 50 pF	26	26	40	2	25	ns	23,24

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TABLE XV. Transition time (t_{TLH}) test -Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
N17	D16	Transition time (t_{TLH}), C_L = 50 pF	26	26	30	2	25	ns	23,24
N18	D17				9				
N19	D18				29				
N20	D19				8				
N21	D20				24				
N22	D21				4				
N23	D22				25				
N24	D23				5				
N25	D24				33				
N26	D25				13				
N27	D26				34				
N28	D27				14				
N29	D28				38				
N30	D29				17				
N31	D30				37				
N32	D31				39				
N33	SDO	Transition time (t_{TLH}), C_L = 50 pF	26	26	20	2	25	ns	23,24

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TABLE XV. Transition time (t_{TLH}) test -Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
N34	D0	Transition time (t_{TLH}), C_L = 150 pF	26	26	28	6	35	ns	23,24
N35	D1				7				
N36	D2				27				
N37	D3				6				
N38	D4				22				
N39	D5				2				
N40	D6				23				
N41	D7				3				
N42	D8				11				
N43	D9				31				
N44	D10				15				
N45	D11				12				
N46	D12				18				
N47	D13				16				
N48	D14				19				
N49	D15				40				
N50	D16	Transition time (t_{TLH}), C_L = 150 pF	26	26	30	6	35	ns	23,24

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TABLE XV. Transition time (t_{TLH}) test -Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N51	D17 Transition time (t_{TLH}), C_L = 150 pF	26	26	9	6	35	ns	23,24
N52	D18			29				
N53	D19			8				
N54	D20			24				
N55	D21			4				
N56	D22			25				
N57	D23			5				
N58	D24			33				
N59	D25			13				
N60	D26			34				
N61	D27			14				
N62	D28			38				
N63	D29			17				
N64	D30			37				
N65	D31			39				
N66	SDO Transition time (t_{TLH}), C_L = 150 pF	26	26	20	6	35	ns	23,24

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TABLE XVI. Transition time (t_{THL}) test.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P1	D0	Transition time (t_{THL}), C_L = 50 pF	26	26	28	1	10	ns	23,24
P2	D1				7				
P3	D2				27				
P4	D3				6				
P5	D4				22				
P6	D5				2				
P7	D6				23				
P8	D7				3				
P9	D8				11				
P10	D9				31				
P11	D10				15				
P12	D11				12				
P13	D12				18				
P14	D13				16				
P15	D14				19				
P16	D15				40				
P17	D16	Transition time (t_{THL}), C_L = 50 pF	26	26	30	1	10	ns	23,24

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TABLE XVI. Transition time (t_{THL}) test -Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P18	D17	Transition time (t_{THL}), C_L = 50 pF	26	26	9	1	10	ns	23,24
P19	D18				29				
P20	D19				8				
P21	D20				24				
P22	D21				4				
P23	D22				25				
P24	D23				5				
P25	D24				33				
P26	D25				13				
P27	D26				34				
P28	D27				14				
P29	D28				38				
P30	D29				17				
P31	D30				37				
P32	D31				39				
P33	SDO	Transition time (t_{THL}), C_L = 50 pF	26	26	20	1	10	ns	23,24

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TABLE XVI. Transition time (t_{THL}) test -Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P34	D0 Transition time (t_{THL}), C_L = 150 pF	26	26	28	2	15	ns	23,24
P35	D1			7				
P36	D2			27				
P37	D3			6				
P38	D4			22				
P39	D5			2				
P40	D6			23				
P41	D7			3				
P42	D8			11				
P43	D9			31				
P44	D10			15				
P45	D11			12				
P46	D12			18				
P47	D13			16				
P48	D14			19				
P49	D15			40				
P50	D16 Transition time (t_{THL}), C_L = 150 pF	26	26	30	2	15	ns	23,24

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TABLE XVI. Transition time (t_{THL}) test -Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P51	D17 Transition time (t_{THL}), C_L = 150 pF	26	26	9	2	15	ns	23,24
P52	D18			29				
P53	D19			8				
P54	D20			24				
P55	D21			4				
P56	D22			25				
P57	D23			5				
P58	D24			33				
P59	D25			13				
P60	D26			34				
P61	D27			14				
P62	D28			38				
P63	D29			17				
P64	D30			37				
P65	D31			39				
P66	SDO Transition time (t_{THL}), C_L = 150 pF	26	26	20	2	15	ns	23,24

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TABLE XVII. Propagation delay time (t_{PLH}) test and SDU input setup and hold time check.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q1	LRC to D0	Propagation delay time (t_{PLH}), C_L = 50 pF	26	26	28	10	40	ns	23,24
Q2	LRC to D1				7				
Q3	LRC to D2				27				
Q4	LRC to D3				6				
Q5	LRC to D4				22				
Q6	LRC to D5				2				
Q7	LRC to D6				23				
Q8	LRC to D7				3				
Q9	LRC to D8				11				
Q10	LRC to D9				31				
Q11	LRC to D10				15				
Q12	LRC to D11				12				
Q13	LRC to D12				18				
Q14	LRC to D13				16				
Q15	LRC to D14				19				
Q16	LRC to D15	Propagation delay time (t_{PLH}), C_L = 50 pF	26	26	40	10	40	ns	23,24

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TABLE XVII. Propagation delay time (t_{PLH}) test and SDU input setup and hold time check -Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q17	LRC to D16	Propagation delay time (t_{PLH}), C_L = 50 pF	26	26	30	10	40	ns	23,24
Q18	IRC to D17				9				
Q19	LRC to D18				29				
Q20	LRC to D19				8				
Q21	LRC to D20				24				
Q22	LRC to D21				4				
Q23	LRC to D22				25				
Q24	LRC to D23				5				
Q25	IRC to D24				33				
Q26	LRC to D25				13				
Q27	LRC to D26				34				
Q28	IRC to D27				14				
Q29	LRC to D28				38				
Q30	LRC to D29				17				
Q31	LRC tp D30				37				
Q32	LRC to D31	Propagation delay time (t_{PLH}), C_L = 50 pF	26	26	39	10	40	ns	23,24

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:43Z
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TABLE XVII. Propagation delay time (t_{PLH}) test and SDU input setup and hold time check -Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q33	URC to SDO	Propagation delay time (t_{PLH}), C_L = 50 pF	Open	35	20	10	40	ns	23,24
Q34	LRC to D0	Propagation delay time (t_{PLH}), C_L = 150 pF	26	26	28		55		
Q35	LRC to D1				7				
Q36	LRC to D2				27				
Q37	LRC to D3				6				
Q38	LRC to D4				22				
Q39	LRC to D5				2				
Q40	LRC to D6				23				
Q41	LRC to D7				3				
Q42	LRC to D8				11				
Q43	LRC to D9				31				
Q44	LRC to D10				15				
Q15	LRC to D11				12				
Q46	LRC to D12				18				
Q47	LRC to D13	Propagation delay time (t_{PLH}), C_L = 150 pF	26	26	16	10	55	ns	23,24

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TABLE XVII. Propagation delay time (t_{PLH}) test and SDU input setup and hold time check -Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q48	IRC to D14	Propagation delay time (t_{PLH}), C_L = 150 pF	26	26	19	10	55	ns	23,24
Q49	IRC to D15				40				
Q50	IRC to D16				30				
Q51	IRC to D17				9				
Q52	LRC to D18				29				
Q53	LRC to D19				8				
Q54	LRC to D20				24				
Q55	LRC to D21				4				
Q56	LRC to D22				25				
Q57	LRC to D23				5				
Q58	LRC to D24				33				
Q59	LRC to D25				13				
Q60	IRC to D26				34				
Q61	LRC to D27				14				
Q62	LRC to D28				38				
Q63	LRC to D29	Propagation delay time (t_{PLH}), C_L = 150 pF	26	26	17	10	55	ns	23,24

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TABLE XVII Propagation delay time (t_{PLH}) test and SDU input setup and hold time check -Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q64	LRC to D30 Propagation delay time (t_{PLH}), C_L = 150 pF	26	26	37	10	55	ns	23,24
Q65	LRC to D31 Propagation delay time (t_{PLH}), C_L = 150 pF	26	26	39	10	55	ns	23,24
Q66	URC to SDO Propagation delay time (t_{PLH}), C_L = 150 pF	Open	35	20	10	55	ns	23,24

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TABLE XVIII. Propagation delay time (t_{PHL}) test and SDU input setup and hold time check.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
R1	IRC to D0	Propagation delay time (t_{PHL}), C_L = 50 pF	26	26	28	10	45	ns	23,24
R2	IRC to D1				7				
R3	IRC to D2				27				
R4	IRC to D3				6				
R5	LRC to D4				22				
R6	IRC to D5				2				
R7	LRC to D6				23				
R8	IRC to D7				3				
R9	IRC to D8				11				
R10	IRC to D9				31				
R11	LRC to D10				15				
R12	LRC to D11				12				
R13	IRC to D12				18				
R14	IRC to D13				16				
R15	IRC to D14				19				
R16	LRC to D15	Propagation delay time (t_{PHL}),	26	26	40	10	45	ns	23,24

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Check the source to verify that this is the current version before use.

TABLE XVIII Propagation delay time (t_{PHL}) test and SDU input setup and hold time check -Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
R17	LRC to D16	Propagation delay time (t_{PHL}), C_L = 50 pF	26	26	30	10	45	ns	23,24
R18	LRC to D17				9				
R19	LRC to D18				29				
R20	LRC to D19				8				
R21	LRC to D20				24				
R22	LRC to D21				4				
R23	LRC to D22				25				
R24	LRC to D23				5				
R25	LRC to D24				33				
R26	LRC to D25				13				
R27	IRC to D26				34				
R28	IRC to D27				14				
R29	IRC to D28				38				
R30	LRC to D29				17				
R31	LRC to D30				37				
R32	LRC to D31	Propagation delay time (t_{PHL}), C_L = 50 pF	26	26	39	10	45	ns	23,24

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:43Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Propagation delay time (t_{PHL}) test and SDU input setup and hold time check -Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
R33	URC to SDO	Propagation delay time (t_{PHL}), C_L = 50 pF	Open	35	20	10	45	ns	23,24
R34	LRC to D0	Propagation delay time (t_{PHL}), C_L = 150 pF	26	26	28		55		
R35	LRC to D1				7				
R36	LRC to D2				27				
R37	LRC to D3				6				
R38	LRC to D4				22				
R39	LRC to D5				2				
R40	LRC to D6				23				
R41	LRC to D7				3				
R42	LRC to D8				11				
R43	LRC to D9				31				
R44	LRC to D10				15				
R45	LRC to D11				12				
R46	LRC to D12				18				
R47	LRC to D13	Propagation delay time (t_{PHL}), C_L = 150 pF	26	26	16	10	55	ns	23,24

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:43Z
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TABLE XVIII. Propagation delay time (t_{PHL}) test and SDU input setup and hold time check -Continued.

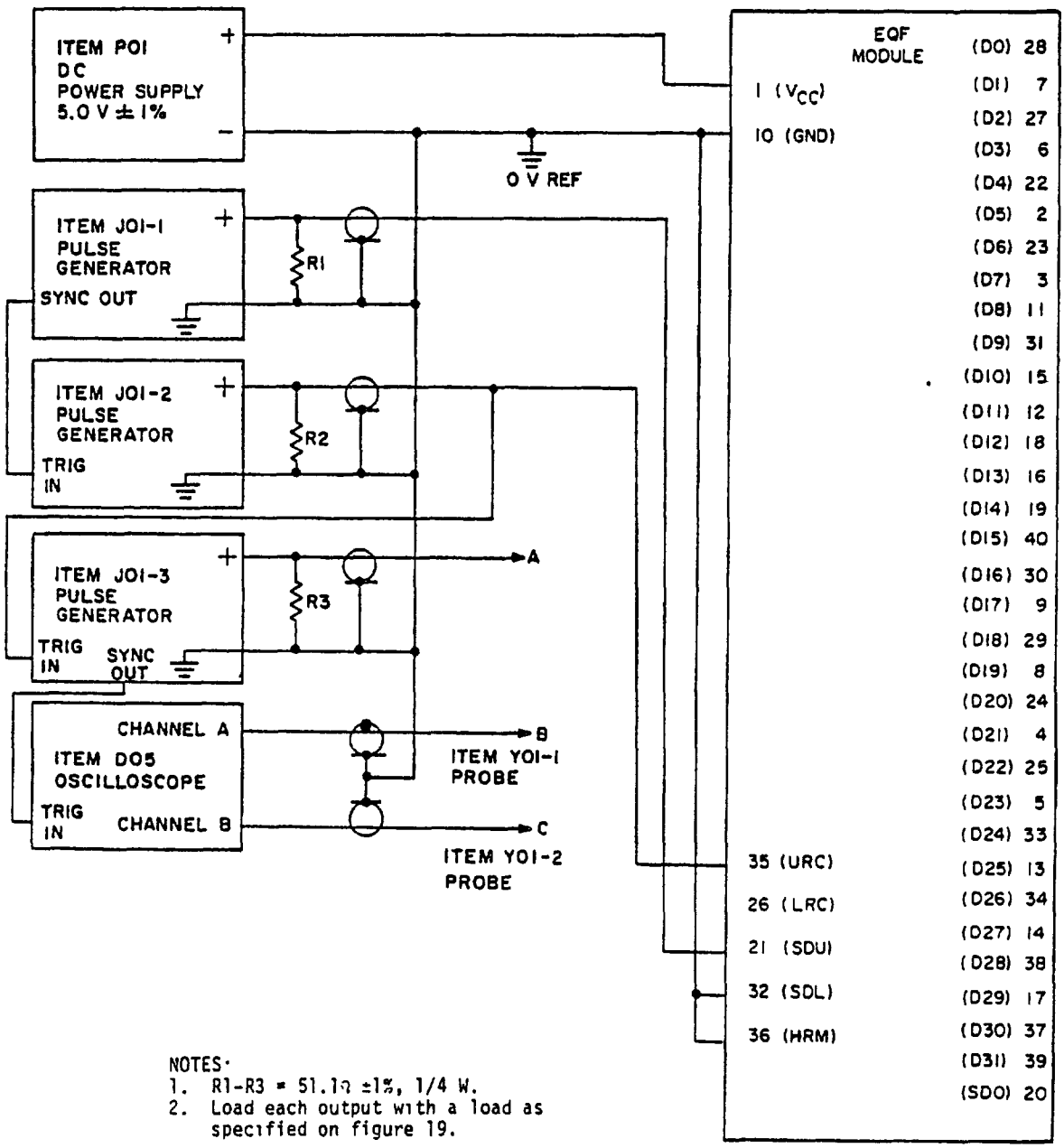
Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
R48	LRC to D14	Propagation delay time (t_{PHL}), C_L = 150 pF	26	26	19	10	55	ns	23,24
R49	LRC to D15				40				
R50	LRC to D16				30				
R51	LRC to D17				9				
R52	LRC to D18				29				
R53	LRC to D19				8				
R54	LRC to D20				24				
R55	LRC to D21				4				
R56	LRC to D22				25				
R57	LRC to D23				5				
R58	LRC to D24				33				
R59	LRC to D25				13				
R60	LRC to D26				34				
R61	LRC to D27				14				
R62	LRC to D28				38				
R63	LRC to D29	Propagation delay time (t_{PHL}), C_L = 150 pF	26	26	17	10	55	ns	23,24

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:43Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Propagation delay time (t_{PHL}) test and SDU input setup and hold time check -Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R64	LRC to D30 Propagation delay time (t_{PHL}), C_L = 150 pF	26	26	37	10	55	ns	23,24
R65	LRC to D31 Propagation delay time (t_{PHL}), C_L = 150 pF	26	26	39	10	55	ns	23,24
R66	URC to SDO Propagation delay time (t_{PHL}), C_L = 150 pF	Open	35	20	10	55	ns	23,24

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NOTES:
1. R1-R3 = 51.1Ω ±1%, 1/4 W.
2. Load each output with a load as specified on figure 19.

FIGURE 23. Transition time (t_{TLH} and t_{THL}), propagation delay time (t_{PLH} and t_{PHL}), and SDU input setup and hold time check test circuit for tables XV through XVIII

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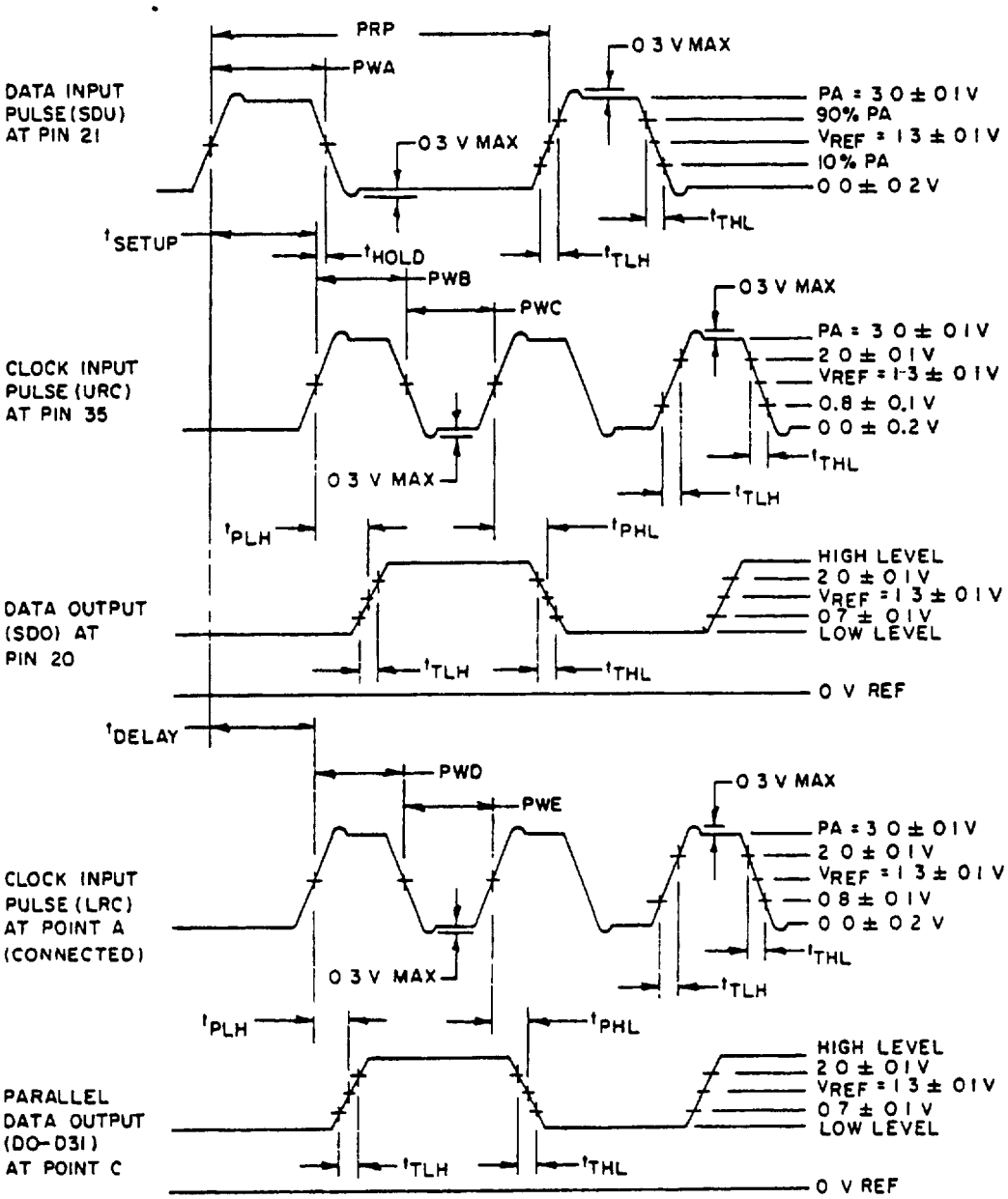


FIGURE 24. Transition time (t_{TLH} and t_{THL}), propagation delay time (t_{PLH} and t_{PHL}), and SDU input setup and hold time check test circuit waveforms for tables XV through XVIII

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NOTES:
1. SDU data input pulse parameters:
   PRP = 500 ±50 ns.
   PWA = 25 ±1 ns.
   $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
2. URC clock input pulse parameters:
   PWB = 50 ±5 ns.
   PWC = 200 ±5 ns.
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
3. LRC clock input pulse parameters:
   PWD = 50 ±5 ns.
   PWE = 200 ±5 ns.
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
4. Item J01-2 is operated in the double pulse mode using the positive output pulse in the dc offset mode. Using item D05, adjust the dc logic level and pulse amplitude of items J01-2 and J01-3 to obtain the required levels for the URC and LRC clock input pulses. Using the delay controls and pulse width controls, adjust the pulse relationship of the data input and clock input pulses so that $t_{setup} = 20 +1.0$ ns, -0.0 ns; $t_{hold} = 5 +1.0$ ns, -0.0 ns; (PWA = $t_{setup} + t_{hold}$); $t_{delay} = 100 \pm 10$ ns.
5. Verify that all outputs have a high level $\geq$ 2.5 V and a low level $\leq$ 0.4 V before making measurements for tables XV through XVIII.
6. The physical length of electrical conductors between resistors R1, R2, and R3 and input pins, between probes Y01-1 and Y01-2 and input pins; and between load circuits and output pins shall be less than 1 inch (25.4 mm).
7. All voltages are referenced to 0 V reference unless otherwise indicated.

FIGURE 24. Transition time (t_{TLH} and t_{THL}), propagation delay time (t_{PLH} and t_{PHL}), and SDU input setup and hold time check test circuit waveforms for tables XV through XVIII - Continued.

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TABLE XIX. URC to LRC clock delay (measuring t_{PD}) check.

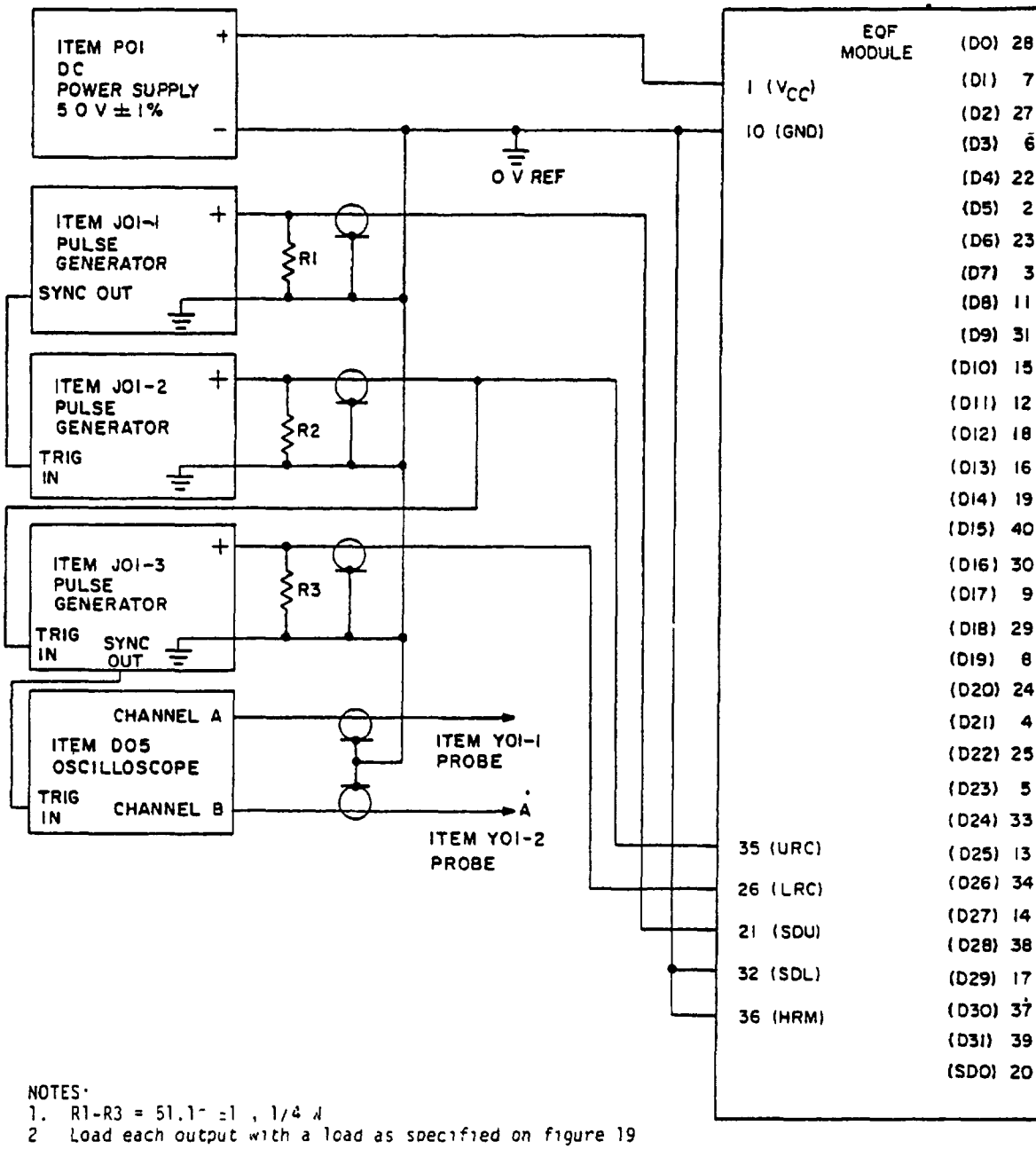
Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
S1	D0	URC to LRC clock delay (measuring t_{PD}) check	28	450	550	ns	25,26
S2	D1		7				
S3	D2		27				
S4	D3		6				
S5	D4		22				
S6	D5		2				
S7	D6		23				
S8	D7		3				
S9	D8		11				
S10	D9		31				
S11	D10		15				
S12	D11		12				
S13	D12		18				
S14	D13		16				
S15	D14		19				
S16	D15		40				
S17	D16	URC to LRC clock delay (measuring t_{PD}) check	30	450	550	ns	25,26

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TABLE XIX. URC to LRC clock delay (measuring t_{PD}) check -Continued.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
S18	D17 URC to LRC clock delay (measuring t_{PD}) check	9	450	550	ns	25,26
S19	D18	29				
S20	D19	8				
S21	D20	24				
S22	D21	4				
S23	D22	25				
S24	D23	5				
S25	D24	33				
S26	D25	13				
S27	D26	34				
S28	D27	14				
S29	D28	38				
S30	D29	17				
S31	D30	37				
S32	D31 URC to LRC clock delay (measuring t_{PD}) check	39	450	550	ns	25,26

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NOTES:
1. R1-R3 = 51.1 ±1 , 1/4 W
2 Load each output with a load as specified on figure 19

FIGURE 25 URC to LRC clock delay (measuring t_{PD}) check test circuit for table XIX.

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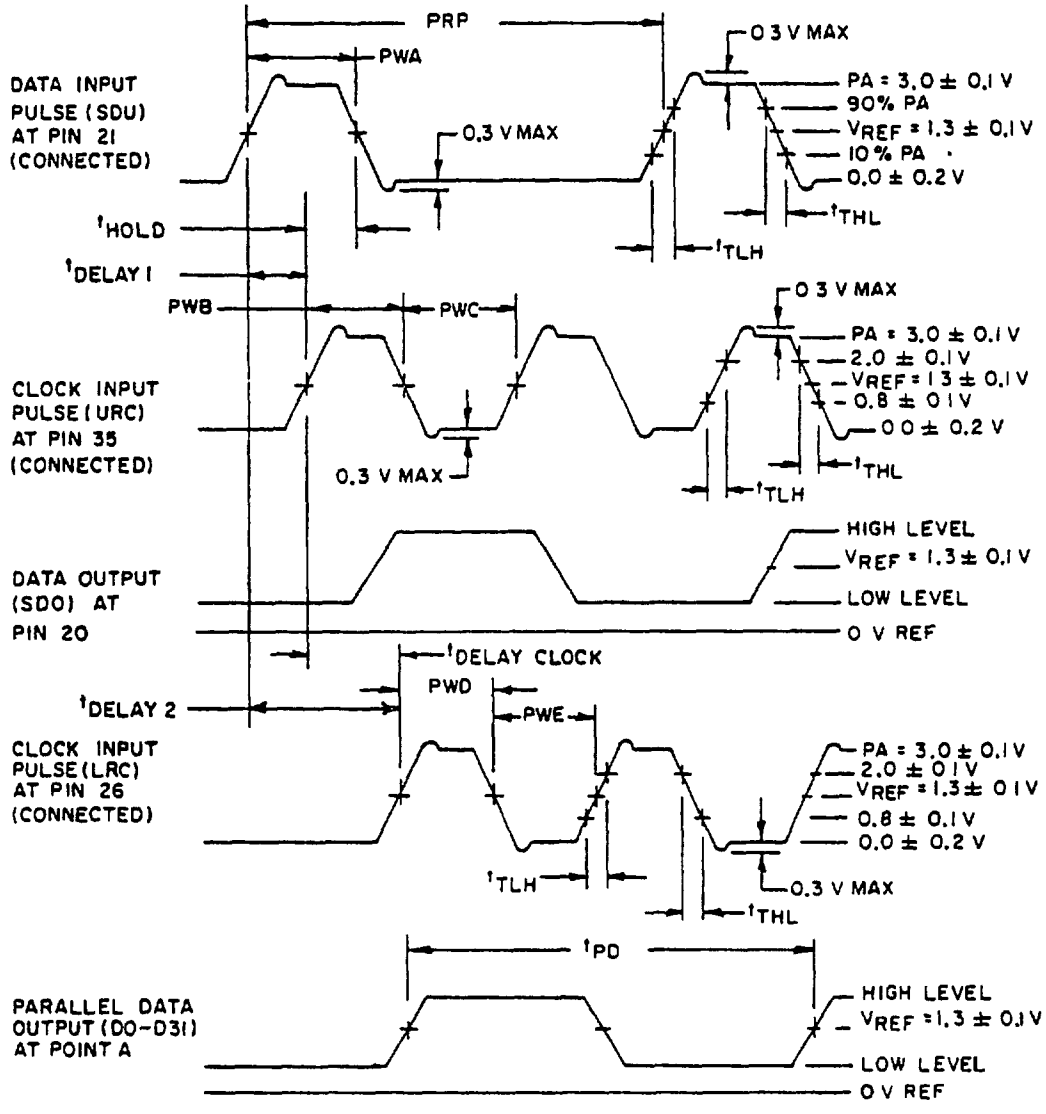


FIGURE 26. URC to LRC clock delay (measuring t_{PD}) check test circuit waveforms for table XIX.

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NOTES:
1. SDU data input pulse parameters:
   PRP = 500 ±50 ns.
   PWA = 50 ±5 ns.
   $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
2. URC clock input pulse parameters:
   PWB = 50 ±5 ns.
   PWC = 200 ±10 ns.
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
   $t_{delay\ 1} = 25$ +2.5 ns, -0.0 ns, $t_{hold} = 25 \pm 2.5$ ns.
3. LRC clock input pulse parameters:
   PWD = 50 ±5 ns.
   PWE = 200 ±10 ns.
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
   $t_{delay\ 2} = 90$ +4 ns, -0 ns.
4. Item J01-2 is operated in the double pulse mode. Items J01-1, J01-2, and J01-3 use the positive output Using item D05 adjust the amplitude and dc offset controls to obtain the required input logic levels of the data and clock input pulses. Using the delay and pulse width controls, adjust the data input and URC clock input pulses so that $t_{delay\ 1} = 25$ +2.5 ns, -0.0 ns. Adjust the data input and LRC clock input pulses so that $t_{delay\ 2} = 90$ +4 ns, -0 ns. (Minimum URC to LRC clock delay = $(t_{delay\ 2} - t_{delay\ 1}) = 65$ ns.)
5. Verify that all outputs have a high level $\geq$ 2.5 V and a low level $\leq$ 0.4 V before making measurements for table XIX.
6. The physical length of electrical conductors between resistors R1, R2, and R3 and input pins; between probes Y01-1 and Y01-2 and input pins, and between load circuits and output pins shall be less than 1 inch (25.4 mm).
7. All voltages are referenced to 0 V reference unless otherwise specified

FIGURE 26. URC to LRC clock delay (measuring t_{pd}) check test circuit waveforms for table XIX - Continued

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TABLE XX. Maximum operating frequency check, minimum clock pulse width check, input data setup and hold time check, and HRM setup and hold time check.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	SDO Maximum operating frequency check, minimum URC clock pulse width check, SDU high level data setup and hold time check (measuring SDO output pulse period t_{PD})	36	21	35	20	130	150	ns	27,28
T2	SDO Maximum operating frequency check, minimum URC clock pulse width check, SDU high level data setup and hold time check (measuring SDO output pulse width t_{PW})					60	80		27,28
T3	SDO Maximum operating frequency check, minimum URC clock pulse width check, SDU low level data setup and hold time check (measuring SDO output pulse period t_{PD})					130	150		27,29
T4	SDO Maximum operating frequency check, minimum URC clock pulse width check, SDU low level data setup and hold time check (measuring SDO output pulse width t_{PW})	36	21	35	20	60	80	ns	27,29

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TABLE XX. Maximum operating frequency check, minimum clock pulse width check, input data setup and hold time check, and IRM setup and hold time check -Continued.

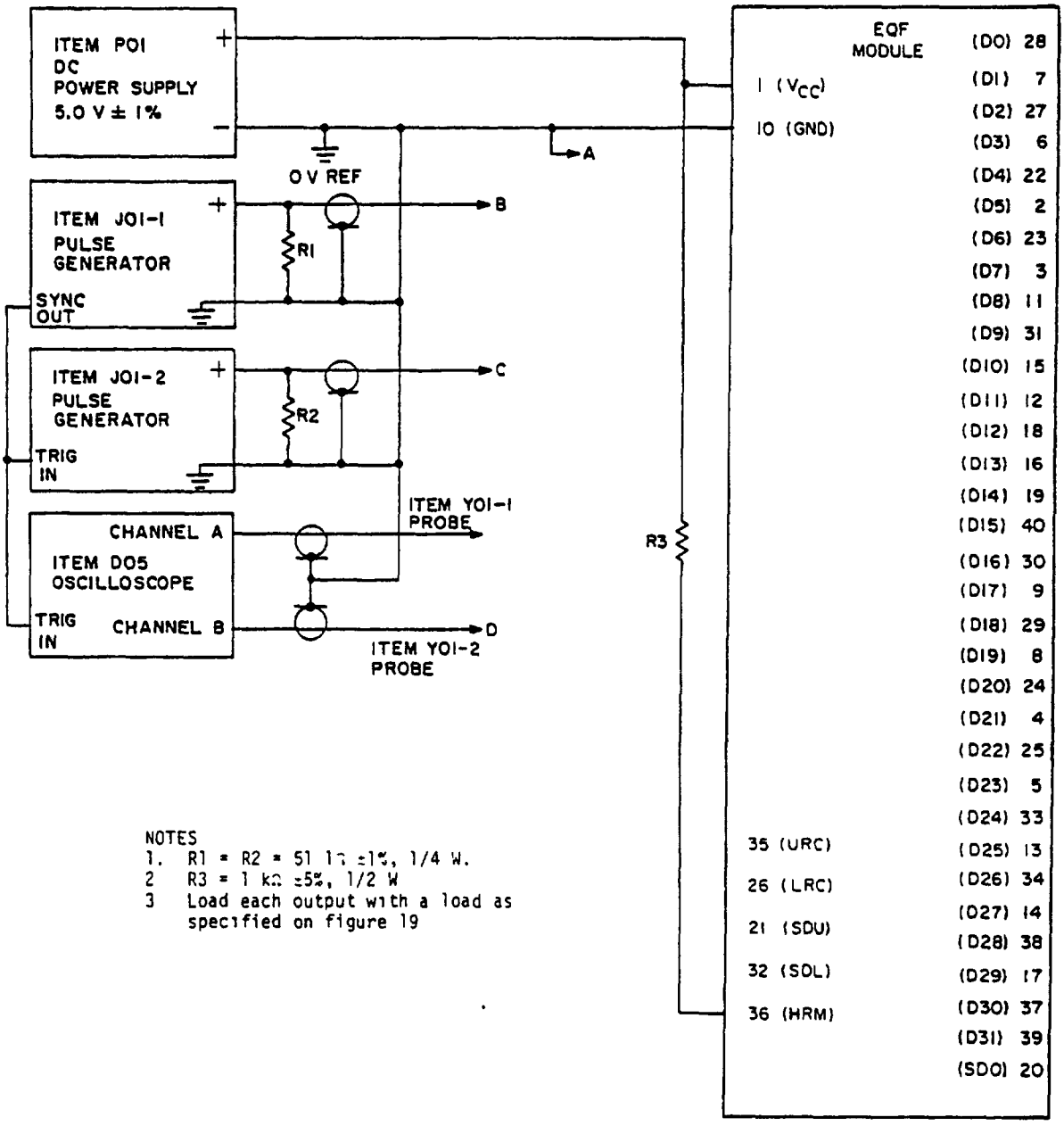
Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T5	D31 Maximum operating frequency check, minimum LRC clock pulse width check, SDL high level data setup and hold time check, (measuring D31 output pulse period t_{PD})	21	32	26	39	130	150	ns	30,31
T6	D31 Maximum operating frequency check, minimum LRC clock pulse width check, SDL high level data setup and hold time check, (measuring D31 output pulse period t_{PW})					60	80		30,31
T7	D31 Maximum operating frequency check, minimum LRC clock pulse width check, SDL low level data setup and hold time check (measuring D31 output pulse period t_{PD})					130	150		30,32
T8	D31 Maximum operating frequency check, minimum LRC clock pulse width check, SDL low level data setup and hold time check (measuring D31 output pulse width t_{PW})	21	32	26	39	60	80	ns	30,32

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TABLE XX Maximum operating frequency check, minimum clock pulse width check, input data setup and hold time check, and HRM setup and hold time check -Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T9	HRM Setup and hold time check for parallel load	Open	Open	Open	39	2.5	4.5	V	33,34
T10	HRM Setup and hold time check for serial shift	Open	Open	Open	39	0.0	0.4	V	33,34

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NOTES
1. R1 = R2 = 51 Ω ±1%, 1/4 W.
2 R3 = 1 kΩ ±5%, 1/2 W
3 Load each output with a load as specified on figure 19

FIGURE 27 SDO maximum operating frequency check, minimum URC clock pulse width check, and SDU input data setup and hold time check test circuit for test conditions T1 - T4, table XX

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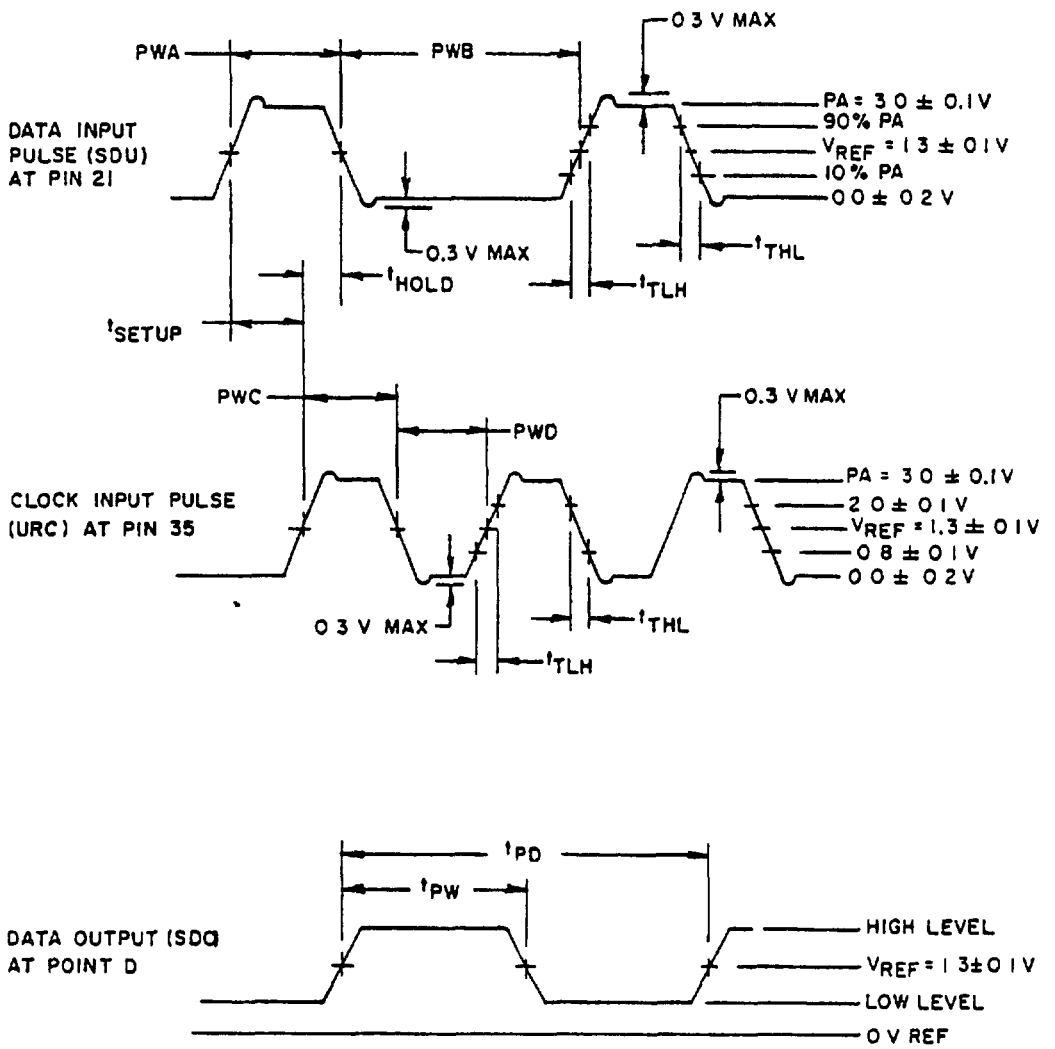


FIGURE 28. SDO maximum operating frequency check, minimum URC clock pulse width check, and SDU high level data setup and hold time check test circuit waveforms for test conditions T1 and T2, table XX.

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NOTES:
1. SDU data input pulse parameters:
   PWA = 25 ±1.0 ns.
   PWB = 115 ±3 ns.
   $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
   $t_{setup} = 20 \pm 1$ ns.
   $t_{hold} = 5$ +1.0 ns, -0.0 ns.
   PWA = $t_{setup} + t_{hold} = 25$ ns
2. URC input pulse parameters:
   PWC = 20 +1 ns, -0.0 ns.
   PWD = 50 ±3 ns.
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
   PWC + PWD = Maximum operating frequency.
3. Item J01-2 is operated in the double pulse mode. Using item D05, adjust the amplitude and dc offset controls to obtain the required input logic levels of the data and upper rank clock (URC) input pulses. Using the delay and pulse width controls, adjust items J01-1 and J01-2 so that the criteria of notes 1 and 2 are met.
4. Verify that all outputs have a high level $\geq$ 2.5 V and a low level $\leq$ 0.4 V before making measurements for table XX.
5. The physical length of electrical conductors between resistors R1, R2, and R3 and input pins between probes Y01-1 and Y01-2 and input pins; and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to 0 V reference unless otherwise specified.

FIGURE 28 SDO maximum operating frequency check, minimum URC clock pulse width check, and SDU high level data setup and hold time check test circuit waveforms for test conditions T1 and T2, table XX - Continued.

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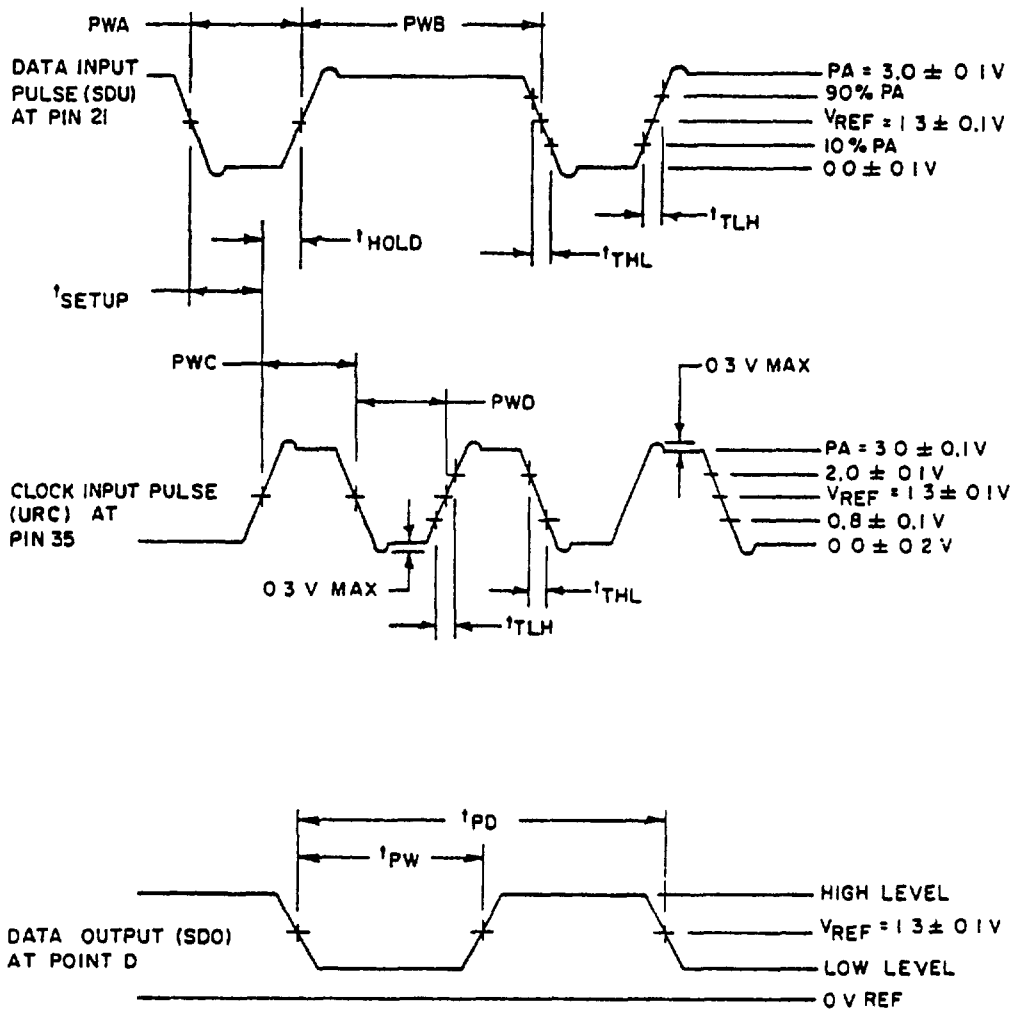


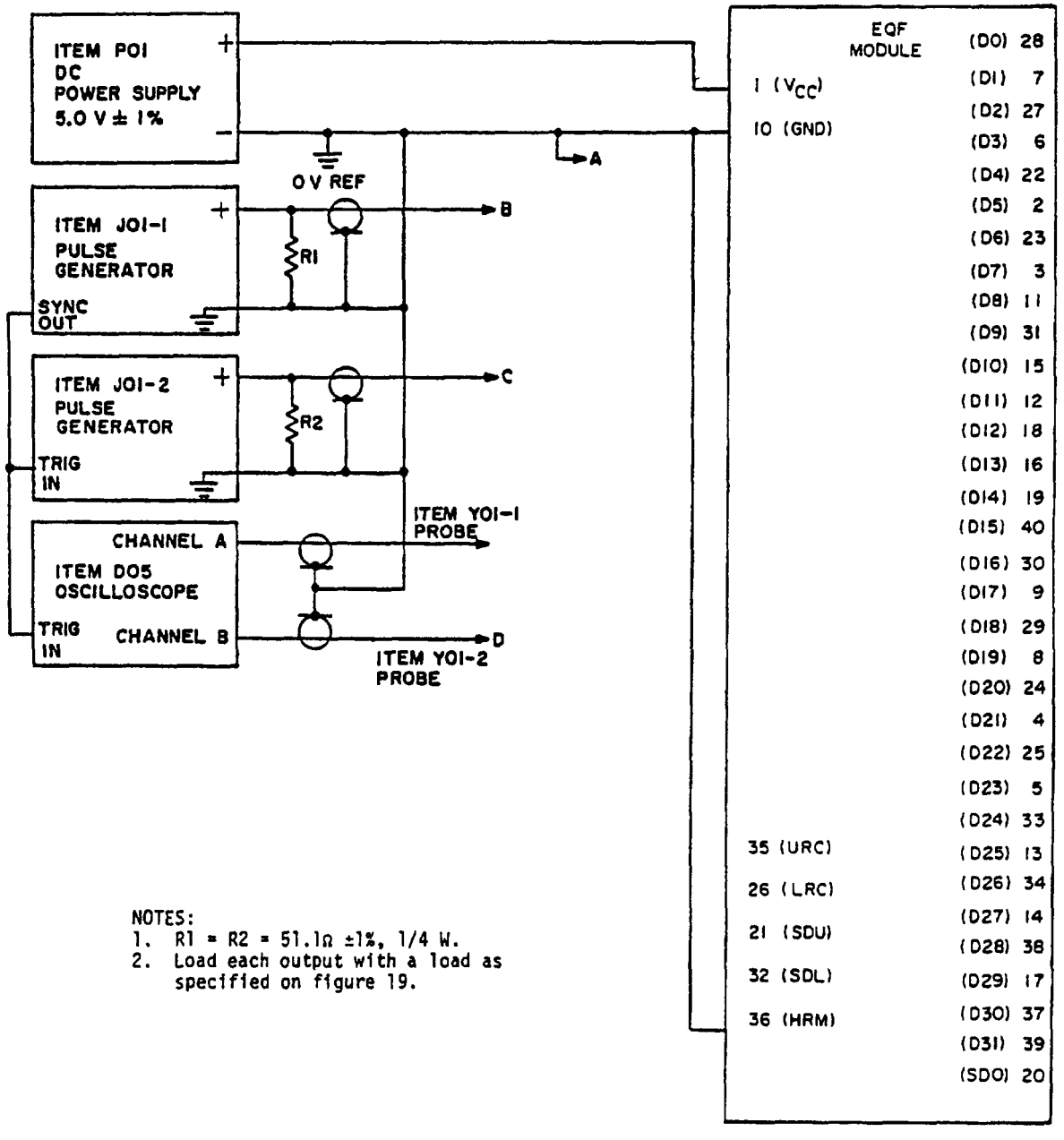
FIGURE 29. SDO maximum operating frequency check, minimum URC clock pulse width check, and SDU low level data setup and hold time check test circuit waveforms for test conditions T3 and T4, table XX.

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NOTES:
1. SDU data input pulse parameters:
   PWA = 25 ±1.0 ns.
   PWB = 115 ±3 ns.
   $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
   $t_{setup} = 20 \pm 1$ ns.
   $t_{hold} = 5$ +1.0 ns, -0.0 ns.
   PWA = $t_{setup} + t_{hold} = 25$ ns.
2. URC input pulse parameters:
   PWC = 20 +1 ns, -0.0 ns.
   PWD = 50 ±3 ns.
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
   PWC + PWD = Maximum operating frequency.
3. Item J01-2 is operated in the double pulse mode. Using item D05, adjust the amplitude and dc offset controls to obtain the required input logic levels of the data and upper rank clock (URC) input pulses. Using the delay and pulse width controls, adjust items J01-1 and J01-2 so that the criteria of notes 1 and 2 are met.
4. Verify that all outputs have a high level $\geq$ 2.5 V and a low level $\leq$ 0.4 V before making measurements for table XX.
5. The physical length of electrical conductors between resistors R1, R2, and R3 and input pins between probes Y01-1 and Y01-2 and input pins, and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to 0 V reference unless otherwise specified.

FIGURE 29. SDO maximum operating frequency check, minimum URC clock pulse width check, and SDU low level data setup and hold time check test circuit waveforms for test conditions T3 and T4, table XX - Continued.

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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/4 W.
2. Load each output with a load as specified on figure 19.

FIGURE 30. D31 maximum operating frequency check, minimum LRC clock pulse width check, and SDL input data setup and hold time check test circuit for test conditions T5 - T8, table XX.

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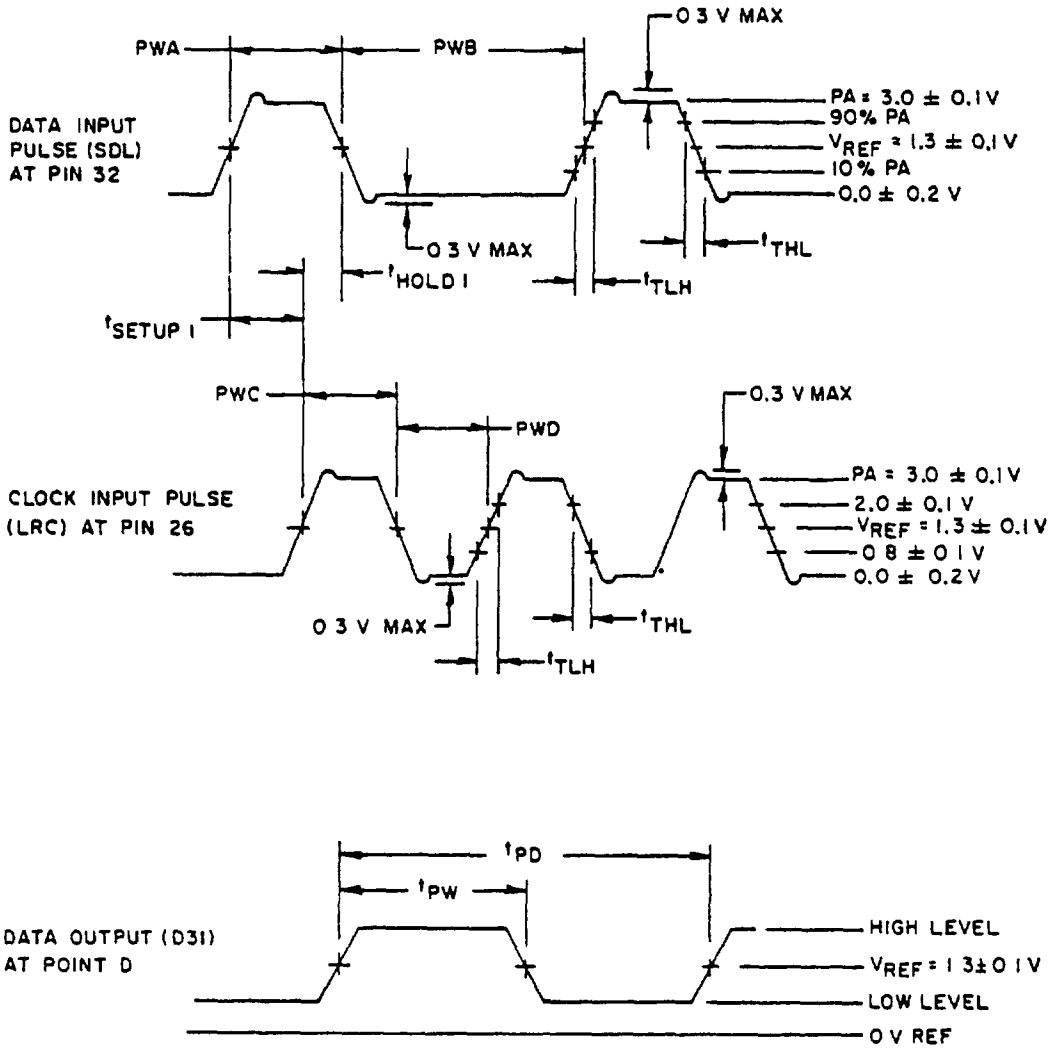


FIGURE 31 D31 maximum operating frequency check, minimum LRC clock pulse width check, and SDL high level data setup and hold time check test circuit waveforms for test conditions T5 and T6, table XX.

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NOTES:
1. SDL data input pulse parameters:
   PWA = 25 ±1.0 ns.
   PWB = 115 ±3 ns.
   $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
   $t_{setup\ 1} = 20 \pm 1$ ns.
   $t_{hold\ 1} = 5\ {}^{+1.0}_{-0.0}$ ns.
   $PWA = t_{setup} + t_{hold} = 25$ ns.
2. LRC input pulse parameters:
   PWC = 20 +1 ns, -0.0 ns.
   PWD = 50 ±3 ns.
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
   PWC + PWD = Maximum operating frequency.
3. Item J01-2 is operated in the double pulse mode. Using item D05, adjust the amplitude and dc offset controls to obtain the required input logic levels of the data and lower rank clock (LRC) input pulses. Using the delay and pulse width controls, adjust items J01-1 and J01-2 so that the criteria of notes 1 and 2 are met.
4. Verify that all outputs have a high level $\geq$ 2.5 V and a low level $\leq$ 0.4 V before making measurements for table XX.
5. The physical length of electrical conductors between resistors R1 and R2 and input pins; between probes Y01-1 and Y01-2 and input pins, and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to 0 V reference unless otherwise specified.

FIGURE 31. D31 maximum operating frequency check, minimum LRC clock pulse width check, and SDL high level data setup and hold time check test circuit waveforms for test conditions T5 and T6, table XX - Continued.

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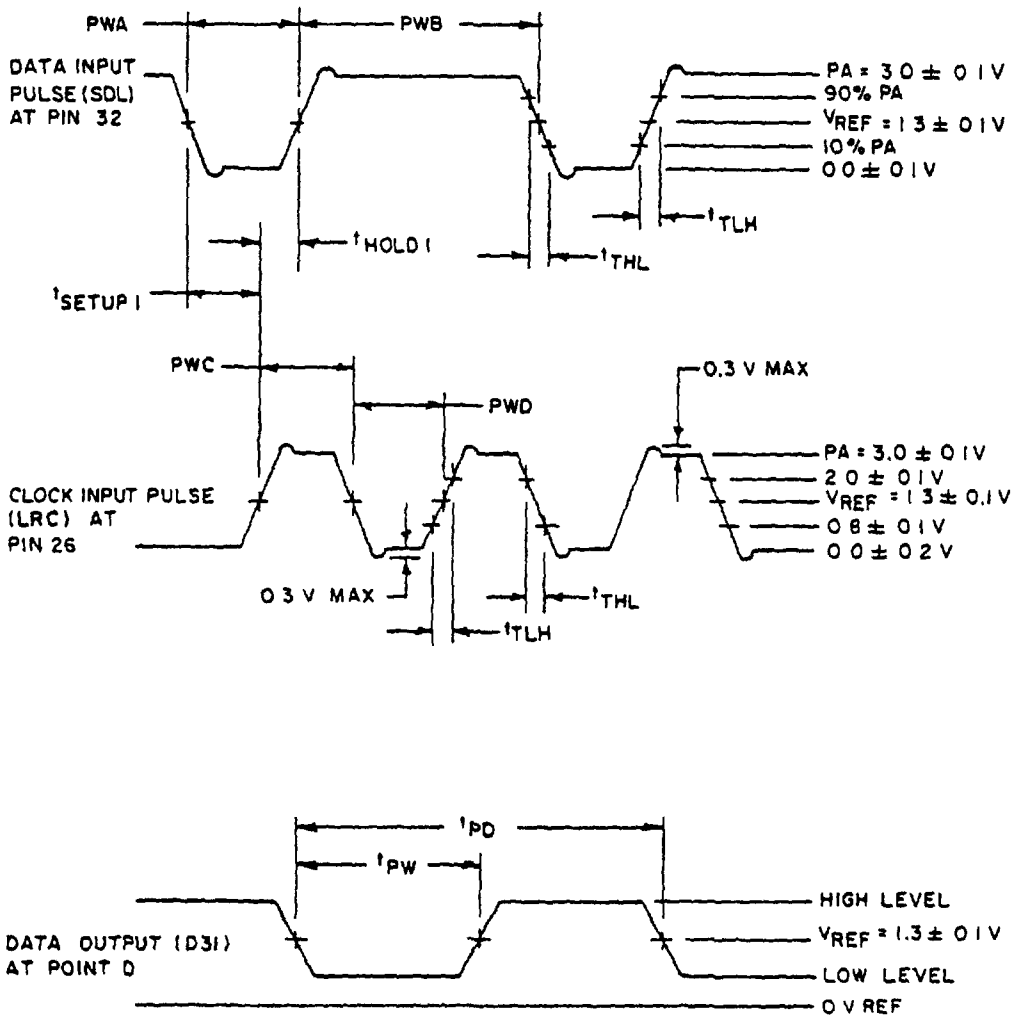


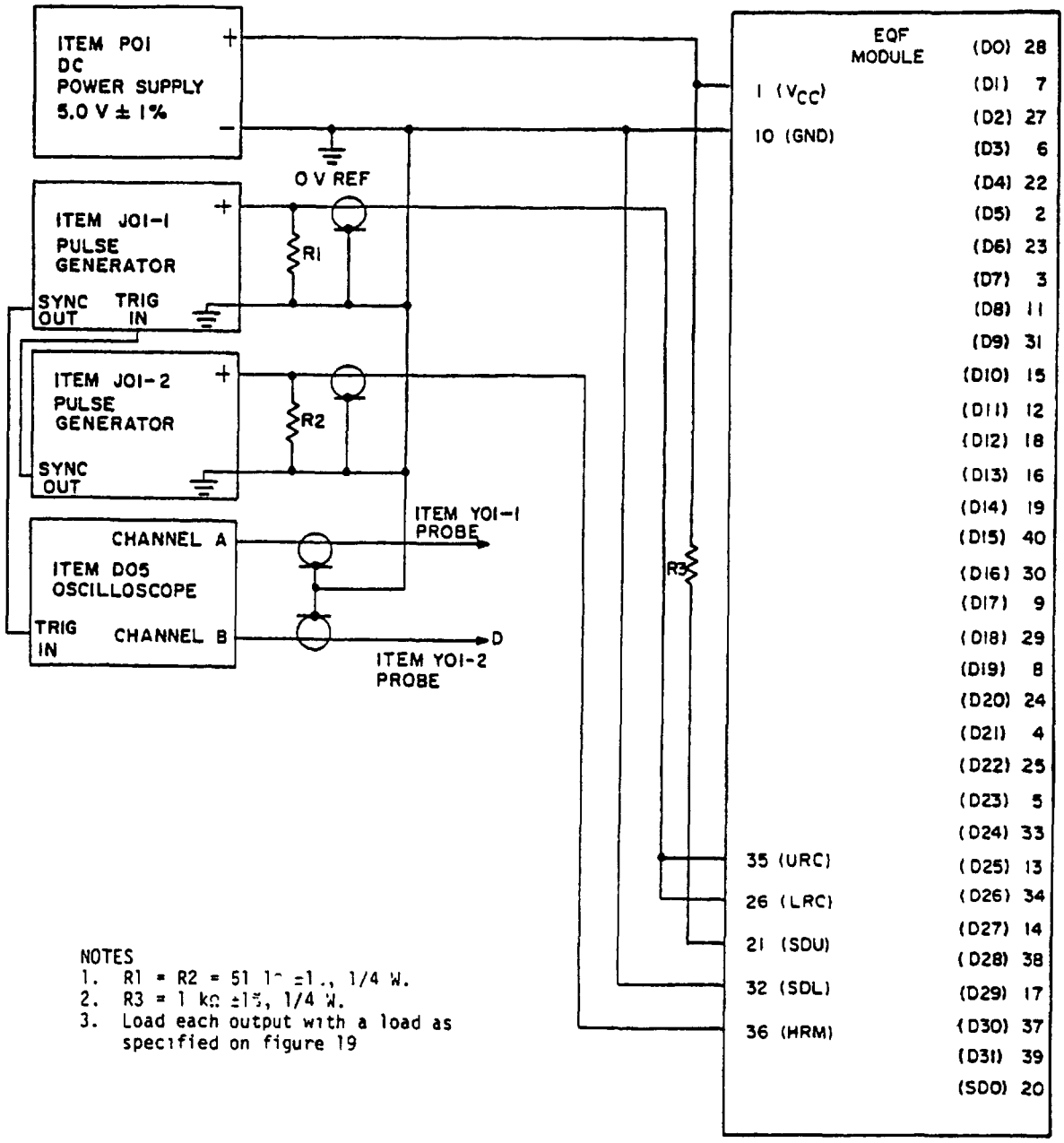
FIGURE 32. D31 maximum operating frequency check, minimum LRC clock pulse width check, and SDL low level data setup and hold time check test circuit waveforms for test conditions T7 and T8, table XX.

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NOTES:
1. SDL data input pulse parameters:
   PWA = 25 ±1.0 ns.
   PWB = 115 ±3 ns.
   $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
   $t_{setup\ 1} = 20 \pm 1$ ns.
   $t_{hold\ 1} = 5\ ^{+1.0}_{-0.0}$ ns.
   $PWA = t_{setup} + t_{hold} = 25$ ns.
2. LRC input pulse parameters:
   PWC = 20 +1 ns, -0.0 ns.
   PWD = 50 ±3 ns.
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
   PWC + PWD = Maximum operating frequency.
3. Item J01-2 is operated in the double pulse mode. Using item D05, adjust the amplitude and dc offset controls to obtain the required input logic levels of the data and lower rank clock (LRC) input pulses. Using the delay and pulse width controls, adjust items J01-1 and J01-2 so that the criteria of notes 1 and 2 are met.
4. Verify that all outputs have a high level $\geq$ 2.5 V and a low level $\leq$ 0.4 V before making measurements for table XX.
5. The physical length of electrical conductors between resistors R1 and R2 and input pins, between probes Y01-1 and Y01-2 and input pins, and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to 0 V reference unless otherwise specified.

FIGURE 32. D31 maximum operating frequency check, minimum LRC clock pulse width check, and SDL low level data setup and hold time check test circuit waveforms for test conditions T7 and T8, table XX - Continued.

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NOTES
1. R1 = R2 = 51 Ω ±1%, 1/4 W.
2. R3 = 1 kΩ ±1%, 1/4 W.
3. Load each output with a load as specified on figure 19

FIGURE 33 HRM setup and hold time check for parallel load and serial shift test circuit for test conditions T9 and T10, table XX

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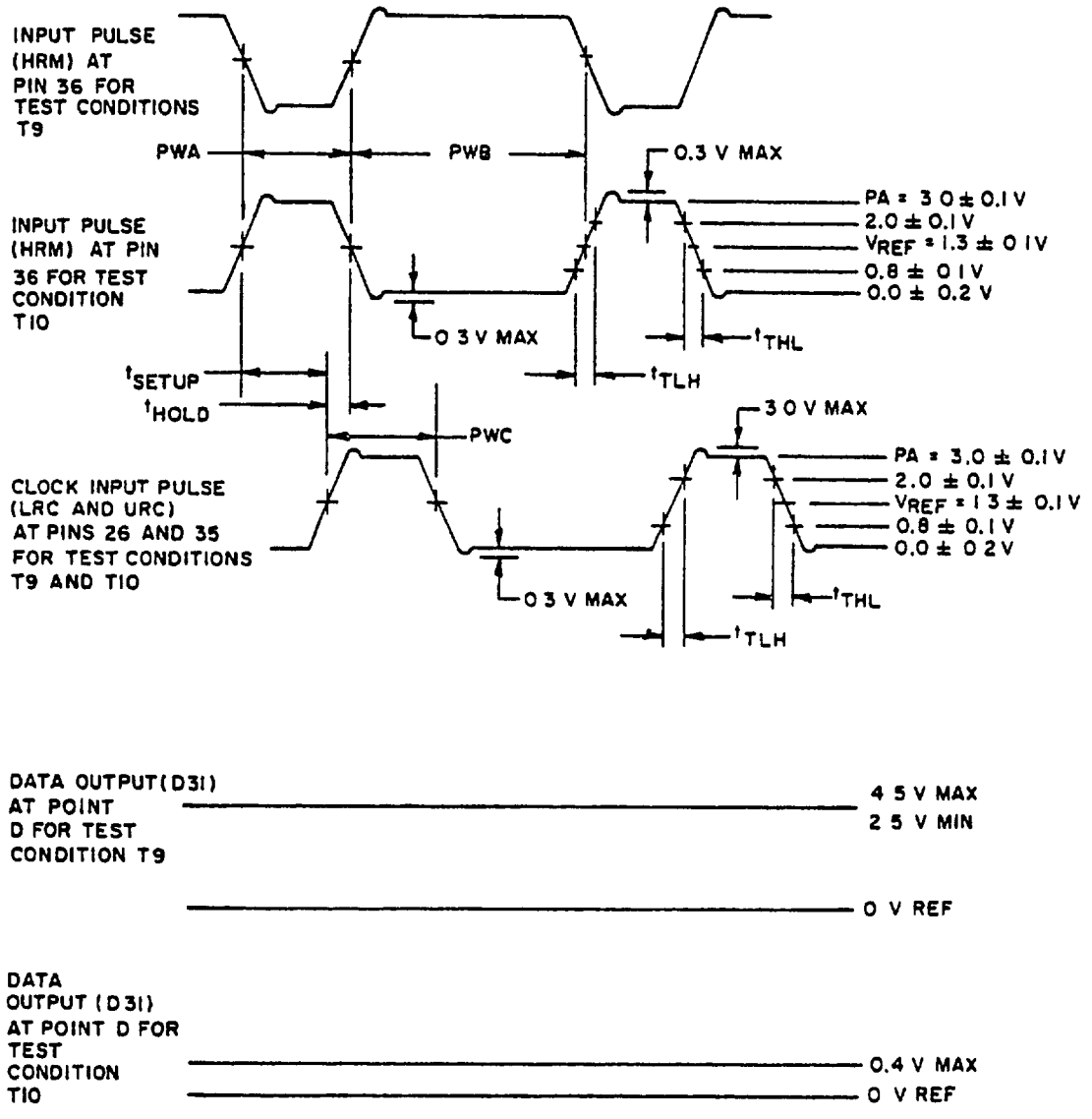


FIGURE 34. HRM setup and hold time check for parallel load and serial shift test circuit waveforms for test conditions T9 and T10, table XX.

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NOTES:
1. HRM input pulse parameters:
   PWA = 45 +2 ns, -0.0 ns.
   PWB = 200 ±10 ns.
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
   $t_{setup} = 40$ +1 ns, -0.0 ns.
   $t_{hold} = 5$ +1 ns, -0.0 ns.
2. URC and LRC input pulse parameters:
   PWC = 25 ±2 ns.
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
3. Using item D05, adjust the amplitude and dc offset controls to obtain the required input logic levels of HRM and clock. Using the delay, pulse width and frequency controls, adjust items J01-1 and J01-2 so that the criteria of notes 1 and 2 are met.
4. Verify that all outputs have a high level $\geq$ 2.5 V and a low level $\leq$ 0.4 V before making measurements for table XX.
5. The physical length of electrical conductors between resistors R1, R2, and R3 and input pins, between probes Y01-1 and Y01-2 and input pins, and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to 0 V reference unless otherwise specified.

FIGURE 34. HRM setup and hold time check for parallel load and serial shift test circuit waveforms for test conditions T9 and T10, table XX - Continued.

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TABLE XXI. Output crosstalk (low-level) test.

Test condition	Parameter		Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
U1	D0	Output crosstalk, low level	Open	32	6	28	B1-B31	B0	21	0	800	mV	35,36, 37,38
U2	D0			21	6	28	B1-B31	B0	32				
U3	D1			32	28	7	B0, B2-B31	B1	21				
U4	D1			21	28	7	B0, B2-B31	B1	32				
U5	D2			32	7	27	B0, B1, B3-B31	B2	21				
U6	D2			21	7	27	B0, B1, B3-B31	B2	32				
U7	D3			32	27	6	B0-B2, B4-B31	B3	21				
U8	D3	Output crosstalk, low level	Open	21	27	6	B0-B2, B4-B31	B3	32	0	800	mV	35,36, 37,38

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TABLE XXI. Output crosstalk (low-level) test -Continued.

Test condition	Parameter	Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
U9	D4 Output crosstalk, low level	Open	32	3	22	B0-B3, B5-B31	B4	21	0	800	mV	35,36, 37,38
U10	D4		21	3	22	B0-B3, B5-B31	B4	32				
U11	D5		32	22	2	B0-B4, B6-B31	B5	21				
U12	D5		21	22	2	B0-B4, B6-B31	B5	32				
U13	D6		32	2	23	B0-B5, B7-B31	B6	21				
U14	D6		21	2	23	B0-B5, B7-B31	B6	32				
U15	D7 Output crosstalk, low level	Open	32	23	3	B0-B6, B8-B31	B7	21	0	800	mV	35,36, 37,38

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TABLE XXI. Output crosstalk (low-level) test -Continued.

Test condition	Parameter	Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
U16	D7 Output crosstalk, low level	Open	21	23	3	B0-B6, B8-B31	B7	32	0	800	mV	35,36, 37,38
U17	D8		32	12	11	B0-B7, B9-B31	B8	21				
U18	D8		21	12	11	B0-B7, B9-B31	B8	32				
U19	D9		32	11	31	B0-B8, B10-B31	B9	21				
U20	D9		21	11	31	B0-B8, B10-B31	B9	32				
U21	D10		32	31	15	B0-B9, B11-B31	B10	21				
U22	D10 Output crosstalk, low level	Open	21	31	15	B0-B9, B11-B31	B10	32	0	800	mV	35,36, 37,38

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TABLE XXI. Output crosstalk (low-level) test -Continued.

Test condition	Parameter	Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
U23	D11 Output crosstalk, low level	Open	32	15	12	B0-B10, B12-B31	B11	21	0	800	mV	35,36, 37,38
U24	D11		21	15	12	B0-B10, B12-B31	B11	32				
U25	D12		32	40	18	B0-B11, B13-B31	B12	21				
U26	D12		21	40	18	B0-B11, B13-B31	B12	32				
U27	D13		32	18	16	B0-B12, B14-B31	B13	21				
U28	D13		21	18	16	B0-B12, B14-B31	B13	32				
U29	D14 Output crosstalk, low level	Open	32	16	19	B0-B13, B15-	B14	21	0	800	mV	35,36, 37,38

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TABLE XXI. Output crosstalk (low-level) test -Continued.

Test condition	Parameter	Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
U30	D14 Output crosstalk, low level	Open	21	16	19	B0-B13, B15-B31	B14	32	0	800	mV	35,36, 37,38
U31	D15		32	19	40	B0-B14, B16-B31	B15	21				
U32	D15		21	19	40	B0-B14, B16-B31	B15	32				
U33	D16		32	8	30	B0-B15, B17-B31	B16	21				
U34	D16		21	8	30	B0-B15, B17-B31	B16	32				
U35	D17		32	30	9	B0-B16, B18-B31	B17	21				
U36	D17 Output crosstalk, low level	Open	21	30	9	B0-B16, B18-B31	B17	32	0	800	mV	35,36, 37,38

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TABLE XXI. Output crosstalk (low-level) test -Continued.

Test condition	Parameter	Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
U37	D18 Output crosstalk, low level	Open	32	9	29	B0-B17, B19-B31	B18	21	0	800	mV	35,36, 37,38
U38	D18		21	9	29	B0-B17, B19-B31	B18	32				
U39	D19		32	29	8	B0-B18, B20-B31	B19	21				
U40	D19		21	29	8	B0-B18, B20-B31	B19	32				
U41	D20		32	5	24	B0-B19, B21-B31	B20	21				
U42	D20		21	5	24	B0-B19, B21-B31	B20	32				
U43	D21 Output crosstalk, low level	Open	32	24	4	B0-B20, B22-B31	B21	21	0	800	mV	35,36, 37,38

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TABLE XXI. Output crosstalk (low-level) test -Continued.

Test condition	Parameter	Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
U44	D21 Output crosstalk, low level	Open	21	24	4	B0-B20, B22-B31	B21	32	0	800	mV	35,36, 37,38
U45	D22		32	4	25	B0-B21, B23-B31	B22	21				
U46	D22		21	4	25	B0-B21, B23-B31	B22	32				
U47	D23		32	25	5	B0-B22, B24-B31	B23	21				
U48	D23		21	25	5	B0-B22, B24-B31	B23	32				
U49	D24		32	14	33	B0-B23, B25-B31	B24	21				
U50	D24 Output crosstalk, low level	Open	21	14	33	B0-B23, B25-B31	B24	32	0	800	mV	35,36, 37,38

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TABLE XXI. Output crosstalk (low-level) test -Continued.

Test condition	Parameter	Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
U51	D25 Output crosstalk, low level	Open	32	33	13	B0-B24, B26-B31	B25	21	0	800	mV	35,36, 37,38
U52	D25		21	33	13	B0-B24, B26-B31	B25	32				
U53	D26		32	13	34	B0-B25, B27-B31	B26	21				
U54	D26		21	13	34	B0-B25, B27-B31	B26	32				
U55	D27		32	34	14	B0-B26, B28-B31	B27	21				
U56	D27		21	34	14	B0-B26, B28-B31	B27	32				
U57	D28 Output crosstalk, low level	Open	32	39	38	B0-B27, B29-B31	B28	21	0	800	mV	35,36, 37,38

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TABLE XXI. Output crosstalk (low-level) test -Continued.

Test condition	Parameter	Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
U58	D28 Output crosstalk, low level	Open	21	39	38	B0-B27, B29-B31	B28	32	0	800	mV	35,36, 37,38
U59	D29		32	38	17	B0-B28, B30-B31	B29	21				
U60	D29		21	38	17	B0-B28, B30-B31	B29	32				
U61	D30		32	17	37	B0-B29, B31	B30	21				
U62	D30		21	17	37	B0-B29, B31	B30	32				
U63	D31		32	37	39	B0-B30	B31	21				
U64	D31		21	37	39	B0-B30	B31	32				
U65	SDO		32	37	20	B0-B30	B31	21				
U66	SDO Output crosstalk, low level	Open	21	37	20	B0-B30	B31	32	0	800	mV	35,36, 37,38

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TABLE XXII. Output crosstalk (high-level) test.

Test condition	Parameter		Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
V1	D0	Output crosstalk, high level	21	Open	6	28	B0	B1-B31	32	0	400	mV	35,36, 37,38
V2	D0		32		6	28	B0	B1-B31	21				
V3	D1		21		28	7	B1	B0, B2-B31	32				
V4	D1		32		28	7	B1	B0, B2-B31	21				
V5	D2		21		7	27	B2	B0, B1, B3-B31	32				
V6	D2		32		7	27	B2	B0, B1, B3-B31	21				
V7	D3		21		27	6	B3	B0-B2, B4-B31	32				
V8	D3	Output crosstalk, high level	32	Open	27	6	B3	B0-B2, B4-B31	21	0	400	mV	35,36, 37,38

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TABLE XXII. Output crosstalk (high-level test -Continued.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Connect G to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
V9	D4	Output crosstalk, high level	21	Open	3	22	B4	B0-B3, B5-B31	32	0	400	mV	35,36 37,38
V10	D4		32		3	22	B4	B0-B3, B5-B31	21				
V11	D5		21		22	2	B5	B0-B4, B6-B31	32				
V12	D5		32		22	2	B5	B0-B4, B6-B31	21				
V13	D6		21		2	23	B6	B0-B5, B7-B31	32				
V14	D6		32		2	23	B6	B0-B5, B7-B31	21				
V15	D7	Output crosstalk, high level	21	Open	23	3	B7	B0-B6, B8-B31	32	0	400	mV	35,36, 37,38

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TABLE XXII. Output crosstalk (high-level) test -Continued.

Test condition	Parameter	Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
V16	D7 Output crosstalk, high level	32	Open	23	3	B7	B0-B6, B8-B31	21	0	400	mV	35,36, 37,38
V17	D8	21		12	11	B8	B0-B7, B9-B31	32				
V18	D8	32		12	11	B8	B0-B7, B9-B31	21				
V19	D9	21		11	31	B9	B0-B8, B10-B31	32				
V20	D9	32		11	31	B9	B0-B8, B10-B31	21				
V21	D10	21		31	15	B10	B0-B9, B11-B31	32				
V22	D10 Output crosstalk, high level	32	Open	31	15	B10	B0-B9, B11-B31	21	0	400	mV	35,36, 37,38

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TABLE XXII. Output crosstalk (high-level) test -Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Connect G to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
V23	D11 Output crosstalk, high level	21	Open	15	12	B11	B0-B10, B12-B31	32	0	400	mV	35,36, 37,38
V24	D11	32		15	12	B11	B0-B10, B12-B31	21				
V25	D12	21		40	18	B12	B0-B11, B13-B31	32				
V26	D12	32		40	18	B12	B0-B11, B13-B31	21				
V27	D13	21		18	16	B13	B0-B12, B14-B31	32				
V28	D13	32		18	16	B13	B0-B12, B14-B31	21				
V29	D14 Output crosstalk, high level	21	Open	16	19	B14	B0-B13, B15-B31	32	0	400	mV	35,36 37,38

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TABLE XXII. Output crosstalk (high-level) test -Continued.

Test condition	Parameter		Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
V30	D14	Output crosstalk, high level	32	Open	16	19	B14	B0-B13, B15-B31	21	0	400	mV	35,36, 37,38
V31	D15		21		19	40	B15	B0-B14, B16-B31	32				
V32	D15		32		19	40	B15	B0-B14, B16-B31	21				
V33	D16		21		8	30	B16	B0-B15, B17-B31	32				
V34	D16		32		8	30	B16	B0-B15, B17-B31	21				
V35	D17		21		30	9	B17	B0-B16, B18-B31	32				
V36	D17	Output crosstalk, high level	32	Open	30	9	B17	B0-B16, B18-B31	21	0	400	mV	35,36, 37,38

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TABLE XXII. Output crosstalk (high-level) test -Continued.

Test condition	Parameter	Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
V37	D18 Output crosstalk, high level	21	Open	9	29	B18	B0-B17, B19-B31	32	0	400	mV	35,36, 37,38
V38	D18	32		9	29	B18	B0-B17, B19-B31	21				
V39	D19	21		29	8	B19	B0-B18, B20-B31	32				
V40	D19	32		29	8	B19	B0-B18, B20-B31	21				
V41	D20	21		5	24	B20	B0-B19, B21-B31	32				
V42	D20	32		5	24	B20	B0-B19, B21-B31	21				
V43	D21 Output crosstalk, high level	21	Open	24	4	B21	B0-B20, B22-B31	32	0	400	mV	35,36, 37,38

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TABLE XXII. Output crosstalk (high-level) test -Continued.

Test condition	Parameter	Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
V44	D21 Output crosstalk, high level	32	Open	24	4	B21	B0-B20, B22-B31	21	0	400	mV	35,36, 37,38
V45	D22	21		4	25	B22	B0-B21, B23-B31	32				
V46	D22	32		4	25	B22	B0-B21, B23-B31	21				
V47	D23	21		25	5	B23	B0-B22, B24-B31	32				
V48	D23	32		25	5	B23	B0-B22, B24-B31	21				
V49	D24	21		14	33	B24	B0-B23, B25-B31	32				
V50	D24 Output crosstalk, high level	32	Open	14	33	B24	B0-B23, B25-B31	21	0	400	mV	35,36, 37,38

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TABLE XXII. Output crosstalk (high-level) test -Continued.

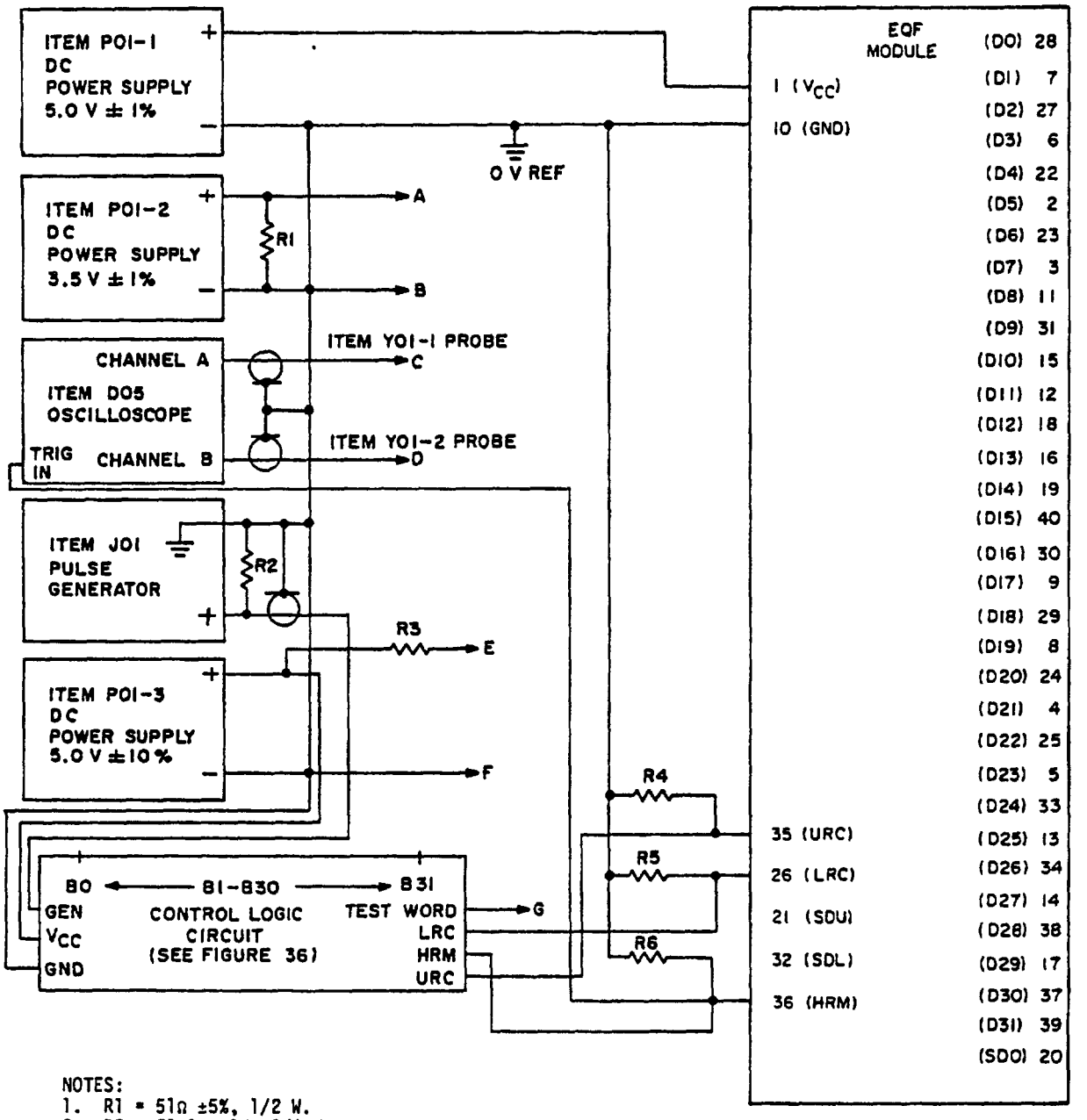
Test condition	Parameter	Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
V51	D25 Output crosstalk, high level	21	Open	33	13	B25	B0-B24, B26-B31	32	0	400	mV	35,36, 37,38
V52	D25	32		33	13	B25	B0-B24, B26-B31	21				
V53	D26	21		13	34	B26	B0-B25, B27-B31	32				
V54	D26	32		13	34	B26	B0-B25, B27-B31	21				
V55	D27	21		34	14	B27	B0-B26, B28-B31	32				
V56	D27	32		34	14	B27	B0-B26, B28-B31	21				
V57	D28 Output crosstalk, high level	21	Open	39	38	B28	B0-B27, B29-B31	32	0	400	mV	35,36 37,38

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TABLE XXII. Output crosstalk (high-level) test -Continued.

Test condition	Parameter		Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
V58	D28	Output crosstalk, high level	32	Open	39	38	B28	B0-B27, B29-B31	21	0	400	mV	35,36 37,38
V59	D29		21		38	17	B29	B0-B28, B30, B31	32				
V60	D29		32		38	17	B29	B0-B28, B30, B31	21				
V61	D30		21		17	37	B30	B0-B29, B31	32				
V62	D30		32		17	37	B30	B0-B29, B31	21				
V63	D31		21		37	39	B31	B0-B30	32				
V64	D31		32		37	39	B31	B0-B30	21				
V65	S00		21		37	20	B31	B0-B30	32				
V66	S00	Output crosstalk, high level	32	Open	37	20	B31	B0-B30	21	0	400	mV	35,36 37,38

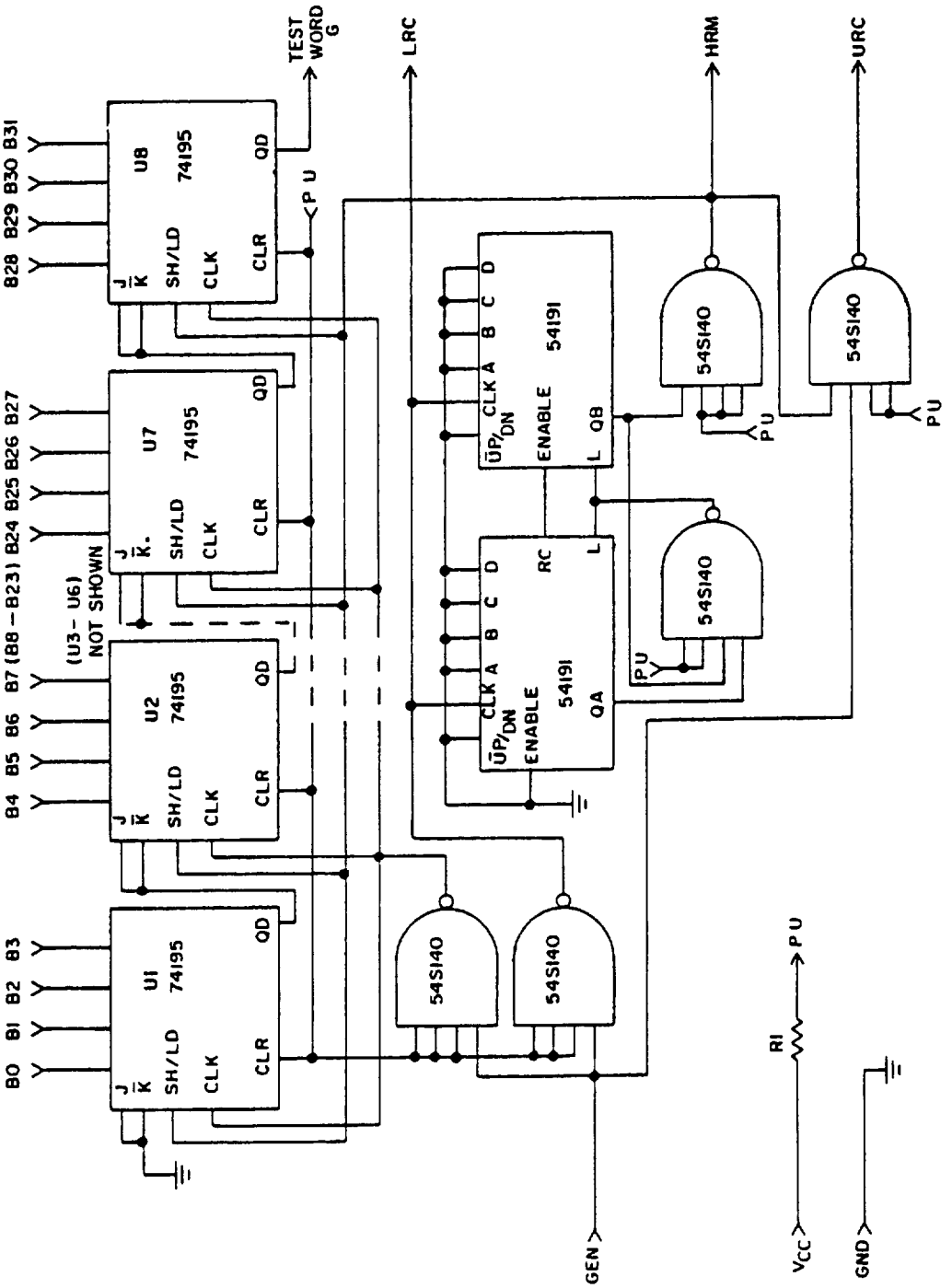
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:43Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. R2 = 51.1Ω ±1%, 1/4 W.
3. R3 = 1.58 kΩ ±1%, 1/8 W.
4. R4, R5, R6 - See notes 5 and 8, figure 37.
5. Load each output with a load as specified on figure 19 with the exception that C_L = 50 pF.

FIGURE 35. Output crosstalk (low-level and high-level) test circuit for tables XXI and XXII.

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NOTE: R1 = 1 kΩ ±5%, 1/2 W.

FIGURE 36. Control logic circuit.

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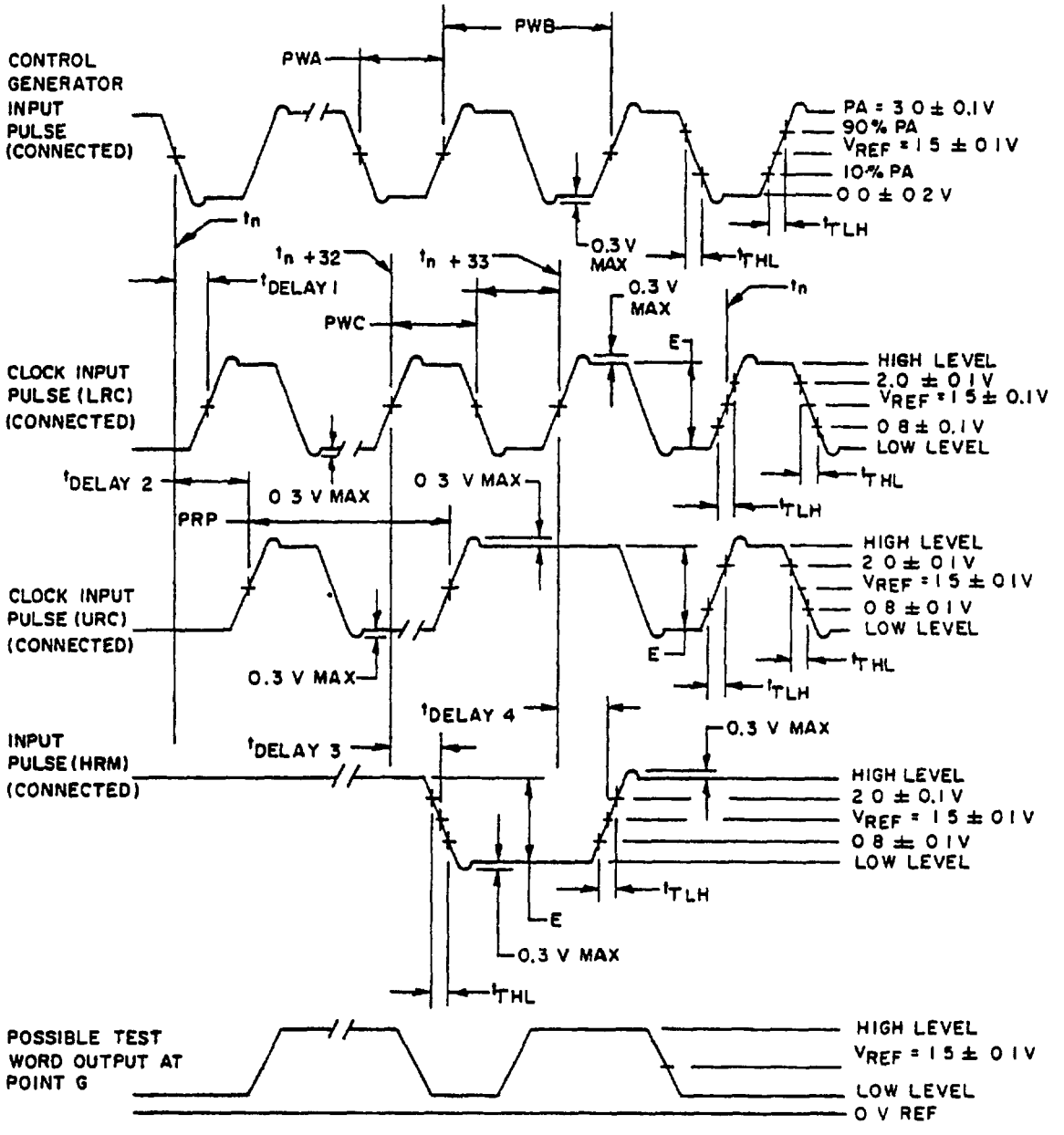


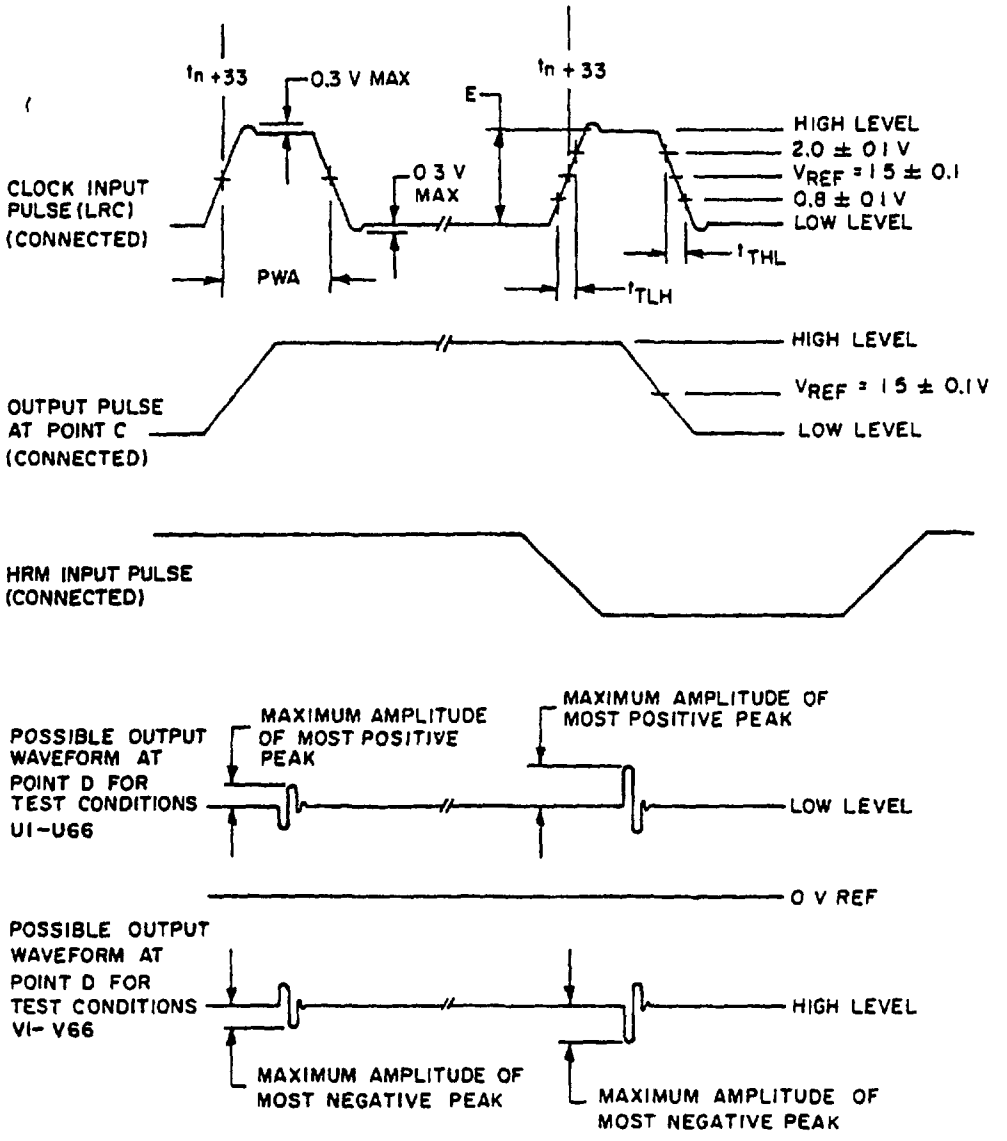
FIGURE 37. Control logic circuit waveforms.

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NOTES.
1. Control generator input pulse parameters:
   PWA = 100 ±10 ns.
   PWB = 200 ±10 ns.
   $t_{TLH} = t_{THL}$ = 10 ±4 ns.
2. LRC clock input pulse parameters:
   PWC = 100 ±10 ns.
   PWD = 200 ±10 ns.
   $t_{TLH} = t_{THL}$ = 4 ns maximum.
   $t_{delay\ 1}$ = 5 ±2 ns.
3. URC clock input pulse parameters:
   PWE = 100 ±10 ns.
   PRP = 300 ±10 ns.
   $t_{delay\ 2}$ = 30 ±8 ns.
   $t_{TLH} = t_{THL}$ = 4 ns maximum.
4. HRM input pulse parameters.
   $t_{delay\ 3}$ = 20 ±10 ns.
   $t_{delay\ 4}$ = 50 ±20 ns.
   $t_{TLH} = t_{THL}$ = 4 ns maximum.
5. The physical length of electrical conductors between the LRC clock, URC clock, and HRM (on the control logic circuit) and input pins shall be less than 6 inches (152.4 mm) Resistors shall be used at the LRC, URC, and HRM input pins and the value of the resistors selected to meet the 0.3 V maximum overshoot requirement.
6. For 32 positive transitions of the LRC or URC clock, the test word shall be shifted into the upper or lower rank of the EQF and a second word (32 bits of logic low or logic high) shall be shifted into the remaining rank of the EQF. At the 33rd bit time (t_n + 33), the data contained in the upper rank of the EQF shall be loaded into the lower rank and the output crosstalk measured. The test word (at t_n + 33) shall be loaded into registers U1 through U8. The 33rd positive clock transition of the URC clock shall be inhibited to prevent shifting the upper rank data.
7. All voltages are referenced to 0 V reference unless otherwise specified.
8. Resistors R4, R5, and R6 are used to terminate the control logic circuit outputs if excessive overshoot is present Values should be selected for optimum effectiveness in the range of 75Ω to 200Ω.

FIGURE 37 Control logic circuit waveforms - Continued.

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NOTES:
1. LRC clock input pulse parameters:
   PWA = 100 ±10 ns.
   $t_{TLH} = t_{THL}$ = 4 ns maximum.
2. The physical length of electrical conductors between resistor R1 and input pins, between probes Y01-1 and Y01-2 and input pins; and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
3. All voltages are referenced to 0 V reference unless otherwise specified.

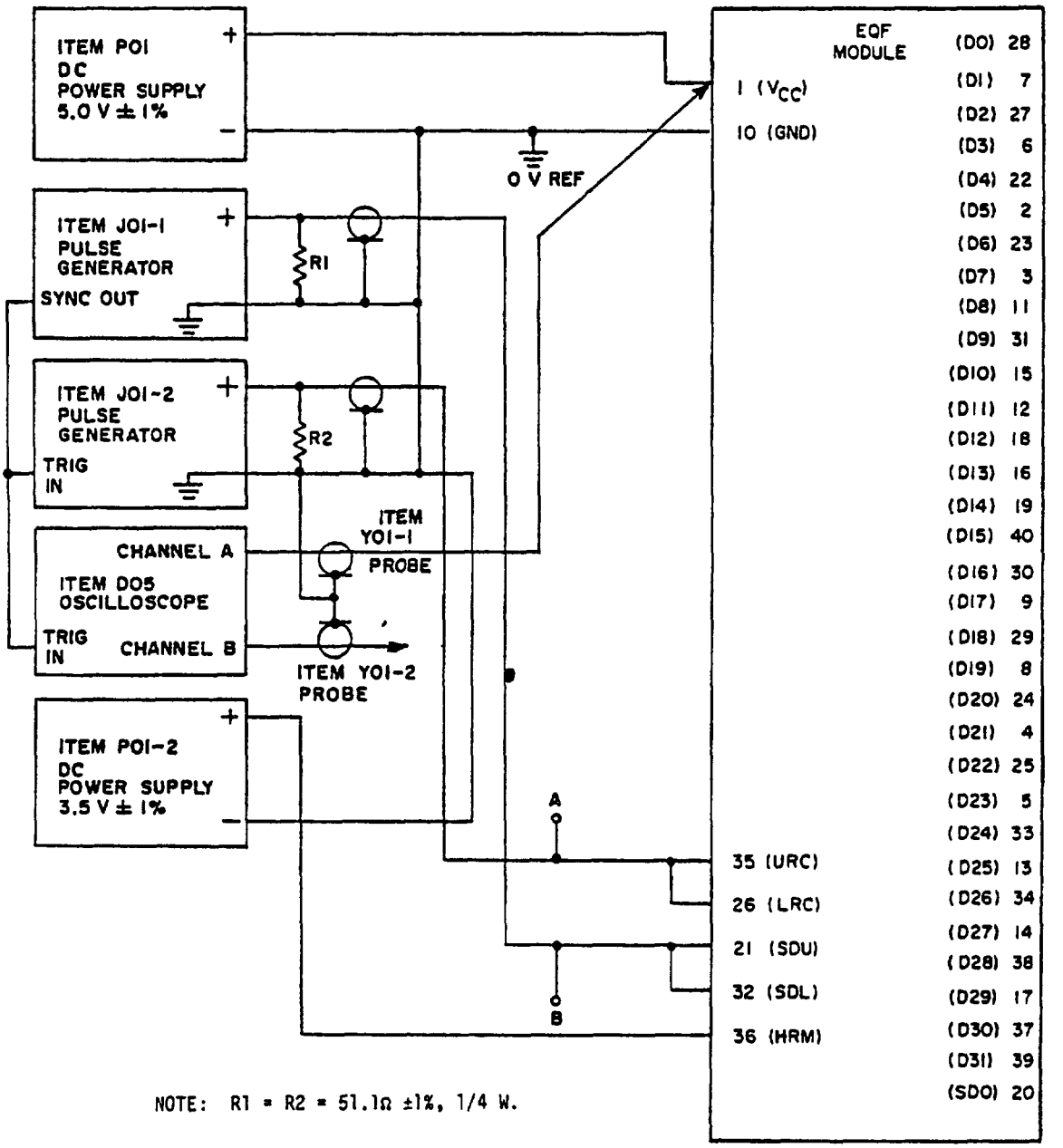
FIGURE 38. Output crosstalk (low-level and high-level) test circuit waveforms for tables XXI and XXII.

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TABLE XXIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
W	Power supply transient amplitude	0	600	mV	39,40

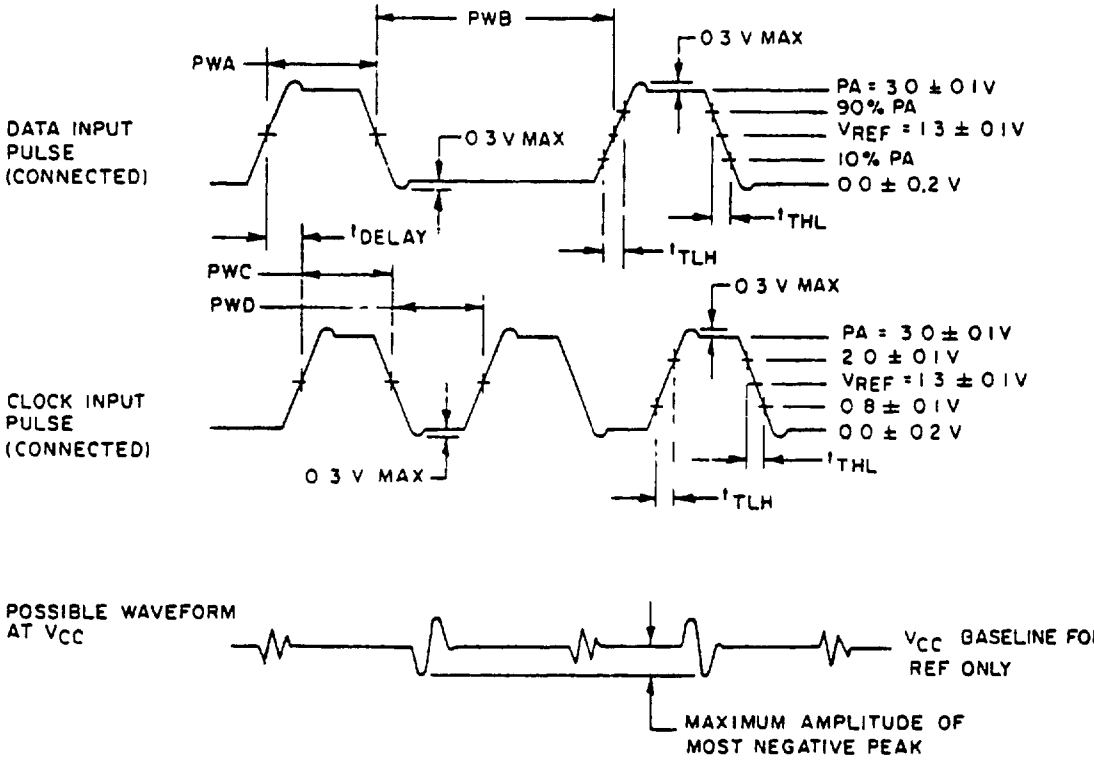
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NOTE: R1 = R2 = 51.1Ω ±1%, 1/4 W.

FIGURE 39. Power supply transient amplitude test circuit for table XXIII.

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NOTES:
1. Data input pulse parameters·
   PWA = 500 ±50 ns.
   PWB = 500 ±50 ns.
   $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
2 Clock input pulse parameters
   PWC = 250 ±25 ns.
   PWD = 250 ±25 ns
   $t_{TLH} = t_{THL} = 10 \pm 1$ ns.
   $t_{delay} = 250 \pm 25$ ns
3 Item J01-2 is operated in the double pulse mode. Using item D05, adjust the controls of items J01-1 and J01-2 to obtain $t_{delay} = 250 \pm 25$ ns and the required pulses for clock and data inputs
4. All voltages are referenced to 0 V reference unless otherwise specified.

FIGURE 40 Power supply transient amplitude test circuit waveforms for table XXIII.

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TABLE XXIV. Power supply filter capacitance test.

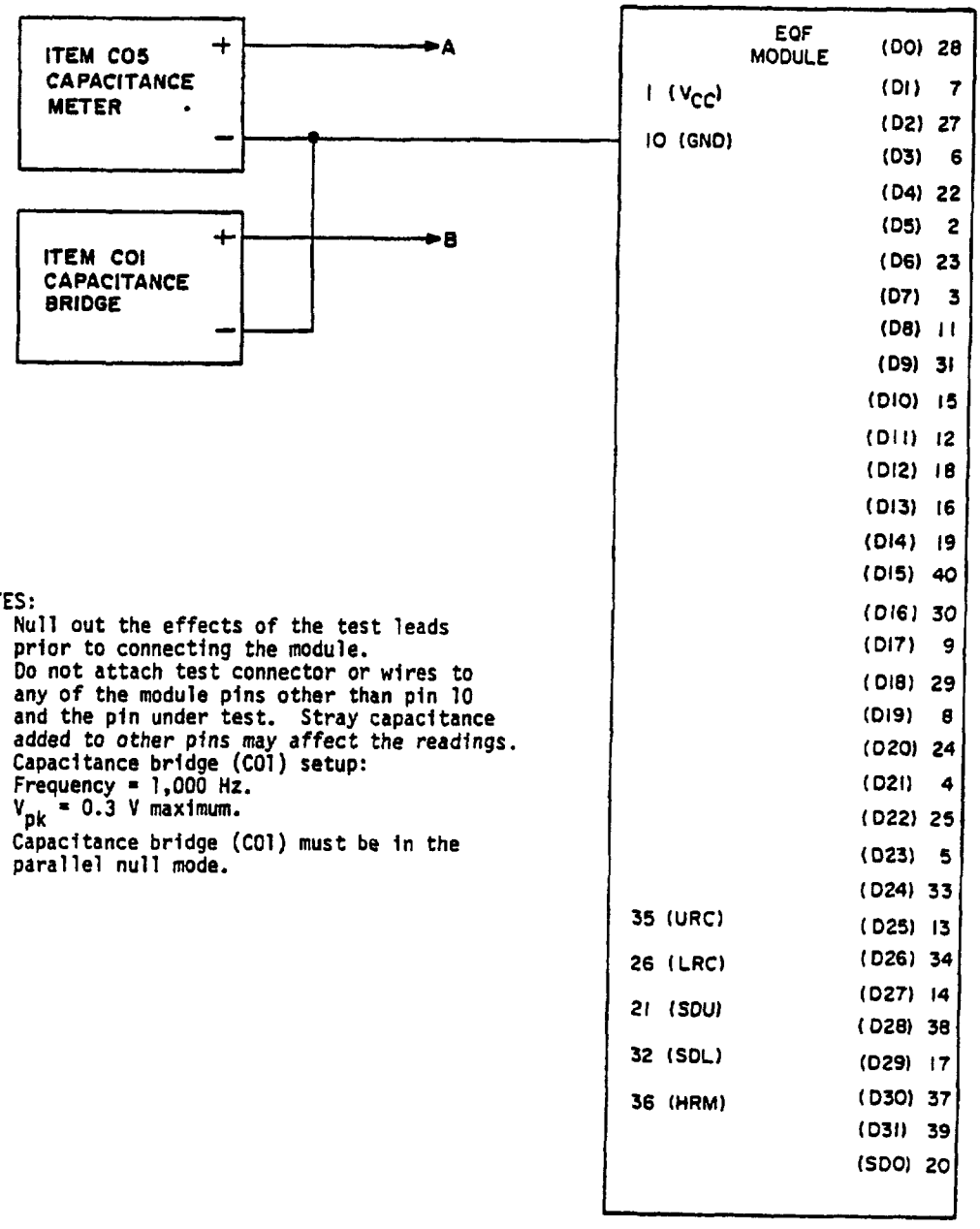
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X	Power supply filter capacitance	Open	1	0.14	0.26	µF	41

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TABLE XXV. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Y1	SDU Input capacitance	21	Open	2	30	pF	41
Y2	SDL	32			30		
Y3	HRM	36			90		
Y4	URC	35			90		
Y5	LRC Input capacitance	26	Open	2	90	pF	41

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (C01) setup:
   Frequency = 1,000 Hz.
   V_{pk} = 0.3 V maximum.
4. Capacitance bridge (C01) must be in the parallel null mode.

FIGURE 41. Power supply filter and input capacitance test circuit for tables XXIV and XXV.

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